Math 230B Notes

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Chapter 1

Differentiation

1.1 The Derivative of a Function

Definition 1.1.1. (Differentiability and the Derivative)
Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, and $c \in I$.

(*i*) We say f is differentiable at c if

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists (that is, it equals a real number). In this case, the quantity $\lim_{x \to c} \frac{f(x)-f(c)}{x-c}$ is called the derivative of f at c and is denoted by

$$f'(c), \quad \frac{df}{dx}(c), \quad \frac{df}{dx}|_{x=c}$$

(*ii*) If $f : I \to \mathbb{R}$ is differentiable at every point $c \in I$, we say f is differentiable (on I).

Remark 1.1.1. Note that

$$\begin{aligned}
f'(c) = L &\iff \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = L \\
&\iff \forall \epsilon > 0 \ \exists \delta > 0 \text{ such that if } 0 < |x - c| < \delta, \text{ then } \left| \frac{f(x) - f(c)}{x - c} - L \right| < \epsilon \\
&\iff \forall \epsilon > 0 \ \exists \delta > 0 \text{ such that if } 0 < |h < \delta, \text{ then } \left| \frac{f(c + h) - f(c)}{h} - L \right| < \epsilon \quad \text{(Let } h = x - c\text{)} \\
&\iff \lim_{h \to 0} \frac{f(c + h) - f(c)}{h} = L
\end{aligned}$$

Remark 1.1.2. Let A denote the collection of all points at which $f : I \to \mathbb{R}$ is differentiable. If $A \neq \emptyset$, the function $f' : A \to \mathbb{R}$ defined by

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \quad \forall c \in A$$

is called the derivative of f.

Example 1.1.1. Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be given by $f(x) = x^2$. Prove that f is differentiable on I and find the derivative.

Proof. $\forall c \in I$,

$$\begin{aligned}
\lim_{x \to c} \frac{f(x) - f(c)}{x - c} &= \lim_{x \to c} \frac{x^2 - c^2}{x - c} \\
&= \lim_{x \to c} x + c \\
&= 2c \qquad \text{(is continuous)}
\end{aligned}$$

So, $\forall c \in I \quad f'(c) = 2c$. Hence,

$$f' : I \to \mathbb{R}, \quad f'(x) = 2x.$$

□

Example 1.1.2. Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be given by $f(x) = x^n$ where $n \in \mathbb{N}, \quad n \geq 3$. Prove that f is differentiable on I and find the derivative.

Proof.

$$\begin{aligned}
\lim_{x\to c} \frac{f(x) - f(c)}{x - c} &= \lim_{x\to c} \frac{x^n - c^n}{x - c} \\
&= \lim_{x\to c} \frac{(x - c)(x^{n-1} + cx^{n-2} + ... + c^{n-1})}{x - c} && \text{(Algebra)} \\
&= \lim_{x\to c} \left[x^{n-1} + cx^{n-1} + ... + c^{n-1}\right] \\
&= c^{n-1} + c \cdot c^{n-2} + ... + c^{n-1} && \text{(Continuity)} \\
&= n \cdot c^{n-1}
\end{aligned}$$

So, $\forall c \in I \quad f'(c) = n \cdot c^{n-1}$. Hence,

$$f' : I \to \mathbb{R}, \quad f'(x) = nx^{n-1}.$$

□

Example 1.1.3. Prove that $f : \mathbb{R} \to \mathbb{R}, \quad f(x) = |x|$ is not differentiable at $c = 0$.

Proof. We need to show that $\lim_{x\to c} \frac{f(x)-f(0)}{x-0}$ does not exist. Note that

$$\frac{f(x) - f(0)}{x - 0} = \frac{|x| - |0|}{x - 0} = \frac{|x|}{x}$$

Let $g(x) = \frac{|x|}{x}$. We want to show $\lim_{x\to 0} g(x)$ does not exist. By the sequential criterion for limits of functions, it is enough to find two sequences (a_n) and (b_n) in $\mathbb{R}\backslash\{0\}$ such that $a_n \to 0$ and $b_n \to 0$, but $\lim g(a_n) \neq \lim g(b_n)$. Let $a_n = -\frac{1}{n}$ and $b_n = \frac{1}{n}$. Clearly, $a_n \to 0$ and $b_n \to 0$. However,

$$\lim_{n\to\infty} g(a_n) = \lim_{n\to\infty} \frac{|a_n|}{a_n} = \lim_{n\to\infty} \frac{|-1/n|}{-1/n} = \lim_{n\to\infty} (-1) = -1$$

$$\lim_{n\to\infty} g(b_n) = \lim_{n\to\infty} \frac{|b_n|}{b_n} = \lim_{n\to\infty} \frac{|1/n|}{1/n} = \lim_{n\to\infty} (1) = 1$$

□

Theorem 1.1.1. (Differentiable $\implies$ Continuous)
Let $I \subseteq \mathbb{R}$ be an interval, $c \in I$, and $f : I \to \mathbb{R}$ be differentiable at c. Then f is continuous at c.

Proof. It is enough to show that $\lim_{x\to c} f(x) = f(c)$ (an interval doesn't have an isolated point). Note that

$$\begin{aligned}
\lim_{x\to c} (f(x) - f(c)) &= \lim_{x\to c} \left[\frac{f(x) - f(c)}{x - c}(x - c)\right] \\
&= \left[\lim_{x\to c} \frac{f(x) - f(c)}{x - c}\right] \left[\lim_{x\to c}(x - c)\right] && \text{(ALT for Functions)} \\
&= f'(c) \cdot 0 = 0.
\end{aligned}$$

So,

$$\begin{aligned}\lim_{x\to c} f(x) &= \lim_{x\to c}[f(x) - f(c) + f(c)] \\ &= \lim_{x\to c}[f(x) - f(c)] + \lim_{x\to c} f(c) \\ &= 0 + f(c) \\ &= f(c).\end{aligned}$$

□

Corollary 1.1.1. If $f : I \to \mathbb{R}$ is not continuous at $c \in I$, then f is not differentiable at c.

Example 1.1.4. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \begin{cases} x^2 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$.

(*i*) Prove f is continuous at 0.

(*ii*) Prove f is discontinuous at all $x \neq 0$.

(*iii*) Prove that f is nondifferentiable at all $x \neq 0$.

(*iv*) Prove that $f'(0) = 0$.

Proof. (*i*) We need to show that

$$\forall \epsilon > 0\ \exists \delta > 0 \text{ such that if } |x - 0| < \delta \text{ then } |f(x) - f(0)| < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\text{if } |x| < \delta \text{ then } |f(x)| < \epsilon \qquad (*)$$

Informal Discussion

Note that

Case 1: if $x \notin \mathbb{Q}$ then $|f(x)| = |0| < \epsilon$ ✓

Case 2: if $x \in \mathbb{Q}$ then $|f(x)| = |x^2| = |x|^2$

So, we want to find δ such that if $|x| < \delta$, then $|x|^2 < \epsilon$. Clearly, $\delta = \sqrt{\epsilon}$ works.

We claim that $(*)$ holds with $\delta = \sqrt{\epsilon}$. See the discussion.

(*ii*) Let $c \neq 0$. Our goal is to show f is discontinuous at c. By the sequential criterion for continuity, it is enough to find a sequence (a_n) such that $a_n \to c$ but $f(a_n) \not\to f(c)$. We proceed by two cases:

Case 1: $c \notin \mathbb{Q}$

$\mathbb{Q}$ is dense in $\mathbb{R}$, so there exists a sequence of rational numbers (r_n) such that $r_n \to c$. We have

$$\left.\begin{array}{l} f(r_n) = r_n^2\ \forall n \\ f(c) = 0 \end{array}\right\} \implies f(r_n) \not\to f(c)$$

$$\left.\begin{array}{l} r_n \to c \\ f(r_n) \not\to f(c) \end{array}\right\} \implies f \text{ is discontinuous at } c.$$

(*iii*) Let $c \neq 0$. By (*ii*), f is not continuous at c. Therefore, f is not differentiable at c.

(*iv*) We need to show $\lim_{x\to c} \frac{f(x)-f(0)}{x-0} = 0$. Note that

$$\frac{f(x) - f(0)}{x - 0} = \frac{f(x)}{x}.$$

Our goal is to show:

$$\forall \epsilon > 0\ \exists \delta > 0 \text{ such that if } 0 < |x - 0| < \delta \text{ then } \left|\frac{f(x)}{x} - 0\right| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\text{if } 0 < |x| < \delta, \text{ then } \left| \frac{f(x)}{x} - 0 \right| < \epsilon \qquad (*)$$

We claim that $(*)$ holds with $\delta = \epsilon$ (or any postive number less than ϵ). Indeed, if $x \in \mathbb{R}$ such that $0 < |x| < \delta = \epsilon$, then

Case 1: $x \notin \mathbb{Q}$

$$\left| \frac{f(x)}{x} \right| = \left| \frac{0}{x} \right| = 0 < \epsilon.$$

Case 2: $x \in \mathbb{Q}$

$$\left| \frac{f(x)}{x} \right| = \left| \frac{x^2}{x} \right| = |x| < \delta = \epsilon.$$

□

Theorem 1.1.2. (Algebraic Differentiability Theorem)
Assume $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ are differentiable at $c \in I$. Then

(*i*) $\forall k \in \mathbb{R}$, kf is differentiable at c and

$$(kf)'(c) = k \cdot f'(x)$$

(*ii*) $f + g$ is differentiable at c and

$$(f + g)'(c) = f'(c) + g'(c)$$

(*iii*) fg is differentiable at c and

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c)$$

(*iv*) $\frac{f}{g}$ is differentiable at c (provided $g(c) \neq 0$) and

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}$$

Proof. Here, we will prove (*ii*) and (*iii*).

(*ii*)

$$\begin{aligned}
\lim_{x \to c} \frac{(f+g)(x) - (f+g)(c)}{x - c} &= \lim_{x \to c} \frac{f(x) + g(x) - f(c) - g(c)}{x - c} \\
&= \lim_{x \to c} \frac{f(x) - f(c)}{x - c} + \lim_{x \to c} \frac{g(x) - g(c)}{x - c} \\
&= f'(c) + g'(c).
\end{aligned}$$

So, $f + g$ is differentiable at c, and $(f + g)'(c) = f'(c) + g'(c)$.

(*iiii*)

$$\begin{aligned}
\lim_{x \to c} \frac{(fg)(x) - (fg)(c)}{x - c} &= \lim_{x \to c} \frac{f(x)g(x) - f(c)g(c)}{x - c} \\
&= \lim_{x \to c} \frac{f(x)g(x) - f(c)g(x) + f(c)g(x) - f(c)g(c)}{x - c} \\
&= \lim_{x \to c} \frac{(f(x) - f(c))\, g(x) + f(c)\, (g(x) - g(c))}{x - c} \\
&= \left[\lim_{x \to c} \frac{f(x) - f(c)}{x - c}\right] \left[\lim_{x \to c} g(x)\right] + \left[\lim_{x \to c} f(c)\right] \left[\lim_{x \to c} \frac{g(x) - g(c)}{x - c}\right] \\
&= f'(c) \cdot g(c) + f(c) \cdot g'(c)
\end{aligned}$$

Thus fg is differentiable at c, and $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$. □

Theorem 1.1.3. (Chain Rule)
Let $I_1 \subseteq \mathbb{R}$ and $I_2 \subseteq \mathbb{R}$ be two intervals. Suppose $f : I_1 \to \mathbb{R}$ and $g : I_2 \to \mathbb{R}$ such that $f(I_1)$ is contained in I_2, f is differentiable at $c \in I_2$, and g is differentiable at $f(c) \in I_2$. Then the function $g \circ f : I_1 \to \mathbb{R}$ is differentiable at $c \in I_1$, and

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c).$$

Informal Discussion

The following is an incorrect proof of the theorem:

$$\begin{aligned}
\lim_{x\to c} \frac{(g\circ f)(x) - (g\circ f)(c)}{x-c} &= \lim_{x\to c} \frac{g(f(x)) - g(f(c))}{x-c} \\
&= \lim_{x\to c} \frac{g(f(x)) - g(f(c))}{f(x)-f(c)} \cdot \frac{f(x)-f(c)}{x-c} \\
&= \left[\lim_{x\to c} \frac{g(f(x)) - g(f(c))}{f(x)-f(c)}\right] \cdot \left[\lim_{x\to c}\lim_{x\to c} \frac{f(x)-f(c)}{x-c}\right] \\
&= g'(f(c)) \cdot f'(c)
\end{aligned}$$

This proof fails because even though $x \to c \implies x \neq c$, it's not necessarily the case that $f(x) \to f(c) \implies f(x) \neq f(c)$. I.e., the algebraic limit theorem for functions fails as $f(x) - f(c)$ might be zero. Dividing by $f(x) - f(c)$ is not legitimate. To see why this fails, consider the case when f is a constant function. We instead use the following idea: Replace $\frac{g(f(x))-g(f(c))}{f(x)-f(c)}$ with a new function $d(f(x))$ such that

(i) $d(f(x)) = \frac{g(f(x))-g(f(c))}{f(x)-f(c)}$ when $f(x) \neq f(c)$

(ii) $d(f(x))$ is defined even when $f(x) = f(c)$

(iii) $d(f(x)) \cdot \frac{f(x)-f(c)}{x-c} = \frac{g(f(x))-g(f(c))}{x-c}$ for all $x \in I_1, x \neq c$

Proof. Let $d : I_2 \to \mathbb{R}$ be defined by

$$d(y) = \begin{cases} \frac{g(y)-g(f(c))}{y-f(c)} & y \neq f(c) \\ \lim\limits_{y\to f(c)} \frac{g(y)-g(f(c))}{y-f(c)} = g'(f(c)) & y = f(c) \end{cases}$$

Clearly, d satisfies requirements (i) and (ii) from above.

Observation 1: d is continuous at $f(c)$. Indeed,

$$\lim_{y\to f(c)} d(y) = \lim_{y\to f(c)} \frac{g(y) - g(f(c))}{y - f(c)} = d(f(c))$$

Observation 2: For all $x \in I_1$ with $x \neq c$, we have

$$d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} = \frac{g(f(x)) - g(f(c))}{x - c}$$

This is true because

Case 1: $f(x) \neq f(c)$

$$\begin{aligned}
d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} &= \frac{g(f(x)) - g(f(c))}{f(x) - f(c)} \cdot \frac{f(x) - f(c)}{x - c} \\
&= \frac{g(f(x)) - g(f(c))}{x - c}
\end{aligned}$$

Case 2: $f(x) = f(c)$

$$\begin{aligned}
LHS = d(f(x)) \cdot \frac{f(x) - f(c)}{x - c} = d(f(c)) \cdot \frac{f(x) - f(c)}{x - c} = g'(f(c)) \cdot 0 = 0 \\
RHS = \frac{g(f(x)) - g(f(c))}{x - c} = \frac{g(f(c)) - g(f(c))}{x - c} = 0
\end{aligned}$$

So, $LHS = RHS = 0$.

We have,

$$\begin{aligned}\lim_{x\to c}\frac{(g\circ f)(x)-(g\circ f)(c)}{x-c} &= \lim_{x\to c}\frac{g(f(x))-g(f(c))}{x-c}\\ &= \lim_{x\to c}\left[d(f(x))\cdot\frac{f(x)-f(c)}{x-c}\right]\\ &= \left[\lim_{x\to c}(d\circ f)(x)\right]\cdot\left[\lim_{x\to c}\frac{f(x)-f(c)}{x-c}\right]\\ &\overset{(*)}{=} (d\circ f)(c)\cdot f'(c)\\ &= d(f(c))\cdot f'(c)\\ &= g'(f(c))\cdot f'(c)\end{aligned}$$

So, $g\circ f$ is differentiable at c and $(g\circ f)'(c) = g'(f(c))\cdot f'(c)$.

$(*)$ Note that f is continuous at c and d is continuous at $f(c)$, so by composition of continuos functions we conclude that $d\circ f$ is continuous at c and

$$\lim_{x\to c}(d\circ f)(c) = (d\circ f)(c).$$

□

Example 1.1.5. Let $f:\mathbb{R}\to\mathbb{R}$ be defined by $f(x) = \begin{cases} x^2\sin\frac{1}{x} & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$

(*i*) Prove that f is differentiable at all $x\neq 0$.

(*ii*) Prove that $f'(0) = 0$

(*iii*) Prove that f' is not continuous at 0.

Proof. (*i*) We have

$$\left.\begin{array}{ll} h_1(x) = 1 & \text{is differentiable on } \mathbb{R}\\ h_2(x) = x & \text{is differentiable on } \mathbb{R}\end{array}\right\} \Longrightarrow \frac{h_1(x)}{h_2(x)} = \frac{1}{x} \text{ is differentiable at all } x\neq 0$$

$$\left.\begin{array}{ll} h_3(x) = \sin x & \text{is differentiable on } \mathbb{R}\\ h_4(x) = \frac{1}{x} & \text{is differentiable at all } x\neq 0\end{array}\right\} \Longrightarrow (h_3\circ h_4)(x) = \sin\frac{1}{x} \text{ is differentiable at all } x\neq 0$$

$$\left.\begin{array}{ll} h_5(x) = x^2 & \text{is differentiable on } \mathbb{R}\\ h_4(x) = \sin\frac{1}{x} & \text{is differentiable at all } x\neq 0\end{array}\right\} \Longrightarrow h_5(x)\cdot h_6(x) = x^2\sin\frac{1}{x} \text{ is differentiable at all } x\neq 0$$

Indeed, it follows from the algebraic differentiation theorem and the chain rule that

$$\begin{aligned}(x^2\sin\frac{1}{x})' &= (x^2)'\cdot\sin\frac{1}{x} + x^2\cdot(\sin\frac{1}{x})'\\ &= 2x\cdot\sin\frac{1}{x} + x^2\left[(\cos\frac{1}{x})(-\frac{1}{x^2})\right]\\ &= 2x\sin\frac{1}{x} - \cos\frac{1}{x}\end{aligned}$$

(*ii*) Note that $f(0) = 0$ does not imply $f'(0) = 0$. When we want to compute f' at any point, in particular at 0, we need to pay attention to the behavior of f in a neighborhood of the point and not just the value of the function at the point. The reason is that $f'(c)$ is defined by taking $\lim_{x\to c}$.

Our goal is to show

$$\lim_{x\to c}\frac{f(x)-f(0)}{x-0} = 0$$

Note that

$$\frac{f(x)-f(0)}{x-0} = \frac{x^2\sin\frac{1}{x}-0}{x-0} = x\sin\frac{1}{x}$$

We want to show

$$\lim_{x\to 0}\left(x\sin\frac{1}{x}\right)=0$$

We have,

$$\left.\begin{array}{l}0\le\left|x\sin\frac{1}{x}\right|\le|x|\\ \lim\limits_{x\to 0}0=0\\ \lim\limits_{x\to 0}|x|=|0|=0\end{array}\right\}\overset{\text{SQZ Thm}}{\Longrightarrow}\lim_{x\to 0}\left|x\sin\frac{1}{x}\right|=0$$

Thus $\lim\limits_{x\to 0}x\sin\frac{1}{x}=0$.

(*iii*) According to parts (*i*) and (*ii*):

$$f':\mathbb{R}\to\mathbb{R},\quad f(x)=\begin{cases}2x\sin\frac{1}{x}-\cos\frac{1}{x} & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$$

By the sequential criterion for continuity, it is enough to find a sequence (a_n) such that

$$a_n\to 0 \text{ but } f'(a_n)\not\to f'(0)$$

Let $a_n=\frac{1}{2n\pi}$. Clearly, $a_n\to 0$. However,

$$\begin{aligned}\lim_{n\to\infty}f'(a_n)&=\lim_{n\to\infty}\left[\frac{2}{2n\pi}\sin(2n\pi)-\cos(2n\pi)\right]\\ &=0-1\\ &\neq 0.\end{aligned}$$

□

1.2 Local Extrema

Definition 1.2.1. (Local Maximum, Local Minimum)
Let $\emptyset \neq A \subseteq (X, d)$, and let $f : A \to \mathbb{R}$.

(*i*) We say that f has a local maximum at $c \in A$ if

$$\exists \delta > 0 \text{ such that } \forall x \in N_\delta(c) \cap A \quad f(x) \leq f(c)$$

(*ii*) We say that f has a local minimum at $c \in A$ if

$$\exists \delta > 0 \text{ such that } \forall x \in N_\delta(c) \cap A \quad f(x) \geq f(c)$$

Lemma 1.2.1. (Order Limit Theorem for Functions)
Suppose $\lim_{x\to c} g(x)$ and $\lim_{x\to c} h(x)$ both exist.

(*i*) If $\exists \delta > 0$ such that $\forall x \in (c-\delta, c) \quad h(x) \leq g(x)$, then $\lim_{x\to c} h(x) \leq \lim_{x\to c} g(x)$

(*ii*) If $\exists \delta > 0$ such that $\forall x \in (c, c+\delta) \quad h(x) \leq g(x)$, then $\lim_{x\to c} h(x) \leq \lim_{x\to c} g(x)$

Proof. Here we will prove (*i*). The proof of (*ii*) is analogous. Let (a_n) be a sequence in $(c-\delta, c)$ such that $a_n \to c$. By the sequential criterion for limits of functions we have

$$a_n \to c \implies \begin{cases} \lim_{n\to\infty} g(a_n) = \lim_{x\to c} g(x) \\ \lim_{n\to\infty} h(a_n) = \lim_{x\to c} h(x) \end{cases} \tag{I}$$

Also note that

$$\begin{aligned} \forall n \quad a_n \in (c-\delta, c) &\implies \forall n \quad h(a_n) \leq g(a_n) \\ &\overset{\text{OLTS}}{\implies} \lim_{n\to\infty} h(a_n) \leq \lim_{n\to\infty} g(a_n) \end{aligned} \tag{II}$$

It follows from $(I), (II)$ that $\lim_{x\to c} h(x) \leq \lim_{x\to c} g(x)$. □

Theorem 1.2.1. (Interior Extremum Theorem)
Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ be a function and $c \in \overline{I}$. Suppose f is differentiable at c. Then

(*i*) If f has a local maximum at c, then $f'(c) = 0$

(*ii*) If f has a local minimum at c, then $f'(c) = 0$

Proof. Here, we will prove (*i*). The proof for (*ii*) is analogous. Suppose f has a local maximum at c.

(i) f has a local maximum at $c \implies \exists \delta_1$ such that $\forall x \in (c-\delta_1, c+\delta_1) \cap I \quad f(x) \leq f(c)$

(ii) c is an interior point of $I \implies \exists \delta_2$ such that $(c-\delta_2, c+\delta_2) \subseteq I$

So, if we let $\delta = \min\{\delta_1, \delta_2\}$, then

$$\forall x \in (c-\delta, c+\delta) \quad f(x) \leq f(c)$$

We have

(*I*) For all $x \in (c-\delta, c)$

$$\left.\begin{matrix} x - c < 0 \\ f(x) \leq f(c) \end{matrix}\right\} \implies \frac{f(x)-f(c)}{x-c} \geq 0$$

$$\overset{OLTF}{\implies} \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \geq \lim_{x\to c} 0$$

$$\implies f'(c) \geq 0.$$

(II) For all $x \in (c, c+\delta)$

$$\left.\begin{array}{l} x-c>0 \\ f(x) \le f(c) \end{array}\right\} \implies \frac{f(x)-f(c)}{x-c} \le 0$$
$$\overset{OLTF}{\implies} \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \le \lim_{x\to c} 0$$
$$\implies f'(c) \le 0.$$

It follows from $(I), (II)$ that $f'(c) = 0$. □

Remark 1.2.1. The following are three techniques that can be used in proving the existence of a solution:

(i) Suppose $h : [a, b] \to \mathbb{R}$ is continuous. Let α be a given real number. One way to show there exists a number c such that $h(c) = \alpha$ is as follows:

$$\text{Prove that } m \le \alpha \le M \text{ where } \begin{cases} m = \min\{h(x) : x \in [a,b]\} \\ M = \max\{h(x) : x \in [a,b]\} \end{cases}$$

(ii) Suppose $g : [a, b] \to \mathbb{R}$ is differentiable. One way to prove that there exists a number c such that $g'(c) = 0$ is as follows:

$$\text{Prove there is a point in } (a,b) \text{ at which } g \text{ has a local maximum or a local minimum}$$

(iii) Suppose $h : [a, b] \to \mathbb{R}$ is differentiable. Let α be a given real number. One way to prove that there exists a number c such that $h'(c) = \alpha$ is as follows:

$$\text{Define } g(x) = h(x) - \alpha x \text{ and prove that there is a point } c \text{ at which } g'(c) = 0$$

Theorem 1.2.2. (Darboux's Theorem)
Suppose $f : [a, b] \to \mathbb{R}$ is differentiable such that $f'(a) < f'(b)$ (or $f'(b) < f'(a)$), and let $\alpha \in \mathbb{R}$ be such that $f'(a) < \alpha < f'(b)$ (or $f'(b) < \alpha < f'(a)$). Then

$$\exists c \in (a,b) \text{ such that } f'(c) = \alpha$$

Proof. Let $g : [a, b] \to \mathbb{R}$ be defined by $g(x) = f(x) - \alpha x$. It follows from the algebraic differentiability theorem that g is differentiable on $[a, b]$, and so it is continuous on $[a, b]$. It is enough to show that

$$\exists c \in (a,b) \text{ such that } g'(c) = 0$$

To this end, it is enough to show that $\exists c \in (a, b)$ at which g has a local minimum. We have

$$\left.\begin{array}{l} g \text{ is continuous on [a,b]} \\ [a,b] \text{ is compact} \end{array}\right\} \implies g \text{ attains its minimum on } [a,b]$$

Let $\hat{c}$ be a point at which g attains a minimum. In what follows we will show that $\hat{c} \in (a, b)$ and so it can be used as the c that we were looking for. Note that (since $g'(x) = f'(x) - \alpha$)

$$g'(a) = f'(a) - \alpha < 0$$
$$g'(b) = f'(b) - \alpha > 0$$

Claim 1: $\hat{c} \neq a$

Assume for contradiction that $\hat{c} = a$. Then

$$\forall x \in [a,b] \quad g(x) \ge g(a)$$

so,

$$\forall x \in [a,b] \quad \begin{cases} g(x) - g(a) \ge 0 \\ x - a > 0 \end{cases}$$

Thus

$$\forall x \in (a,b) \quad \frac{g(x)-g(a)}{x-a} \ge 0$$

Thus

$$\lim_{x\to c}\frac{g(x)-g(a)}{x-a}\geq\lim_{x\to a}0$$

That is, $g'(a)\geq 0$. This contradicts the fact that $g'(a)<0$.

Claim 2: $\hat{c}\neq b$

Assume for contradiction that $\hat{c}=b$. In a similar manner to claim 1:

$$\forall x\in[a,b]\quad g(x)\geq g(b)\implies\forall x\in[a,b]\quad\begin{cases}g(x)-g(b)\geq 0\\ x-b<0\end{cases}$$
$$\implies\forall x\in[a,b]\quad\frac{g(x)-g(b)}{x-b}\leq 0$$

Thus,

$$\lim_{x\to c}\frac{g(x)-g(b)}{x-b}\leq\lim_{x\to b}0$$

That is,

$$g'(b)\leq 0.$$

This contradicts the fact that $g'(b)>0$.

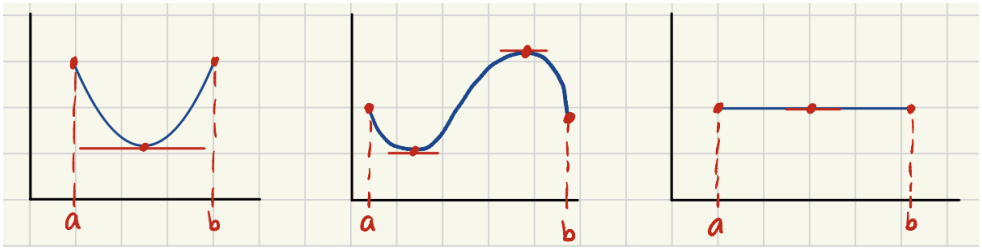
Example 1.2.1. Does there exist a differentiable function $f:[-1,1]\to\mathbb{R}$ whose derivative is $H:[-1,1]\to\mathbb{R}$ defined by

$$H(x)=\begin{cases}1 & 0\leq x\leq 1\\ 0 & -1\leq x\leq 0\end{cases}\ ?$$

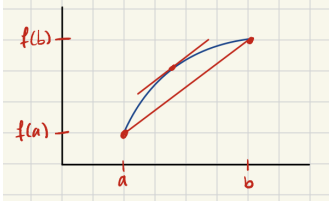
No! H does not have the intermediate value property. So, it cannot be the derivative of any differentiable function.

The following are some geometric conjectures involving the derivative of a function.

Conjecture 1.2.1.
Suppose $f:[a,b]\to\mathbb{R}$ is differentiable. Suppose $f(a)=f(b)$. Then there exists a point $c\in(a,b)$ at which the tangent line is horizontal. I.e., there exists $c\in(a,b)$ such that $f'(c)=0$.



Conjecture 1.2.2.
Suppose $f:[a,b]\to\mathbb{R}$ is differentiable. Then there exists a point $c\in(a,b)$ at which the tangent line is parallel to the line through the endpoints $(a,f(a))$ and $(b,f(b))$. I.e., there exists $c\in(a,b)$ such that $f'(c)=\frac{f(b)-f(a)}{b-a}$.



Conjecture 1.2.3.
Suppose $\vec{r}:[a,b]\to\mathbb{R}^2$, $\vec{r}(t)=(f(t),g(t))$ is a differentiable path in $\mathbb{R}^2$. Then there exists a point $\vec{r}(c)$ on the curve at which the tangent line is parallel to the line through the endpoints $\vec{r}(a)$ and $\vec{r}(b)$. Let's

try to find a mathematical formula for this statement:

$*$) The direction vector for the tangent line at the point $\vec{r}(c)$: $\vec{r}'(c) = (f'(c), g'(c))$

$*$) The direction vector for the line through the endpoints: $(f(b) - f(a), g(b) - g(a))$

So, assuming these vectors are nonzero, the claim of the conjecture can be described mathematically as

$$\exists c \exists \lambda \in \mathbb{R}\backslash\{0\} \text{ such that } (f'(c), g'(c)) = \lambda\,(f(b) - f(a), g(b) - g(a))$$

Note that

$$\begin{aligned}
(f'(c), g'(c)) = \lambda\,(f(b) - f(a), g(b) - g(a)) & \\
\implies & \begin{cases} f'(c) = \lambda\,(f(b) - f(a)) \\ g'(c) = \lambda\,(g(b) - g(a)) \end{cases} \\
\implies & \lambda f'(c)\,[g(b) - g(a)] = \lambda g'(c)\,[f(b) - f(a)] \\
\implies & f'(c)\,[f(b) - f(a)] = g'(c)\,[g(b) - g(a)]
\end{aligned}$$

1.3 Mean Value Theorems

We now study three theorems that make the previous geometric observations precise.

Theorem 1.3.1. (Rolle's Theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous. Let f be differentiable on (a, b). Suppose $f(a) = f(b)$. Then there exists a point $c \in (a, b)$ such that $f'(c) = 0$.

Proof. It is enough to show that there exists a point $c \in (a, c)$ at which f has a local maximum or a local minimum. We have

$$\left.\begin{array}{l} f \text{ is continuous} \\ [a,b] \text{ is compact} \end{array}\right\} \overset{EVT}{\Longrightarrow} f \text{ attains its maximum and minimum on } [a,b]$$

We consider two cases:

Case 1: Both $\max\limits_{a\le x\le b} f(x)$ and $\min\limits_{a\le x\le b} f(x)$ occur at the endpoints.
In this case, it follows from the assumption $f(a) = f(b)$ that $\max\limits_{a\le x\le b} f(x) = \min\limits_{a\le x\le b} f(x)$. So, f is a constant function on $[a, b]$. Hence

$$\forall x \in [a,b] \quad f'(x) = 0$$

So, we may choose c to be any point we like in (a, b).

Case 2: Either $\max\limits_{a\le x\le b} f(x)$ or $\min\limits_{a\le x\le b} f(x)$ occurs at a point $c \in (a, b)$.
It follows from the interior extreme value theorem that $f'(c) = 0$.

□

Theorem 1.3.2. (Mean Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ be continuous and let f be differentiable on (a, b). Then there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.

Proof. Let $g : [a, b] \to \mathbb{R}$ be defined by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}x$$

Note that

$*$) Algebraic continuity theorem $\implies$ g is continuous on $[a, b]$

$*$) Algebraic differentiability theorem $\implies$ g is differentiable on (a, b)

$*$) $g(a) = f(a) - \frac{f(b)-f(a)}{b-a}a = \frac{bf(a)-af(a)-af(b)+af(a)}{b-a} = \frac{bf(a)-af(a)}{b-a}$

$*$) $g(b) = f(b) - \frac{f(b)-f(a)}{b-a}b = \frac{bf(a)-af(b)}{b-a}$

g is continuous on $[a, b]$, differentiable on (a, b), and $g(a) = g(b)$. By Rolle's theorem,

$$\exists c \in (a,b) \text{ such that } g'(c) = 0$$

Note that $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$, so

$$\begin{aligned} g'(c) = 0 &\iff f'(c) - \frac{f(b) - f(a)}{b - a} = 0 \\ &\iff f'(c) = \frac{f(b) - f(a)}{b - a} \end{aligned}$$

□

Theorem 1.3.3. (Generalized Mean Value Theorem)
Let $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ be continuous functions that are differentiable on (a, b). Then there

exists a point $c \in (a,b)$ such that

$$f'(c)\,[g(b)-g(a)] = g'(c)\,[f(b)-f(a)]$$

Proof. Let $h(x) = [f(b)-f(a)]\,g(x) - [g(b)-g(a)]\,f(x)$. It follows from the assumptions, the algebraic continuity theorem, and the algebraic differentiability theorem that h is continuous on $[a,b]$ and differentiable on (a,b). Therefore, by the mean value theorem,

$$\exists c \in (a,b) \text{ such that } h'(c) = \frac{h(b)-h(a)}{b-a} \qquad (*)$$

Note that

$$\begin{aligned} h(b) &= [f(b)-f(a)]\,g(b) - [g(b)-g(a)]\,f(b) \\ &= f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) \\ &= g(a)f(b) - f(a)g(b) \\ h(a) &= [f(b)-f(a)]\,g(a) - [g(b)-g(a)]\,f(a) \\ &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - g(b)f(a) \end{aligned}$$

So $h(a) = h(b)$. Hence it follows from $(*)$ that $\exists c \in (a,b)$ such that $h'(c) = 0$ Now note that

$$\begin{aligned} h'(x) &= [f(b)-f(a)]\,g'(x) - [g(b)-g(a)]\,f'(x) \\ \implies h'(c) &= [f(b)-f(a)]\,g'(c) - [g(b)-g(a)]\,f'(c) \end{aligned}$$

Therefore,

$$\exists c \in (a,b) \text{ such that } [f(b)-f(a)]\,g'(c) - [g(b)-g(a)]\,f'(c) = 0$$

That is,

$$\exists c \in (a,b) \text{ such that } [f(b)-f(a)]\,g'(c) = [g(b)-g(a)]\,f'(c)$$

□

Remark 1.3.1. If g' is neer zero in (a,b), then we may rewrite the claim of general mean value theorem as follows:

$$\exists c \in (a,b) \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$$

Theorem 1.3.4.
Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable such that $f'(x) = 0\ \forall x \in I$. Then f is a constant function on I, that is, there exists $k \in \mathbb{R}$ such that $\forall x \in I, f(x) = k$.

Proof. Let $x, y \in I$ with $x < y$. It is enough to show that $f(x) = f(y)$. To this end, we wil apply the mean value theorem to f on the interval $[x,y]$:

$$\begin{aligned} \exists c \in (x,y) \text{ such that } f'(c) = \frac{f(y)-f(x)}{y-x} & \\ \implies 0 = \frac{f(y)-f(x)}{y-x} & \\ \implies 0 = f(y) - f(x) & \\ \implies f(x) = f(y) & \end{aligned}$$

□

Remark 1.3.2. Consider $f : A \to \mathbb{R}$ where $A = (-1,0) \cup (2,3)$ and $f(x) = \begin{cases} 1 & x \in (-1,0) \\ -1 & x \in (2,3) \end{cases}$ Then $\forall x \in A\ \ f'(x) = 0$, but f is not a constant function on A. The theorem above doesn't apply since A is not an interval.

Theorem 1.3.5.
Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be differentiable such that $f'(x) = g'(x)\ \forall x \in I$. Then there exists $k \in \mathbb{R}$ such that $\forall x \in I,\ f(x) = g(x) + k$.

Proof. Let $h = f - g$. We have

$$\begin{aligned} &\forall x \in I \quad h'(x) = (f-g)'(x) = f'(x) - g'(x) = 0 \\ &\overset{1.3.4}{\implies} \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad h(x) = k \\ &\implies \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad f(x) - g(x) = k \\ &\implies \exists k \in \mathbb{R} \text{ such that } \forall x \in I \quad f(x) = g(x) + k \end{aligned}$$

□

Theorem 1.3.6.
Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be differentiable. Then

(i) f is increasing $\iff \forall c \in I \quad f'(c) \geq 0$

(ii) f is decreasing $\iff \forall c \in I \quad f'(c) \leq 0$

Proof.
Here, we will prove (i). The proof of (ii) is analogous.
$(\implies)$ Suppose f is increasing on I. Let $c \in I$. Note that for all $x \in I, x \neq c$ we have $\frac{f(x)-f(c)}{x-c} \geq 0$. Indeed,

$$\text{if } x > c \text{ then } \begin{cases} x - c > 0 \\ f(x) \geq f(c) \end{cases} \implies \frac{f(x)-f(c)}{x-c} \geq 0$$

$$\text{if } x < c \text{ then } \begin{cases} x - c < 0 \\ f(x) \leq f(c) \end{cases} \implies \frac{f(x)-f(c)}{x-c} \geq 0$$

It follows from the order limit theorem for functions that

$$\lim_{x \to c} \frac{f(x)-f(c)}{x-c} \geq \lim_{x \to c} 0$$

Hence, $f'(c) \geq 0$ as desired.

$(\impliedby)$ Suppose $\forall c \in I \quad f'(c) \geq 0$. Let $x_1, x_2 \in I$ with $x_1 < x_2$. It is enough to show that $f(x_1) \leq f(x_2)$. To this end, we apply the mean value theorem to the function f on $[x_1, x_2]$:

$$\exists c \in (x_1, x_2) \text{ such that } f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

So, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$. Thus $f(x_2) - f(x_1) \geq 0$, that is, $f(x_1) \leq f(x_2)$ as desired. □

Theorem 1.3.7. (L'Hĉpital's Rule)
Let $I \subseteq \mathbb{R}$ be an interval, and $a \in I$. Let $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ be continuous. Suppose f and g are differentiable at all points in $I \backslash \{a\}$ and $f(a) = g(a) = 0$, $g'(x) \neq 0\ \forall x \in I \backslash \{a\}$ and $\lim_{x \to a} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}$. Then $\lim_{x \to a} \frac{f(x)}{g(x)} = L$.

Proof. Our goal is to show that

$$\forall \epsilon > 0\ \exists \delta > 0 \text{ such that if } 0 < |x - a| < \delta \text{ (with } x \in I) \text{ then } \left| \frac{f(x)}{g(x)} - L \right| < \epsilon$$

Let $\epsilon > 0$. Our goal is to find $\delta > 0$ such that

$$\text{if } 0 < |x - a| < \delta \text{ (with } x \in I) \text{ then } \left| \frac{f(x)}{g(x)} - L \right| < \epsilon \qquad (*)$$

Since by assumption $\lim_{x\to a}\frac{f'(x)}{g'(x)} = L$, for the gien $\epsilon > 0$, there exists $\hat{\delta} > 0$ such that

$$\text{if } 0 < |x-a| < \hat{\delta} \text{ (with } x \in I) \text{ then } \left|\frac{f'(x)}{g'(x)} - L\right| < \epsilon$$

We claim that this $\hat{\delta}$ satisfies $(*)$. The reason is as follows:
Suppose $x \in I$ such that $0 < |x-a| < \hat{\delta}$. In what follows we will show that $\left|\frac{f(x)}{g(x)} - L\right| < \epsilon$. We consider two cases:

Case 1: $x > a \quad \left(x \in (a, a+\hat{\delta})\right)$

We apply the general mean value theorem to f and g on the interval $[a, x]$:

$$\exists c \in (a,x) \text{ such that } \frac{f'(x)}{g'(x)} = \frac{f(x)-f(a)}{g(x)-g(a)}$$

Since $f(a) = g(a) = 0$, we conclude that

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}$$

It follows that

$$\left|\frac{f(x)}{g(x)} - L\right| = \left|\frac{f'(x)}{g'(x)} - L\right| < \epsilon$$

(the latter inequality is true because $0 < |c-a| \leq |x-a| \leq \hat{\delta}$)

Case 2: $x < a \quad \left(x \in (a-\hat{\delta}, a)\right)$

We apply the general mean value theorem to f and g on $[x, a]$:

$$\exists c \in (x,a) \text{ such that } \frac{f'(c)}{g'(c)} = \frac{f(a)-f(x)}{g(a)-g(x)}$$

Since $f(a) = g(a) = 0$, we conclude that

$$\frac{f'(c)}{g'(c)} = \frac{f(x)}{g(x)}$$

It follows that

$$\left|\frac{f(x)}{g(x)} - L\right| = \left|\frac{f'(x)}{g'(x)} - L\right| < \epsilon$$

(the latter inequality is true because $0 < |c-a| \leq |x-a| \leq \hat{\delta}$)

□

1.4 Taylor Polynomials

Consider $f : I \to \mathbb{R}$ given by $f(x) = (x - x_0)^k$. What do the n derivatives look like? If $n = 0$, then by the mean value theorem we have

$$\left.\begin{array}{l} f : I \to \mathbb{R} \text{ is differentiable} \\ f(x_0) = 0 \end{array}\right\} \implies f(x) = f'(c)(x - x_0)$$

Observation 1.4.1. Let k be a natural number. Let x_0 be a fixed number.

$*)$ $\frac{d}{dx}\left[(x-x_0)^k\right] = k(x-x_0)^{k-1}$

$*)$ $\frac{d^2}{dx^2}\left[(x-x_0)^k\right] = \frac{d}{dx}\left[k(x-x_0)^{k-1}\right] = k(k-1)(x-x_0)^{k-2}$

$*)$ $\frac{d^3}{dx^3}\left[(x-x_0)^k\right] = \frac{d}{dx}\left[k(k-1)(x-x_0)^{k-2}\right] = k(k-1)(k-2)(x-x_0)^{k-3}$

$\vdots$

$*)$ $\frac{d^k}{dx^k}\left[(x-x_0)^k\right] = k(k-1)\dots(2)(1)(x-x_0)^{k-k} = k!$

If $j < k$

$*)$ $\frac{d^j}{dx^j}\left[(x-x_0)^k\right] = k(k-1)\dots(k-(j-1))\,(x-x_0)^{k-j}$

Thus we have

$$\frac{d^j}{dx^j}\left[(x-x_0)^k\right] = \begin{cases} k(k-1)\dots(k-j+1)(x-x_0)^{k-j} & \text{if } j < k \\ k! & \text{if } j = k \\ 0 & \text{if } j > k \end{cases}$$

$$\frac{d^j}{dx^j}\left[(x-x_0)^k\right]\Big|_{x=x_0} = \begin{cases} 0 & \text{if } j < k \\ k! & \text{if } j = k \\ 0 & \text{if } j > k \end{cases}$$

Theorem 1.4.1. (Corollary of the General Mean Value Theorem)
Let $I \subseteq \mathbb{R}$ be an open interval, $x_0 \in I$, and $n \in \mathbb{N} \cup \{0\}$. Let $f : I \to \mathbb{R}$ have $n+1$ derivatives. Suppose $f^{(k)}(x_0) = 0 \quad \forall 0 \le k \le n$. Then for each point $x \neq x_0$ in the interval I, there exists a point c_{x,x_0} strictly between x and x_0 such that

$$f(x) = \frac{f^{(n+1)}(c_{x,x_0})}{(n+1)!}(x - x_0)^{n+1}$$

Proof. Here we will prove the claim for the case where $x > x_0$. The proof for $x < x_0$ is completely analogous. Let $g : I \to \mathbb{R}$ be defined by $g(t) = (t - x_0)^{n+1}$. Note that

$$g^{(k)}(x_0) = 0 \quad \forall 0 \le k \le n$$
$$g^{(n+1)}(t) = (n+1)! \quad \forall t \in I$$

Now, we apply the general mean value theorem to f and g on the interval $[x_0, x]$:

$$\begin{aligned} &\exists x_1 \in (x_0, x) \text{ such that } \frac{f'(x_1)}{g'(x_1)} = \frac{f(x) - f(x_0)}{g(x) - g(x_0)} \\ &\implies \frac{f'(x_1)}{g'(x_1)} = \frac{f(x)}{g(x)} \end{aligned} \tag{I}$$

Next, we apply the general mean value theorem to f' and g' on the interval $[x_0, x_1]$:

$$\begin{aligned} &\exists x_2 \in (x_0, x_1) \text{ such that } \frac{f''(x_2)}{g''(x_2)} = \frac{f'(x_1) - f'(x_0)}{g'(x_1) - g'(x_0)} \\ &\implies \frac{f''(x_2)}{g''(x_2)} = \frac{f'(x_1)}{g'(x_1)} \\ &\overset{(I)}{\implies} \frac{f''(x_2)}{g''(x_2)} = \frac{f(x)}{g(x)} \end{aligned}$$

Continuing in this way, we will obtain $x_{n+1} \in (x_0, x)$ such that

$$\frac{f^{(n+1)}(x_{n+1})}{g^{(n+1)}(x_{n+1})} = \frac{f(x)}{g(x)}$$

So,

$$\frac{f^{(n+1)}(x_{n+1})}{(n+1)!} = \frac{f(x)}{(x-x_0)^{n+1}}$$

Thus

$$f(x) = \frac{f^{(n+1)}(x_{n+1})}{(n+1)!}(x-x_0)^{n+1}$$

(We can use x_{n+1} as the c we were looking for) □

Question: What are the nicest functions that we know? Which functions are the easiest to work with?

Answer: Polynomials

General Question: Given a function f, is it possible to find a "good" approximation for f among polynomials?

Setup:

$*$) Let I be a nonempty open interval in $\mathbb{R}$
$*$) Let n be a nonnegative integer
$*$) Suppose $f : I \to \mathbb{R}$ has n derivatives and $x_0 \in I$
$*$) Suppose that we want to use the values

$$f(x_0), f'(x_0), ..., f^{(n)}(x_0)$$

to construct a polynomial approximation for f

What is the best we could hope for? Find a polynomial such that

$$\begin{aligned} p(x_0) &= f(x_0) \\ p'(x_0) &= f'(x_0) \\ &\vdots \\ p^{(n)}(x_0) &= f^{(n)}(x_0) \end{aligned}$$

Observation 1.4.2. Let x_0 be a fixed real number. A general polynomial of degree at most n can be expressed in powers of $(x - x_0)$ in the form

$$p(x) = c_0 + c_1(x-x_0) + ... + c_n(x-x_0)^n$$

Example 1.4.1. Consider $p(x) = x^2 - 3x - 1$. Let $x_0 = 1$. We can express $p(x)$ in powers of $x - 1$:

$$\begin{aligned} p(x) = x^2 - 3x - 1 &= [(x-1)+1]^2 - 3[(x-1)+1] - 1 \\ &= (x-1)^2 + 2(x-1) + 1 - 3(x-1) - 3 - 1 \\ &= (x-1)^2 - (x-1) - 3 \end{aligned}$$

Theorem 1.4.2. (Uniqueness of the Approximating Polynomial)
Let $I \subseteq \mathbb{R}$ be an open interval and $n \in \mathbb{N}$. Suppose $f : I \to \mathbb{R}$ has n derivatives ad $x_0 \in I$. Then there exists a unique polynomial $p(x)$ of degree at most n such that

$$\forall 0 \le l \le n \;\; p^{(l)}(x_0) = f^{(l)}(x_0), \text{ with } \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

Proof. Let $p(x)$ be a general polynomial of degree at most n:

$$p(x) = c_0 + c_1(x-x_0) + ... + c_n(x-x_0)^n$$

Our goal is to show that

$$\text{If } \forall 0 \le l \le n \;\; p^{(l)}(x_0) = f^{(l)}(x_0) \text{ then } p(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

Note that $p(x_0) = c_0$. Also, for $1 \le l \le n$ we have

$$\begin{aligned} p^{(l)}(x) &= \frac{d^l}{dx^l}\left[c_0 + \sum_{k=1}^{n} c_k(x - x_0)^k\right] \\ &= \frac{d^l}{dx^l}\left[\sum_{k=1}^{n} c_k(x - x)^k\right] \\ &= \sum_{k=1}^{n} c_k \frac{d^l}{dx^l}\left[(x - x_0)^k\right] \end{aligned}$$

Hence,

$$p^{(l)}(x_0) = \sum_{k=1}^{n} c_k \frac{d^l}{dx^l}\left[(x - x_0)^k\right]\Big|_{x=x_0} = c_l \cdot l!$$

Therefore,

$$\forall 1 \le l \le n \;\; p^{(l)}(x_0) = c_l \cdot l!$$

We conclude that

$$\begin{aligned} p \text{ agrees with } f \text{ to order } n \text{ at } x_0 &\iff \begin{cases} p(x_0) = f(x_0) \\ p^{(l)}(x_0) = f^{(l)}(x_0) \;\forall 1 \le l \le n \end{cases} \\ &\iff \begin{cases} c_0 = f(x_0) \\ l!c_l = f^{(l)}(x_0) \;\forall 1 \le l \le n \end{cases} \\ &\iff \begin{cases} c_0 = f(x_0) \\ c_l = \frac{f^{(l)}(x_0)}{l!} \;\forall 1 \le l \le n \end{cases} \\ &\iff p(x) = \sum_{k=0}^{n} c_k(x - x_0)^k \\ &= c_0 + \sum_{k=1}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \\ &= f(x_0) + \sum_{k=1}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \\ &= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \end{aligned}$$

□

Note . n^{th} Taylor Polynomial centered at 0 is called n^{th} Maclaurin Polynomial

Big Lesson: There is exactly one polynomial of degree at most n that satisfies

$$\begin{aligned} p(x_0) &= f(x_0) \\ p'(x_0) &= f'(x_0) \\ &\vdots \\ p^{(n)}(x_0) &= f^{(n)}(x_0) \end{aligned}$$

This polynomial is called the nth Taylor polynomial for f centered at x_0, and is given by

$$T_{n,x_0}(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \ldots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

Theorem 1.4.3. (Taylor's Theorem with Lagarange Remainder)
Let $I \subseteq \mathbb{R}$ be an open interval, $x_0 \in I$, and $n \in \mathbb{N} \cup \{0\}$. Let $f : I \to \mathbb{R}$ have $n+1$ derivatives. Then for each point $x \neq x_0$ in I, there is a point c strictly between x and x_0 such that

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{n!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$$

Remark 1.4.1. Note that clearly the above equality holds at $x = x_0$ too (for any value of c). Recall that for any fixed number R, $\lim_{n\to\infty} \frac{R^{n+1}}{(n+1)!} = 0$, however $f^{(n+1)}(c)$ may become very large.

Proof. Let $F_{n,x_0} = f(x) - T_{n,x_0}(x)$. Our goal is to show that

$$R_{n,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$$

for some c between x and x_0. Note that

$$\left.\begin{array}{l} f \text{ has } n+1 \text{ derivatives} \\ (i)\ T_{n,x+0} \text{ is a polynomial of degree } n, \text{ so it has } n+1 \text{ derivatives} \\ R_{n,x_0} = f - T \end{array}\right\} \implies R_{n,x_0} \text{ has } n+1 \text{ derivatives}$$

$(ii)\ \forall 0 \leq k \leq n \quad R^{(k)}_{n,x_0}(x_0) = f^{(k)}(x_0) - T^{(k)}_{n,x_0}(x_0) = 0$

$(i), (ii),$ Theorem 1.4.1 $\implies$ For each point $x \neq x_0$ in I, we have

$$R_{n,x_0}(x) = \frac{R^{(n+1)}_{n,x_0}(c)}{(n+1)!}(x - x_0)^{n+1} \text{ for some } c \text{ strictly between } x \text{ and } x_0 \qquad (I)$$

Now note that

$$R^{(n+1)}_{n,x_0}(c) = f^{(n+1)}(c) - T^{(n+1)}_{n,x_0}(c) = f^{(n+1)}(c) \qquad (II)$$

$$(I), (II) \implies R_{n,x_0} = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{(n+1)}$$

□

Chapter 2

Integration

2.1 The Riemann-Stieltjes Integral

Definition 2.1.1. (Almost Disjoint Intervals)
We say two intervals I and J are almost disjoint if either $I \cap J = \emptyset$ or $I \cap J$ has a single element.

Definition 2.1.2. (Partition)

First Viewpoint: A partition P of an interval $[a, b]$ is a finite set of points in $[a, b]$ that include both a and b. We always list the points of a partition $P = \{x_0, ..., x_n\}$ in increasing order:

$$x_0 = a < x_1 < x_2 < ... < b = x_n$$

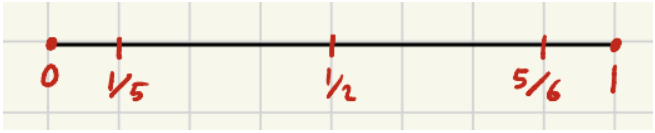
Second Viewpoint: A partition P of an interval $[a, b]$ is a finite collection of almost disjoint (nonempty) compact intervals whose union is $[a, b]$:

$$P = I_1, ..., I_n \text{ where } I_1 = [x_0, x_1], I_2 = [x_1, x_2], ..., I_n = [x_{n-1}, x_n] \quad (\text{with } x_0 = a, x_1 = b)$$

Example 2.1.1.

$$P = \left\{0, \frac{1}{5}, \frac{1}{2}, \frac{5}{6}, 1\right\}$$
$$P = \left\{[0, \frac{1}{5}], [\frac{1}{5}, \frac{1}{2}], [\frac{1}{2}, \frac{5}{6}], [\frac{5}{6}, 1]\right\}$$

are both partitions of $[0, 1]$.



Notation . Let $f : [a, b] \to \mathbb{R}$ be a bounded function. Let $P = \{x_0 = a, ..., x_n = b\}$ be a partition of $[a, b]$. We let

$$m = \inf\{f(x) : x \in [a, b]\}$$
$$M = \sup\{f(x) : x \in [a, b]\}$$
$$\forall 1 \leq k \leq n \quad m_k = \inf\{f(x) : x \in [x_{k-1}, x_k]\}$$
$$\forall 1 \leq k \leq n \quad M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\}$$

The existence of m, M, m_k, M_k as real numbers is guaranteed due to the assumption that f is bounded on $[a, b]$.

Remark 2.1.1. Suppose A and B are nonempty, bounded sets in $\mathbb{R}$ such that $A \subseteq B$. Then

(i) $\inf A \leq \sup A$

(ii) $\inf B \leq \sup B$

(iii) $\sup A \leq \sup B$

(iv) $\inf A \geq \inf B$

As a result

$$\forall 1 \leq k \leq n \quad m \leq m_k \leq M_k \leq M$$

Remark 2.1.2. If $f : [a,b] \to \mathbb{R}$ is continuous, then f attains its maximum and minimum over each compact subinterval. In this case,

$$m_k = \min\{f(x) : x \in [x_{k-1}, x_k]\}$$
$$M_k = \max\{f(x) : x \in [x_{k-1}, x_k]\}$$

Definition 2.1.3. (Lower Sum, Upper Sum)
Let $f : [a,b] \to \mathbb{R}$ be bounded and let $\alpha : [a,b] \to \mathbb{R}$ be increasing. Let $P = \{x_0, ..., x_n\}$ be a partition of $[a,b]$. Let $\Delta_{\alpha_k} = \alpha(x_k) - \alpha(x_{k-1})$.

(i) The lower Riemann-Stieltjes sum of f (R.S. sum of f) with respect to α for the partition P is defined by

$$L(f,\alpha,P) = \sum_{k=1}^{n} m_k\left(\alpha(x_k) - \alpha(x_{k-1})\right) = \sum_{k=1}^{n} m_k \Delta\alpha_k$$

(ii) The upper Riemann-Stieltjes sum of f (R.S sum of f) with respect α for the partition P is defined by

$$U(f,\alpha,P) = \sum_{k=1}^{n} M_k\left(\alpha(x_k) - \alpha(x_{k-1})\right) = \sum_{k=1}^{n} M_k \Delta\alpha_k$$

Note . Note that

$$m(\alpha(b) - \alpha(a)) \leq L(f,\alpha,P) \leq U(f,\alpha,P) \leq M(\alpha(b) - \alpha(a))$$

Indeed,

$$\begin{aligned} L(f,\alpha,P) = \sum_{k=1}^{n} m_k \Delta\alpha_k &\geq \sum_{k=1}^{n} m(\alpha(x_k) - \alpha(x_{k-1})) \\ &= m\sum_{k=1}^{n} \alpha(x_k) - \alpha(x_{k-1}) \\ &= m\left[\alpha(x_1) - \alpha(x_0) + \alpha(x_2) - \alpha(x_1) + ... + \alpha(x_n) - \alpha(x_{n-1})\right] \\ &= m\left[\alpha(x_n) - \alpha(x_0)\right] \\ &= m\left[\alpha(b) - \alpha(a)\right] \end{aligned}$$

Similarly,

$$\begin{aligned} U(f,\alpha,P) = \sum_{k=1}^{n} M_k \Delta\alpha_k &\leq \sum_{k=1}^{n} M(\alpha(x_k) - \alpha(x_{k-1})) \\ &= M\sum_{k=1}^{n} \alpha(x_k) - \alpha(x_{k-1}) \\ &= M\left[\alpha(x_1) - \alpha(x_0) + \alpha(x_2) - \alpha(x_1) + ... + \alpha(x_n) - \alpha(x_{n-1})\right] \\ &= M\left[\alpha(x_n) - \alpha(x_0)\right] \\ &= M\left[\alpha(b) - \alpha(a)\right] \end{aligned}$$

Notation . $\Pi[a,b]$, or Π for short, denotes the collection of all the possible partitions of $[a,b]$.

Definition 2.1.4. (Upper Riemann-Stieltjes Integral, Lower Riemann-Stieltjes Integral)
Let $f:[a,b]\to\mathbb{R}$ be bounded and $\alpha:[a,b]\to\mathbb{R}$ be increasing.

(*i*) The upper Riemann-Stieltjes integral of f with respect to α (on $[a,b]$) is defined by

$$U(f,\alpha)=\overline{\int_a^b} f\,d\alpha=\inf\{U(f,\alpha,P):P\in\Pi\}$$

(Note that the set $\{U(f,\alpha,P):P\in\Pi\}$ of all upper sums is bounded below by $m(\alpha(b)-\alpha(a))$, so the infimum is a real number)

(*ii*) The lower Riemann-Stieltjes integral of f with respect to α (on $[a,b]$) is defined by

$$L(f,\alpha)=\underline{\int_a^b} f\,d\alpha=\sup\{L(f,\alpha,P):P\in\Pi\}$$

Definition 2.1.5. (Riemann-Stieltjes Integrable Function)
Let $\alpha:[a,b]\to\mathbb{R}$ be an increasing function. A function $f:[a,b]\to\mathbb{R}$ is said to be Riemann-Stieltjes integrable (on $[a,b]$) with respect to α if

(*i*) f is bounded

(*ii*) $L(f,\alpha)=U(f,\alpha)$

In this case, the Riemann-Stieltjes integral of f with respect to α, denoted by

$$\int_a^b f\,d\alpha \text{ or } \int_a^b f(x)\,d\alpha \text{ or } \int_{[a,b]} f\,d\alpha$$

is the common value of $L(f,\alpha)$ and $U(f,\alpha)$. That is,

$$\int_a^b f\,d\alpha=L(f,\alpha)=U(f,\alpha)$$

Example 2.1.2. Let c be a fixed real number. Prove that the constant function $f(x)=c$ on $[a,b]$ is R.S. integrable and

$$\int_a^b f\,d\alpha=c(b-a)$$

Proof. For any partition $P=\{x_0,...,x_n\}$ of $[a,b]$ we have

$$\forall 1\le k\le n\quad m_k=\inf\{f(x):x\in[x_{k-1},x_k]\}=\inf\{c\}=c$$
$$M_k=\sup\{f(x):x\in[x_{k-1},x_k]\}=\sup\{c\}=c$$

Therefore,

$$\begin{aligned}L(f,\alpha,P)&=\sum_{k=1}^n m_k(\alpha(x_k)-\alpha(x_{k-1}))\\&=\sum_{k=1}^n c(\alpha(x_k)-\alpha(x_{k-1}))\\&=c\sum_{k=1}^n \alpha(x_k)-\alpha(x_{k-1})\\&=c[\alpha(b)-\alpha(a)]\end{aligned}$$

and

$$\begin{aligned} U(f,\alpha,P) &= \sum_{k=1}^{n} M_k(\alpha(x_k)-\alpha(x_{k-1})) \\ &= \sum_{k=1}^{n} c(\alpha(x_k)-\alpha(x_{k-1})) \\ &= c\sum_{k=1}^{n} \alpha(x_k)-\alpha(x_{k-1}) \\ &= c[\alpha(b)-\alpha(a)] \end{aligned}$$

Hence,

$$\begin{aligned} L(f,\alpha) &= \sup\{L(f,\alpha,P) : P\in\Pi\} = \sup\{c(\alpha(b)-\alpha(a))\} = c(\alpha(b)-\alpha(a)) \\ U(f,\alpha) &= \inf\{U(f,\alpha,P) : P\in\Pi\} = \inf\{c(\alpha(b)-\alpha(a))\} = c(\alpha(b)-\alpha(a)) \end{aligned}$$

Since $L(f,\alpha)=U(f,\alpha)=c(\alpha(b)-\alpha(a))$, we conclude that f is R.S. integrable with respect to α and

$$\int_a^b f d\alpha = c[\alpha(b)-\alpha(a)]$$

□

Example 2.1.3. Let $f:[a,b]\to\mathbb{R}$ be bounded and let $\alpha:[a,b]\to\mathbb{R}$ be a constant function ($\alpha(x)=c$). Prove that $\int_a^b f d\alpha = 0$.

Proof. For any partition $P=\{x_0,...,x_n\}$ of $[a,b]$,

$$\begin{aligned} L(f,\alpha,P) &= \sum_{k=1}^{n} m_k\left[\alpha(x_k)-\alpha(x_{k-1})\right] = \sum_{k=1}^{n} m_k\cdot 0 = 0 \\ U(f,\alpha,P) &= \sum_{k=1}^{n} M_k\left[\alpha(x_k)-\alpha(x_{k-1})\right] = \sum_{k=1}^{n} M_k\cdot 0 = 0 \end{aligned}$$

Therefore,

$$\begin{aligned} L(f,\alpha) &= \sup\{L(f,\alpha,P) : P\in\Pi\} = 0 \\ U(f,\alpha) &= \inf\{U(f,\alpha,P) : P\in\Pi\} = 0 \end{aligned}$$

So $L(f,\alpha)=U(f,\alpha)=0 \implies \int_a^b f d\alpha = 0$. □

Example 2.1.4. Let $f:[a,b]\to\mathbb{R}$ be defined by

$$f(x)=\begin{cases}1 & \text{if } x\in\mathbb{Q}\\ 0 & \text{if } x\notin\mathbb{Q}\end{cases}$$

Prove that

(*i*) if $\alpha:[a,b]\to\mathbb{R}$ is constant, then $\int_a^b f d\alpha = 0$ (proved in the last example)

(*ii*) if $\alpha:[a,b]\to\mathbb{R}$ is increasing and $\alpha(a)\neq\alpha(b)$, then $f\notin R_\alpha[a,b]$

Proof. (*ii*) For any partition $P=\{x_0,...,x_n\}$ of $[a,b]$ we have

$$\begin{aligned} \forall 1\le k\le n \quad m_k &= \inf\{f(x) : x\in[x_{k-1},x_k]\} = \inf\{0,1\} = 0 \\ M_k &= \sup\{f(x) : x\in[x_{k-1},x_k]\} = \sup\{0,1\} = 1 \end{aligned}$$

Therefore

$$L(f,\alpha,P)=\sum_{k=1}^{n} m_k\left[\alpha(x_k)-\alpha(x_{k-1})\right]=0\cdot[\alpha(b)-\alpha(a)]=0$$

$$U(f,\alpha,P)=\sum_{k=1}^{n} M_k\left[\alpha(x_k)-\alpha(x_{k-1})\right]=1\cdot[\alpha(b)-\alpha(a)]=\alpha(b)-\alpha(a)$$

Hence

$$\left.\begin{array}{l} L(f,\alpha)=\sup\{L(f,\alpha,P):P\in\Pi\}=0 \\ U(f,\alpha)=\inf\{U(f,\alpha,P):P\in\Pi\}=\alpha(b)-\alpha(a)\end{array}\right\}\implies L(f,\alpha)\neq U(f,\alpha)\implies f\notin R_\alpha[a,b]$$

□

Definition 2.1.6. (Refinement of a Partition)

First Viewpoint: A partition $Q=\{z_0,...,z_m\}$ of $[a,b]$ is a refinement of a partition $P=\{x_0,...,x_n\}$ of $[a,b]$ if $P\subseteq Q$. That is, if Q contains all points of P.

Second Viewpoint: A partition $Q=J_1,...,J_m$ of $[a,b]$ is a refinement of a partition $P=\{I_0,...,I_n\}$ of $[a,b]$ if every interval I_k of P is an almost disjoint union of one or more intervals of Q.

Example 2.1.5. Consider the following partitions of $[0,1]$:

$$P=\{0,\frac{1}{2},1\}$$
$$Q=\{0,\frac{1}{3},\frac{1}{2},1\}$$

Then Q is a refinement of P since $P\subseteq Q$.

Remark 2.1.3. Let P and Q be any two partitions of $[a,b]$. Then $P\cup Q$ will be a refinement of both P and Q because $P\subseteq P\cup Q$ and $Q\subseteq P\cup Q$. So, any two partitions of $[a,b]$ have a common refinement.

Example 2.1.6. Consider the following partitions of $[0,1]$:

$$P=\{0,\frac{1}{2},1\}$$
$$Q=\{0,\frac{1}{3},\frac{2}{3},1\}$$

P is not a refinement of Q and Q is not a refinement of P, but

$$P\cup Q=\{0,\frac{1}{3},\frac{1}{2},\frac{2}{3},1\}$$

is a refinement of both P and Q.

Definition 2.1.7. (Size of a Partition)
Let $P=\{x_0,...,x_n\}$ be a partition of $[a,b]$. The size of P, denoted $||P||$, is defined by

$$\begin{aligned}||P|| &= \max\{|x_k-x_{k-1}|:1\leq k\leq n\} \\ &= \max\{|x_1-x_0|,|x_2-x_1|,...,|x_n-x_{n-1}|\}\end{aligned}$$

2.2 9 Useful Theorems of Integrability

Theorem 2.2.1. (Inequalities of Refinements)
Let $f : [a, b] \to R$ be bounded and let $\alpha : [a, b] \to \mathbb{R}$ be increasing. Let P be a partition of $[a, b]$ and let Q be a refinement of P. Then

(i) $L(f, \alpha, P) \leq L(f, \alpha, Q)$

(ii) $U(f, \alpha, P) \geq U(f, \alpha, Q)$

Proof. Here, we will prove (i). The proof of (ii) is completely analogous. We proceed by induction on $l = \text{card}(Q \backslash P)$ (the number of points in $Q \backslash P$). Let $P = \{x_0, ..., x_n\}$.

Base Case: If $l = 0$, then

$$\left.\begin{array}{l} P \subseteq Q \\ \text{card}Q = \text{card}P \end{array}\right\} \implies P = Q \implies L(f, \alpha, P) = L(f, \alpha, Q)$$

If $l = 1$, then Q has exactly one extra point. Let's call this point z, so $\{z\} = Q \backslash P$. Note that

$$\left.\begin{array}{l} z \in [a, b] \\ P \text{ is a partition of } [a, b] \end{array}\right\} \implies \exists 1 \leq i \leq n \text{ such that } z \in (x_{i-1}, x_i)$$

Let

$$m_i' = \inf \{f(x) : x \in [x_{i-1}, z]\}$$
$$m_i'' = \inf \{f(x) : x \in [z, x_i]\}$$

Recall that if $A \subseteq B$, then $\inf A \geq \inf B$. Hence $m_i' \geq m_i$ and $m_i'' \geq m_i$ (where $m_i = \inf \{f(x) : x \in [x_{i-1}, x_i]\}$). We have

$$\begin{aligned} L(f, \alpha, P) &= \sum_{k=1}^{n} m_k \left[\alpha(x_k) - \alpha(x_{k-1})\right] \\ &= \left[\sum_{k=1, k\neq i}^{n} m_k \left(\alpha(x_k) - \alpha(x_{k-1})\right)\right] + m_i \left(\alpha(x_i) - \alpha(x_{i-1})\right) \\ &= \left[\sum_{k=1, k\neq i}^{n} m_k \left(\alpha(x_k) - \alpha(x_{k-1})\right)\right] + m_i \left(\alpha(z) - \alpha(x_{i-1})\right) + m_i \left(\alpha(x_i) - \alpha(z)\right) \\ &\leq \left[\sum_{k=1, k\neq i}^{n} m_k \left(\alpha(x_k) - \alpha(x_{k-1})\right)\right] + m_i' \left(\alpha(z) - \alpha(x_{i-1})\right) + m_i'' \left(\alpha(x_i) - \alpha(z)\right) \\ &= L(f, \alpha, Q) \end{aligned}$$

so

$$L(f, \alpha, P) \leq L(f, \alpha, Q)$$

Inductive Step: Suppose the claim is true for $l = r \geq 1$. We want to show the claim holds for $l = r + 1$. Suppose $\text{card}(Q \backslash P) = r + 1$. Let

$$Q \backslash P = \{z_1, ..., z_r, z_{r+1}\}$$

Let

$$\hat{Q} = P \cup \{z_1, ..., z_r\}$$

We have

$$L(f, \alpha, P) \overset{\text{hypoth.}}{\leq} L(f, \alpha, \hat{Q}) \overset{\text{base case}}{\leq} L(f, \alpha, Q)$$

So,

$$L(f, \alpha, P) \leq L(f, \alpha, Q)$$

□

Theorem 2.2.2. (Lower Sums are Smaller than Upper Sums)
Let $f : [a,b] \to \mathbb{R}$ be bounded and $\alpha : [a,b] \to \mathbb{R}$ be increasing. Let P_1 and P_2 be any two partitions of $[a,b]$. Then $L(f,\alpha,P_1) \leq U(f,\alpha,P_2)$.

Proof. Let $Q = P_1 \cup P_2$ be the common refinement of P_1 and P_2. We have

$$L(f,\alpha,P_1) \leq L(f,\alpha,Q) \leq U(f,\alpha,Q) \leq U(f,\alpha,P_2)$$

□

Theorem 2.2.3. (The Lower R.S. Integral is less than the Upper R.S. Integral)
Let $f : [a,b] \to \mathbb{R}$ be bounded and $\alpha : [a,b] \to \mathbb{R}$ be increasing. Then $L(f,\alpha) \leq U(f,\alpha)$.

Proof. Note that if A and B are nonempty subsets of $\mathbb{R}$ such that $\forall a \in A \quad \forall b \in B \quad a \leq b$, then $\sup A \leq \inf B$. Let $A = \{L(f,\alpha,P) : P \in \Pi\}$ and $B = \{U(f,\alpha,P) : P \in \Pi\}$. By Thm 2.2.2, $a \leq b$ for every $a \in A$ and $b \in B$. So, it follows that $\sup A \leq \sup B$, that is $L(f,\alpha) \leq U(f,\alpha)$. □

Theorem 2.2.4. (Cauchy Criterion for R.S. Integrability)
Let $f : [a,b] \to \mathbb{R}$ be bounded and let $\alpha : [a,b] \to \mathbb{R}$ be increasing.

$$f \in \mathbb{R}_\alpha[a,b] \iff \forall \epsilon > 0 \;\; \exists P_\epsilon \in \Pi \text{ such that } U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon.$$

Proof. ($\Longleftarrow$) Suppose $f : [a,b]$ is bounded and $\alpha : [a,b] \to \mathbb{R}$ is increasing, and

$$\forall \epsilon > 0 \; \exists P_\epsilon \in \Pi \text{ such that } U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

We want to show $f \in R_\alpha[a,b]$. That is, we wat to show $L(f,\alpha) = U(f,\alpha)$. Since $U(f,\alpha) - L(f,\alpha) \geq 0$, it is enough to show that

$$\forall \epsilon > 0 \quad U(f,\alpha) - L(f,\alpha) < \epsilon$$

Let $\epsilon > 0$ be given. By assumption, there exists $P_\epsilon \in \Pi$ such that

$$U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

We have

$$U(f,\alpha) = \inf\{U(f,\alpha,P) : P \in \Pi\} \leq U(f,\alpha,P_\epsilon)$$
$$L(f,\alpha) = \sup\{L(f,\alpha,P) : P \in \Pi\} \geq L(f,\alpha,P_\epsilon)$$

Hence,

$$L(f,\alpha,P_\epsilon) \leq L(f,\alpha) \overset{Thm 2.2.3}{\leq} U(f,\alpha) \leq U(f,\alpha,P_\epsilon)$$

The interval $[L(f,\alpha), U(f,\alpha)]$ is contained in the interval $[L(f,\alpha,P_\epsilon), U(f,\alpha,P_\epsilon)]$. Thus

$$U(f,\alpha) - L(f,\alpha) \leq U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

as desired.

($\Longleftarrow$) Suppose $f \in R_\alpha[a,b]$. Our goal is to show

$$\forall \epsilon > 0 \quad \exists P_\epsilon \in \Pi \text{ such that } U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

Let *epsilon* > 0 be given. Our goal is to find $P_\epsilon \in \Pi$ such that

$$U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) < \epsilon$$

Note that

$$U(f,\alpha) = \inf\{U(f,\alpha,P) : P \in \Pi\} \implies \exists P_1 \in \Pi \text{ such that } U(f,\alpha,P_1) < U(f,\alpha) + \frac{\epsilon}{2}$$
$$L(f,\alpha) = \sup\{L(f,\alpha,P) : P \in \Pi\} \implies \exists P_2 \in \Pi \text{ such that } L(f,\alpha) - \frac{\epsilon}{2} < L(f,\alpha,P_2)$$

Let $P_\epsilon = P_1 \cup P_2$. We claim that this partition can be used to prove the claim.

$$L(f,\alpha) - \frac{\epsilon}{2} < L(f,\alpha,P_2) \leq L(f,\alpha,P_\epsilon) \leq U(f,\alpha,P_\epsilon) \leq U(f,\alpha,P_1) < U(f,\alpha) + \frac{\epsilon}{2}$$

Thus

$$\begin{aligned} U(f,\alpha,P_\epsilon) - L(f,\alpha,P_\epsilon) &< [(U(f,\alpha)+\epsilon/2) - (L(f,\alpha)-\epsilon/2)] \\ &= U(f,\alpha) - L(f,\alpha) + \epsilon \\ &= 0 + \epsilon \\ &= \epsilon \end{aligned}$$

□

Theorem 2.2.5. (Refinement and Tag Stability for R.S. Integrals)
Suppose $f : [a,b] \to \mathbb{R}$ is bounded and $\alpha : [a,b] \to \mathbb{R}$ is increasing. Let $\epsilon > 0$ and let $P = \{x_0, ..., xn\}$ be a partition of $[a,b]$ such that $U(f\alpha, P) - L(f,\alpha,P) < \epsilon$. Then

(*i*) If Q is a refinement of P, then $U(f,\alpha,Q) - L(f,\alpha,Q) < \epsilon$.

(*ii*) If $\forall 1 \le k \le n$ t_k and s_k are arbitrary points in $[x_{k-1}, x_k]$, then

$$\sum_{k=1}^{n} |f(s_k) - f(t_k)| \, \Delta\alpha_k < \epsilon$$

(*iii*) If $f \in R_\alpha[a,b]$ and $\forall 1 \le k \le n$ $s_k \in [x_{k-1}, x_k]$, then

$$\left| \sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \int_a^b f d\alpha \right| < \epsilon$$

Proof. (*i*) We have

$$L(f,\alpha,P) \le L(f,\alpha,Q) \le U(f,\alpha,Q) \le U(f,\alpha,P)$$

Therefore,

$$U(f,\alpha,Q) - L(f,\alpha,Q) \le U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

(*ii*) For each $1 \le k \le n$, we have

$$\begin{aligned} &m_k \le f(s_k) \le M_k \\ &m_k \le f(t_k) \le M_k \iff -M_k \le -f(t_k) \le -m_k \end{aligned}$$

so

$$\begin{aligned} &m_k - M_k \le f(s_k) - f(t_k) \le M_k - m_k \\ \implies &-(M_k - m_k) \le f(s_k) - f(t_k) \le M_k - m_k \\ \implies &|f(s_k) - f(t_k)| \le M_k - m_k \end{aligned}$$

Hence,

$$\sum_{k=1}^{n} |f(s_k) - f(t_k)| \, \Delta\alpha_k \le \sum_{k=1}^{n} (M_k - m_k) \, \Delta\alpha_k = U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

(*iii*) $\forall 1 \le k \le n$ $m_k \le f(s_k) \le M_k$ so,

$$\sum_{k=1}^{n} m_k \Delta\alpha_k \le \sum_{k=1}^{n} f(s_k)\Delta\alpha_k \le \sum_{k=1}^{n} M_k \Delta\alpha_k$$

Therefore,

$$L(f,\alpha,P) \le \sum_{k=1}^{n} f(s_k)\Delta\alpha_k \le U(f,\alpha,P)$$

Also, note that

$$L(f,\alpha,P) \le \int_a^b f d\alpha \le U(f,\alpha,P)$$

Hence,

$$\left| \sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \int_a^b f d\alpha \right| \le U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

□

Remark 2.2.1. Let $f : [a,b] \to \mathbb{R}$ be bounded, $\alpha : [a,b] \to \mathbb{R}$ be increasing, and $P = \{x_0, ..., x_n\}$ be a partition of $[a,b]$.

* If for every $1 \leq k \leq n$ we choose a point $t_k \in [x_{k-1}, x_k]$, then we call the pair $(P, (t_k)_{1\leq k\leq n})$ a tagged partition.
* We sometimes refer to $(t_k)_{1\leq k\leq n}$ as a tag of a partition.
* A sum of the form
$$S\left(f, \alpha, (P, (t_k)_{1\leq k\leq n})\right) = \sum_{k=1}^{n} f(t_k)\Delta\alpha_k$$
is called a Riemann-Stieltjes sum of f with respect to α (The Riemann-Stieltjes sum of f with respect to α for the tagged partition $(P, (t_k)_{1\leq k\leq n})$).
* As a direct consequence of Theorem 2.2.5 and the Cauchy Criterion for Riemann-Stieltjes integrability, we have
$$f \in R_\alpha[a,b]$$
$$\implies \forall \epsilon > 0 \;\; \exists P_\epsilon \in \Pi \text{ such that } \forall P_\epsilon \subseteq P \;\; \forall \text{tag}(t_k) \text{ of } P \;\; \left| S(f, \alpha, (P, (t_k)_{1\leq k\leq n})) - \int_a^b f d\alpha \right| < \epsilon$$
The converse is true in the following sense:
Suppose $f : [a,b] \to \mathbb{R}$ is bounded and $\alpha : [a,b] \to \mathbb{R}$ is increasing. Then the following two statements are equivalent:

(*i*) $f \in R_\alpha[a,b]$

(*ii*) There exists a number A such that
$$\forall \epsilon > 0 \;\; \exists P_\epsilon \in \Pi \text{ such that } \forall P_\epsilon \subseteq P \;\; \forall \text{tag}(t_k) \text{ of } P \;\; |S(f, \alpha, (P, (t_k)_{1\leq k\leq n})) - A| < \epsilon$$

Lemma 2.2.1. Let $f : [a,b] \to \mathbb{R}$ be a bounded function. Let $P = \{x_0, ..., x_n\}$ be a partition of $[a,b]$. Then
$$\forall 1 \leq k \leq n \quad \sup\{|f(s) - f(t)| : s, t \in [x_{k-1}, x_k]\} = M_k - m_k$$

Proof. Let $k \in \{1, ..., n\}$. We need to show

(*i*) $\forall s, t \in [x_{k-1}, x_k] \; |f(s) - f(t)| \leq M_k - m_k$

(*ii*) $\forall \epsilon > 0 \;\; \exists \hat{s}, \hat{t} \in [x_{k-1}, x_k]$ such that $M_k - m_k < \left|f(\hat{s}) - f(\hat{t})\right|$

We proved (*i*) in our discussion of theorem 2.2.5. It remains to show (*ii*):
Let $\epsilon > 0$ be given. Then
$$m_k = \inf\{f(t) : [x_{k-1}, x_k]\} \implies \exists \hat{t} \in [x_{k-1}, x_k] \text{ such that } f(\hat{t}) < m_k + \frac{\epsilon}{2} \implies -m_k - \frac{\epsilon}{2} < -f(\hat{t})$$
$$M_k = \sup\{f(t) : [x_{k-1}, x_k]\} \implies \exists \hat{s} \in [x_{k-1}, x_k] \text{ such that } M_k - \frac{\epsilon}{2} < f(\hat{s})$$
Therefore, adding the inequalities gives us
$$M_k - m_k - \epsilon < f(\hat{s}) - f(\hat{t}) \leq \left|f(\hat{s}) - f(\hat{t})\right|$$
□

Theorem 2.2.6. (Continuity Implies Integrability)
Let $f : [a,b] \to \mathbb{R}$ be continuous and let $\alpha : [a,b] \to \mathbb{R}$ be increasing. Then $f \in R_\alpha[a,b]$.

Proof. First, note that
$$\left.\begin{array}{l} f : [a,b] \to \mathbb{R} \text{ is continuous} \\ [a,b] \text{ is compact} \end{array}\right\} \overset{\text{EVT}}{\implies} f \text{ is bounded on } [a,b]$$

Now, according to the Cauchy Criterion of R.S. integrability, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f,\alpha,P) - L(f,\alpha,P) < \epsilon$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(f,\alpha,P) - L(f,\alpha,P) < \epsilon \qquad (*)$$

We have

$$\left.\begin{array}{l} f \text{ is continuous on } [a,b] \\ [a,b] \text{ is compact} \end{array}\right\} \implies f \text{ is uniformly continuous on } [a,b]$$

So, for the given *epsilon*,

$$\exists \delta \text{ such that } \forall s,t \in [a,b] \quad |s-t| < \delta \implies |f(s)-f(t)| < \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Let P be any partition of $[a,b]$ such that $\|P\| < \delta$. We claim that $(*)$ holds for such a P. Indeed, $\forall k \in \{1,...,n\} \quad \forall s,t \in [x_{k-1},x_k]$ we have

$$|s-t| < \delta \implies |f(t)-f(s)| < \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Thus

$$\sup_{s,t\in[x_{k-1},x_k]} |f(s)-f(t)| \leq \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Hence, by lemma 2.2.1,

$$M_k - m_k \leq \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]}$$

Therefore,

$$\begin{aligned} U(f,\alpha,P) - L(f,\alpha,P) &= \sum_{k=1}^{n} M_k - m_k \Delta\alpha_k \\ &\leq \sum_{k=1}^{n} \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]} \Delta\alpha_k \\ &= \frac{\epsilon}{2[\alpha(b)-\alpha(a)+1]} [\alpha(b)-\alpha(a)] \\ &\leq \frac{\epsilon}{2} \\ &< \epsilon \end{aligned}$$

□

Example 2.2.1. $f : [0,10] \to \mathbb{R}$ defined by $f(x) = \begin{cases} \frac{sinx}{x} & x > 0 \\ 1 & x = 0 \end{cases}$ is continuous.
$\alpha : [0,10] \to \mathbb{R}$ defined by $\alpha(x) = \lfloor x \rfloor$ is an increasing function.
Hence, by theorem 2.2.6,

$$\int_0^{10} f(x) d\lfloor x \rfloor \text{ exists}$$

Lemma 2.2.2. Let $\alpha : [a,b] \to \mathbb{R}$ be increasing and continuous and let $\alpha(a) < \alpha(b)$. Then for each $n \in \mathbb{N}$, there exists a partition $P = \{x_0,...,x_n\}$ such that

$$\forall 1 \leq k \leq n \quad \Delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1}) = \frac{\alpha(b)-\alpha(a)}{n}$$

Proof. Let $n \in \mathbb{N}$. Divide the interval $[\alpha(a),\alpha(b)]$ into n subintervals of equal length $\frac{\alpha(b)-\alpha(a)}{n}$. For each $1 < k < n$, we have $y_k \in (\alpha(a),\alpha(b))$, hence by the intermediate value theorem

$$\exists x_k \in (a,b) \text{ such that } y_k = \alpha(x_k)$$

Since α is increasing, we have $a < x_1 < x_2 < ... < x_{n-1} < b$. Now, if we define $x_0 = a$ and $x_n = b$, then $P = \{x_0, ..., x_n\}$ will be a partition of $[a, b]$ such that

$$\forall 1 \leq k \leq n \quad \delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1}) = y_k - y_{k-1} = \frac{\alpha(b) - \alpha(a)}{n}$$

□

Theorem 2.2.7. (Monotone Functions are Riemann-Stieltjes Integrable)
Suppose $\alpha : [a, b] \to \mathbb{R}$ is increasing and continuous.

(i) If $f : [a, b] \to \mathbb{R}$ is increasing, then $f \in R_\alpha[a, b]$

(ii) If $f : [a, b] \to \mathbb{R}$ is decreasing, then $f \in R_\alpha[a, b]$

Proof. We prove (i) ((ii) is analogous). First note that

$$\forall x \in [a, b] \quad f(a) \leq f(x) \leq f(b) \implies f \text{ is bounded on } [a, b]$$

If $\alpha(a) = \alpha(b)$, then we previously proved that $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = 0$. So it remains to prove the claim for the case where $\alpha(a) \neq \alpha(b)$. According to the Cauchy Criterion for integrability, in order to show $f \in R_\alpha[a, b]$, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in Pi$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon$$

Choose $n \in N$ large enough so that

$$\frac{\alpha(b) - \alpha(a)}{n}[f(b) - f(a)] < \epsilon.$$

Let $\tilde{P} = \{x_0, ..., x_n\}$ be a partition of $[a, b]$ such that

$$\begin{aligned} \forall 1 \leq k \leq n \quad \delta\alpha_k &= \alpha(x_k) - \alpha(x_{k-1}) \\ &= \frac{\alpha(b) - \alpha(a)}{n} \end{aligned}$$

We claim that $\tilde{P}$ can be used as the P we are looking for. Indeed,

$$\begin{aligned} U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) &= \sum_{k=1}^{n} (M_k - m_k)\delta\alpha_k \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{k=1}^{n} (M_k - m_k) \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{k=1}^{n} [f(x_k) - f(x_{k-1})] \quad \text{(for all } k \ M_k = \sup f(x) = f(x_k) \text{ and } m_k = f(x_{k-1})) \\ &= \frac{\alpha(b) - \alpha(a)}{n}[f(b) - f(a)] \\ &< \epsilon \end{aligned}$$

as desired. □

Example 2.2.2. $\alpha : [0, 2] \to \mathbb{R}$ given by $\alpha(x) = x^2$ is continuous and increasing and $f : [0, 2] \to \mathbb{R}$ given by $f(x) = \lfloor x^2 \rfloor$ is increasing. By theorem 2.2.7, $\int_0^2 \lfloor x^2 \rfloor d(x^2)$ exists.

Remark 2.2.2. Recall that to show $f \in R_\alpha[a, b]$, it is enough to show

$$\forall \epsilon > 0 \quad \exists P \in \Pi \text{ such that } U(f, \alpha, P) - L(f, \alpha, P) < \epsilon,$$

Also, note that

$$U(f,\alpha,P)-L(f,\alpha,P)=\sum_{k=1}^{n}(M_k-m_k)\delta\alpha_k.$$

So, we would like to make $\sum(M_k-m_k)\delta\alpha_k$ small.

$*$) Sometimes, this can be done by ensuring that M_k-m_k is small

$$M_k-m_k<\epsilon \implies \sum(M_k-m_k)\delta\alpha_k<\epsilon\sum\delta\alpha_k=\epsilon[\alpha(b)-\alpha(a)]$$

$*$) Sometimes, this can be done by making $\delta\alpha_k$ small

$$\sum(M_k-m_k)\delta\alpha_k \overset{\text{Lemma2.2.1}}{\leq} \sum 2\tilde{M}\delta\alpha_k=2\tilde{M}\sum\delta\alpha_k$$

where $\tilde{M}=\sup|f(x)|$.

Theorem 2.2.8. (Finite Discontinuities Implies R.S. Integrability)
Let $f:[a,b]\to\mathbb{R}$ be bounded and suppose f has finitely many discontinuities: $y_1<y_2<...<y_n$. Suppose $\alpha:[a,b]\to\mathbb{R}$ is increasing and α is continuous at $y_1<y_2<...<y_n$. Then $f\in R_\alpha[a,b]$.

Proof. According to the Cauchy criterion for Riemann-Stieltjes integrability, it is enough to show that

$$\forall\epsilon>0 \ \ \exists P\in\Pi \text{ such that } U(f,\alpha,P)-L(f,\alpha,P)<\epsilon$$

Let $\epsilon>0$ be given. Our goal is to find P such that

$$U(f,\alpha,P)-L(f,\alpha,P)<\epsilon$$

Let $\tilde{M}=\sup\limits_{a\leq x\leq b}|f(x)|$. Let $\hat{\epsilon}=\frac{\epsilon}{[\alpha(b)-\alpha(a)]+2\tilde{M}+1}$. We make two claims:

Claim 1: There exists finitely many disjoint intervals $[u_!,v_1],...,[u_m,v_m]$ such that

(i) $\forall 1\leq j\leq m \ \ y_j\in[u_j,v_j]$.
(ii) $\forall 1\leq j\leq m$ if $y_j\notin\{a,b\}$ then $y_j\in(u_j,v_j)$.
(iii) $\forall 1\leq j\leq m \ \ \alpha(v_j)-\alpha(u_j)<\frac{\hat{\epsilon}}{m}$ and so $\sum_{j=1}^{m}\alpha(v_j)-\alpha(u_j)<\epsilon$.

Claim 2: Let $K=[a,b]\setminus\bigcup_{j=1}^{m}(u_j,v_j)$. Then f is uniformly continuous on K.

We will prove claim 1 and claim 2 shortly, for now assume they hold. By claim 2,

$$\exists\delta>0 \text{ such that } \forall s,t\in K \text{ if } |s-t|<\delta \text{ then } |f(s)-f(t)|<\hat{\epsilon}.$$

Now, we form a partition $\tilde{P}$ of $[a,b]$ as follows:

(i) $\forall 1\leq j\leq m \ \ u_j,v_j\in\tilde{P}$.

(ii) $\forall 1\leq j\leq m$ no point of the segment (u_j,v_j) is in $\tilde{P}$.

(iii) If $1\leq k\leq n$ is such that $x_{k-1}\notin\{u_1,...,u_m\}$, then we choose x_k such that $x_k-x_{k-1}<\delta$.

We claim that $\tilde{P}$ can be used as the P that we were looking for. Indeed, let

$$A=\{k:x_{k-1}\notin\{u_1,...,u_m\}\},B=\{1,...,n\}\setminus A.$$

Note that

$*$) if $k\in A$, then $x_k-x_{k-1}<\delta$, so

$$\begin{aligned}
&\forall s,t\in[x_{k-1},x_k] \ \ |s-t|<\epsilon\\
\implies &\forall s,t\in[x_{k-1},x_k] \ \ |f(s)-f(t)|<\hat{\epsilon}\\
\implies &\sup_{s,t\in[x_{k-1},x_k]}|f(s)-f(t)|\leq\hat{\epsilon}\\
\implies &M_k-m_k\leq\hat{\epsilon}
\end{aligned}$$

$*$) if $k \in B$, then

$$M_k - m_k = \sup_{s,t \in [x_{k-1}, x_k]} |f(s) - f(t)| \leq 2\tilde{M}$$

Therefore,

$$\begin{aligned} U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) &= \sum_{k=1}^{n} (M_k - m_k)\Delta\alpha_k \\ &= \sum_{k \in A} (M_k - m_k)\Delta\alpha_k + \sum_{k \in B} (M_k - m_k)\Delta\alpha_k \\ &\leq \sum_{k \in A} \hat{\epsilon}\Delta\alpha_k + \sum_{k \in B} 2\tilde{M}\Delta\alpha_k \\ &\leq \tilde{\epsilon} \sum_{k=1}^{n} \Delta\alpha_k + 2\tilde{M} \sum_{k \in B} \Delta\alpha_k \\ &\leq \tilde{\epsilon}[\alpha(b) - \alpha(a)] + 2\tilde{M}\hat{\epsilon} \\ &= [\alpha(b) - \alpha(a) + 2\tilde{M}]\hat{\epsilon} \\ &< \epsilon \end{aligned}$$

Proof of Claim 1: Since for each $q \leq j \leq m$ α is continuous at y_j, we can choose $\delta_j > 0$ such that

$$\text{if } |y - y_j| < \delta_j \text{ then } |\alpha(y) - \alpha(y_j)| < \frac{\hat{\epsilon}}{2M}.$$

Now let

$$\tilde{\delta} = \frac{\min}{4} \{\delta_1, \delta_2, ..., \delta_m, y_2 - y_1, y_3 - y_2, y_4 - y_3, ..., y_m - y_{m-1}\}.$$

For each $1 \leq j \leq m$, we define

(i) If $y_j \notin \{a, b\}$, then $[u_j, v_j] = [y_j - \tilde{\delta}, y_j + \tilde{\delta}]$.

(ii) If $y_j = a$, then $[u_j, v_j] = [a, a + \tilde{\delta}]$.

(iii) If $y_j = b$, then $[u_j, v_j] = [b - \tilde{\delta}, b]$.

Clearly, these intervals satisfy all the requirements. In particular,

$$\alpha(v_j) - \alpha(u_j) = |\alpha(v_j) - \alpha(u_j)| \leq |\alpha(v_j) - \alpha(y_j)| + |\alpha(y_j) - \alpha(u_j)| < \frac{\hat{\epsilon}}{2M} + \frac{\hat{\epsilon}}{2M} = \frac{\hat{\epsilon}}{M}$$

Proof of Claim 2: Note that $\bigcup_{j=1}^{m} (u_j, v_j)$ is open. Hence,

$$K = [a, b] \backslash \bigcup_{j=1}^{m} (u_j, v_j) = [a, b] \cap [\bigcup_{j=1}^{m} (u_j, v_j)]^c \implies K \text{ is closed.}$$

$$\left.\begin{array}{l} K \subseteq [a, b] \\ K \text{ is closed} \\ [a, b] \text{ is compact} \end{array}\right\} \implies K \text{ is compact.}$$

Since $f : K \to \mathbb{R}$ is continuous and K is compact, we can conclude that f is uniformly continuous on K.

□

Theorem 2.2.9. (Applying Continuous Functions Preserves Integrability)
Suppose $f \in R_\alpha[a, b]$ such that $\forall x \in [a, b] \;\; m \leq f(x) \leq M$ and suppose $\varphi : [m, M] \to \mathbb{R}$ is continuous. Then $\varphi \circ f \in R_\alpha[a, b]$.

Proof. First, note that a composition of bounded functions is bounded, so $h = \varphi \circ f$ is bounded. According to the Cauchy criterion, in order to show $h \in R_\alpha[a, b]$ it is enough to show that

$$\forall \epsilon > 0 \; \exists P \in \Pi \text{ such that } U(h, \alpha, P) - L(h, \alpha, P) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(h, \alpha, P) - L(h, \alpha, P) < \epsilon.$$

Let $\tilde{M} = \sup_{a \le x \le b} |h(x)|$. Let $\hat{\epsilon} = \frac{\epsilon}{[\alpha(b)-\alpha(a)]+2\tilde{M}+1}$. We have

$$\left.\begin{array}{l} \varphi \text{ is continuous on } [m, M] \\ [m, M] \text{ is compact} \end{array}\right\} \implies \varphi \text{ is uniformly continuous on } [m, M].$$

so

$$\exists 0 < \delta < \hat{\epsilon} \text{ such that } \forall s, t \in [m, M] \text{ if } |s - t| < \delta \text{ then } |\varphi(s) - \varphi(t)| < \hat{\epsilon}$$

Also,

$$f \in R_\alpha[a, b] \implies \exists \tilde{P} \text{ such that } U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) = \sum_{k=1}^{n} (M_k - m_k)\, \Delta\alpha_k < \delta^2$$

We claim that this $\tilde{P}$ can be used as the P we are looking for. Indeed, let

$$\forall 1 \le k \le n \quad m_k^* = \inf \{h(x) : x \in [x_{k-1}, x_k]\}, \quad M_k^* = \sup \{h(x) : x \in [x_{k-1}, x_k]\}.$$

Note that

$$U(h, \alpha, \tilde{P}) - L(h, \alpha, \tilde{P}) = \sum_{k=1}^{n} (M_k^* - m_k^*)\, \Delta\alpha_k$$

In what follows, we will show that the above sum is less than ϵ. Divide the numbers $1, ..., n$ into two cases:

$$A = \{k : M_k - m_k < \delta\} \quad B = \{k : M_k - m_k \ge \delta\}$$

We have

$$U(h, \alpha, \tilde{P}) - L(h, \alpha, \tilde{P}) = \sum_{k=1}^{n} (M_k^* - m_k^*)\, \Delta\alpha_k = \sum_{k \in A} (M_k^* - m_k^*)\, \Delta\alpha_k + \sum_{k \in B} (M_k^* - m_k^*)\, \Delta\alpha_k \qquad \text{(I)}$$

$*$) If $k \in A$:

$$\begin{aligned} M_k - m_k < \delta &\implies \sup_{x \in [x_{k-1}, x_k]} |f(x) - f(y)| < \delta \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |f(x) - f(y)| < \delta \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |\varphi(f(x)) - \varphi(f(y))| < \hat{\epsilon} \\ &\implies \forall x, y \in [x_{k-1}, x_k] \quad |h(x) - h(y)| < \hat{\epsilon} \\ &\implies \sup_{x, y \in [x_{k-1}, x_k]} |h(x) - h(y)| \le \hat{\epsilon} \\ &\implies M_k^* - m_k^* \le \hat{\epsilon} \end{aligned} \qquad \text{(II)}$$

$*$) If $k \in B$, then

$$\delta \sum_{k \in B} \Delta\alpha_k = \sum_{k \in B} \delta \Delta\alpha_k \le \sum_{k \in B} (M_k - m_k)\, \Delta\alpha_k \le \sum_{k=1}^{n} (M_k - m_k)\, \Delta\alpha_k = U(f, \alpha, \tilde{P}) - L(f, \alpha, \tilde{P}) < \delta^2$$

$$\implies \delta \sum_{k \in B} \Delta\alpha_k < \delta^2 \implies \sum_{k \in B} \Delta\alpha_k < \delta \implies \sum_{k \in B} \Delta\alpha_k < \hat{\epsilon} \qquad \text{(III)}$$

It folows from (I), (II), and (III) that

$$\begin{aligned}
\sum_{k=1}^{n}(M_k^*-m_k^*)\,\Delta\alpha_k &= \sum_{k\in A}(M_k^*-m_k^*)\,\Delta\alpha_k+\sum_{k\in B}(M_k^*-m_k^*)\,\Delta\alpha_k\\
&\le \sum_{k\in A}\hat{\epsilon}\Delta\alpha_k+\sum_{k\in B}2\tilde{M}\Delta\alpha_k\\
&= \hat{\epsilon}\sum_{k\in A}\Delta\alpha_k+2\tilde{M}\sum_{k\in B}\Delta\alpha_k\\
&\le \hat{\epsilon}\sum_{k=1}^{n}\Delta\alpha_k+2\tilde{M}\hat{\epsilon}\\
&= \hat{\epsilon}[\alpha(b)-\alpha(a)+2\tilde{M}]\\
&= \frac{\epsilon}{\alpha(b)-\alpha(a)+2\tilde{M}+1}[\alpha(b)-\alpha(a)+2\tilde{M}]\\
&< \epsilon
\end{aligned}$$

□

Example 2.2.3. If $f\in R_\alpha[a,b]$, then $|f|\in R_\alpha[a,b]$.
If $f\in R_\alpha[a,b]$, then $f^2\in R_\alpha[a,b]$.

2.3 The Sequential Criterion of Integrability

Theorem 2.3.1. (The Sequential Criterion of Integrability)
Suppose $f : [a, b] \to \mathbb{R}$ is bounded and $\alpha : [a, b] \to \mathbb{R}$ is increasing.

(i) If $f \in R_\alpha[a, b]$, then there exists a sequence of partitions $(P_n)_{n\geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0.$$

(ii) If there exists a sequence of partitions $(P_n)_{n\geq 1}$ in $\Pi[a, b]$ such that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0,$$

then $f \in R_\alpha[a, b]$ and

$$\int_a^b f d\alpha = \lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} L(f, \alpha, P_n).$$

Proof. (i) $f \in R_\alpha[a, b] \iff \forall \epsilon > 0, \quad \exists P_\epsilon \in \Pi$ such that $U(f, \alpha, P_\epsilon) - L(f, \alpha, P_\epsilon) < \epsilon$. In particular,

$$\begin{aligned}
\epsilon = 1 &\implies \exists P_1 \in \Pi \text{ such that } U(f, \alpha, P_1) - L(f, \alpha, P_1) < 1 \\
\epsilon = \frac{1}{2} &\implies \exists P_2 \in \Pi \text{ such that } U(f, \alpha, P_2) - L(f, \alpha, P_2) < \frac{1}{2} \\
\epsilon = \frac{1}{3} &\implies \exists P_3 \in \Pi \text{ such that } U(f, \alpha, P_3) - L(f, \alpha, P_3) < \frac{1}{3} \\
&\vdots
\end{aligned}$$

In this way, we obtain a sequence $(P_n)_{n\geq 1}$ of partitions such that

$$\forall n \geq 1 \quad U(f, \alpha, P_n) - L(f, \alpha, P_n) < \frac{1}{n}.$$

It follows from the squeeze theorem that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0.$$

(ii) According the Cauchy Criterion, it is enough to show that

$$\forall \epsilon > 0 \quad \exists P \in \Pi[a, b] \text{ such that } U(f, \alpha, P_n) - L(f, \alpha, P_n) < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $P \in \Pi$ such that

$$U(f, \alpha, P) - L(f, \alpha, P) < \epsilon.$$

Since

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0,$$

there exists $N \in \mathbb{N}$ such that

$$\forall n \in \mathbb{N} \quad U(f, \alpha, P_n) - L(f, \alpha, P_n) < \epsilon.$$

In particular, P_{N+1} can be used as the P we were looking for. It remains to show that $\int_a^b f d\alpha = \lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} L(f, \alpha, P_n)$. We have

$$0 \leq U(f, \alpha, P_n) - U(f, \alpha) \leq U(f, \alpha, P_n) - L(f, \alpha, P_n) \leq U(f, \alpha, P_n) - L(f, \alpha, P_n) \to 0 \text{ as } n \to \infty.$$

If follows from the squeeze theorem that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - U(f, \alpha)] = 0.$$

Thus

$$\lim_{n\to\infty} U(f, \alpha, P_n) = U(f, \alpha) = \int_a^b f d\alpha.$$

Similary,

$$0 \leq L(f,\alpha) - L(f,\alpha,P_n) \leq U(f,\alpha) - L(f,\alpha,P_n) \leq U(f,\alpha,P_n) - L(f,\alpha,P_n) \to 0 \text{ as } n \to \infty$$
$$\overset{\text{SQZ Thm}}{\Longrightarrow} \lim_{n\to\infty} [L(f,\alpha) - L(f,\alpha,P_n)]$$
$$\Longrightarrow \lim_{n\to\infty} L(f,\alpha,P_n) = L(f,\alpha) = \int_a^b f d\alpha.$$

□

Theorem 2.3.2. (Algebraic Limit Theorem for Riemann-Stieltjes Integrals)
Suppose $f, g \in R_\alpha[a,b]$. Then

(i) $\forall k \in \mathbb{R} \quad kf \in R_\alpha[a,b]$ with $\int_a^b kf d\alpha = k\int_a^b f d\alpha$.

(ii) $f+g \in R_\alpha[a,b]$ with $\int_a^b f+g d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$.

(iii) $f^2 \in R_\alpha[a,b]$, and $fg \in R_\alpha[a,b]$.

(iv) If $g \neq 0$ on $[a,b]$ and $\frac{1}{g}$ is bounded on $[a,b]$, then $\frac{1}{g} \in R_\alpha[a,b]$ and $\frac{f}{g} \in R_\alpha[a,b]$.

Remark 2.3.1. (1) As a result of (i) and (ii), the set $R_\alpha[a,b]$ is a vector space. Moreover, the function

$$T : R_\alpha[a,b] \to \mathbb{R}$$
$$T(f) = \int_a^b f d\alpha$$

is linear.

(2) The function $g : [0,1] \to \mathbb{R}$ defined by

$$g(x) = \begin{cases} x & 0 < x \leq 1 \\ 1 & x = 0 \end{cases}$$

is an example of a bounded function which is never zero, but $\frac{1}{g}$ is unbounded.

Notation . Let $h : A \subseteq \mathbb{R} \to \mathbb{R}$ be a bounded function.

(i) We denote $\sup\{h(x) : x \in A\}$ by $\sup_A h$ or $\sup_A h(x)$ or $\sup_{x\in A} h(x)$.

(ii) We denote $\inf\{h(x) : x \in A\}$ by $\inf_A h$ or $\inf_A h(x)$ or $\inf_{x\in A} h(x)$.

Lemma 2.3.1. Let $A \subseteq \mathbb{R}$. Let $f : A \to \mathbb{R}$ and $g : A \to \mathbb{R}$ be bounded fucntions. Then

(i) $\sup_A(f+g) \leq \sup_A f + \sup_A g$.

(ii) $\inf_A(f+g) \geq \inf_A f + \inf_A g$.

(iii) $\forall k \geq 0 \quad \sup_A(kf) = k\sup_A f$ and $\inf_A(kf) = k\inf_A f$.

(iv) $\forall k \leq 0 \quad \sup_A(kf) = k\inf_A f$ and $\inf_A(kf) = k\sup_A f$.

(v) $\sup\{|f(x) - f(y)| : x, y \in A\} = \sup_A f - \inf_A f$.

(vi) If $\exists k > 0$ such that $\forall z, w \in A \quad |f(z) - f(w)| \leq k|g(z) - g(w)|$, then

$$\sup_A f - \inf_A f \leq k\left[\sup_A g - \inf_A g\right]$$

Lemma 2.3.2. Let $f : [a,b] \to \mathbb{R}$ and $g : [a,b] \to \mathbb{R}$ be bounded, $\alpha : [a,b] \to \mathbb{R}$ be icreasing, and $P \in \Pi[a,b]$. Then

(i) $U(f+g,\alpha,P) \leq U(f,\alpha,P) + U(g,\alpha,P)$.

(ii) $L(f+g,\alpha,P) \geq L(f,\alpha,P) + L(g,\alpha,P)$.

(iii) $\forall k \geq 0 \;\; U(kf,\alpha,P) = kU(f,\alpha,P)$ and $L(kf,\alpha,P) = kL(f,\alpha,P)$.

(iv) $\forall k < 0 \;\; U(kf,\alpha,P) = kL(f,\alpha,P)$ and $L(kf,\alpha,P) = kU(f,\alpha,P)$.

Proof. Proof of the Algebraic Limit Theorem for Integrals.

(i) $\forall k \in \mathbb{R}, \quad kf \in R_\alpha[a,b]$ with $\int_a^b kf d\alpha = k\int_a^b f d\alpha$:
Note that

$$f \in R_\alpha[a,b] \implies f \text{ is bounded on } [a,b] \implies kf \text{ is bounded on } [a,b].$$

Since $f \in R_\alpha[a,b]$, there exists a sequence $(P_n)n \geq 1$ in $\Pi[a,b]$ such that

$$\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] = 0, \int_a^b f d\alpha = \lim_{n\to\infty} U(f,\alpha,P_n) = \lim_{n\to\infty} L(f,\alpha,P_n).$$

In what follows, we'll show that

$$\lim_{n\to\infty}[U(kf,\alpha,P_n) - L(kf,\alpha,P_n)] = 0$$

and so $kf \in R_\alpha[a,b]$. We may consider two cases:

Case 1: $k \geq 0$

$$\begin{aligned}\lim_{n\to\infty}[U(kf,\alpha,P_n) - L(kf,\alpha,P_n)] &= \lim_{n\to\infty}[kU(f,\alpha,P_n) - kL(f,\alpha,P_n)] \\ &= k\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] \\ &= 0.\end{aligned}$$

Case 2: $k < 0$

$$\begin{aligned}\lim_{n\to\infty}[U(kf,\alpha,P_n) - L(kf,\alpha,P_n)] &= \lim_{n\to\infty}[kL(f,\alpha,P_n) - kU(f,\alpha,P_n)] \\ &= -k\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] \\ &= 0.\end{aligned}$$

Moreover, if $k \geq 0$, then

$$\int_a^b kf d\alpha = \lim_{n\to\infty} U(kf,\alpha,P_n) = \lim_{n\to\infty} k \cdot U(f,\alpha,P_n) = k \cdot \int_a^b f d\alpha.$$

Similarly, for $k < 0$,

$$\int_a^b kf d\alpha = \lim_{n\to\infty} U(kf,\alpha,P_n) = \lim_{n\to\infty} k \cdot L(f,\alpha,P_n) = k \cdot \int_a^b f d\alpha.$$

(ii) $f+g \in R_\alpha[a,b]$ with $\int_a^b f+g d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$:
Note that

$$\left.\begin{array}{l} f \in R_\alpha[a,b] \implies f \text{ is bounded on } [a,b] \\ g \in R_\alpha[a,b] \implies g \text{ is bounded on } [a,b] \end{array}\right\} \implies f+g \text{ is bounded on } [a,b].$$

Since $f \in R_\alpha[a,b]$ there exists $(P_n)_{n\geq 1}$ in $\Pi[a,b]$ such that

$$\lim_{n\to\infty}[U(f,\alpha,P_n) - L(f,\alpha,P_n)] = 0, \quad \int_a^b f d\alpha = \lim_{n\to\infty} U(f,\alpha,P_n) = \lim_{n\to\infty} L(f,\alpha,P_n).$$

Since $g \in R_\alpha[a,b]$ there exists $(Q_n)_{n\geq 1}$ in $\Pi[a,b]$ such that

$$\lim_{n\to\infty}[U(g,\alpha,Q_n) - L(g,\alpha,Q_n)] = 0, \quad \int_a^b g d\alpha = \lim_{n\to\infty} U(g,\alpha,Q_n) = \lim_{n\to\infty} L(g,\alpha,Q_n).$$

In what follows, we will show that

$$\lim_{n\to\infty}[U(f+g,\alpha,P_n \cup Q_n) - L(f+g,\alpha,P_n \cup Q_n)] = 0$$

and so $f+g \in R_\alpha[a,b]$. We have

$$\begin{aligned} L(f+g,\alpha,P_n\cup Q_n) &\geq L(f,\alpha,P_n\cup Q_n)+L(g,\alpha,P_n\cup Q_n)\\ &\geq L(f,\alpha,P_n)+L(g,\alpha,Q_n)\end{aligned}$$

and

$$\begin{aligned} U(f+g,\alpha,P_n\cup Q_n) &\leq U(f,\alpha,P_n\cup Q_n)+U(g,\alpha,P_n\cup Q_n)\\ &\leq U(f,\alpha,P_n)+U(g,\alpha,Q_n)\end{aligned}$$

so

$$\begin{aligned} L(f,\alpha,P_n)+L(g,\alpha,Q_n) &\leq L(f+g,\alpha,P_n\cup Q_n)\\ &\leq U(f+g,\alpha,P_n\cup Q_n)\\ &\leq U(f,\alpha,P_n)+U(g,\alpha,Q_n).\end{aligned}$$

Therefore,

$$\begin{aligned} 0 &\leq U(f+g,\alpha,P_n\cup Q_n)-L(f+g,\alpha,P_n\cup Q_n)\\ &\leq [U(f,\alpha,P_n)-L(f,\alpha,P_n)]+[U(f,\alpha,Q_n)-L(f,\alpha,Q_n)]\to 0+0=0.\end{aligned}$$

By the squeeze theorem,

$$\lim_{n\to\infty}[U(f+g,\alpha,P_n\cup Q_n)-L(f+g,\alpha,P_n\cup Q_n)]=0.$$

Moreover,

$$\begin{aligned} \int_a^b f+g\,d\alpha &= \lim_{n\to\infty}[U(f+g,\alpha,P_n\cup Q_n)]\\ &\leq \lim_{n\to\infty}[U(f,\alpha,P_n)+U(g,\alpha,P_n)]=\int_a^b f\,d\alpha+\int_a^b g\,d\alpha\\ \int_a^b f+g\,d\alpha &= \lim_{n\to\infty}[L(f+g,\alpha,P_n\cup Q_n)]\\ &\geq \lim_{n\to\infty}[L(f,\alpha,P_n)+L(g,\alpha,P_n)]=\int_a^b f\,d\alpha+\int_a^b g\,d\alpha.\end{aligned}$$

This implies $\int_a^b f+g\,d\alpha=\int_a^b f\,d\alpha+\int_a^b g\,d\alpha$.

(iii) $f^2\in R_\alpha[a,b]$:
Note that $\phi:\mathbb{R}\to\mathbb{R}$, $\phi(x)=x^2$ is continuous, so since $f\in R_\alpha[a,b]$, $\phi\circ f\in R_\alpha[a,b] \implies f^2\in R_\alpha[a,b]$.
$fg\in R_\alpha[a,b]$:
Note that

$$\left.\begin{aligned}(f+g)^2&=f^2+2fg+g^2\\(f-g)^2&=f^2-2fg+g^2\end{aligned}\right\}\implies 4fg=(f+g)^2-(f-g)^2\implies fg=\frac{1}{4}\left[(f+g)^2-(f-g)^2\right].$$

Therefore,

$$\begin{aligned} f,g\in R_\alpha[a,b] &\overset{(i)}{\implies} f,g,-g\in R_\alpha[a,b]\\ &\overset{(ii)}{\implies} f+g,f-g\in R_\alpha[a,b]\\ &\overset{(iii)}{\implies} (f+g)^2,(f-g)^2\in R_\alpha[a,b]\\ &\overset{(i)}{\implies} (f+g)^2,-(f-g)^2\in R_\alpha[a,b]\\ &\overset{(ii)}{\implies} (f+g)^2-(f-g)^2\in R_\alpha[a,b]\\ &\overset{(i)}{\implies} \frac{1}{4}\left[(f+g)^2-(f+g)^2\right]\in R_\alpha[a,b]\\ &\implies fg\in R_\alpha[a,b].\end{aligned}$$

□

Theorem 2.3.3. (Order Properties of Riemann-Stieltjes Integrals)
Supose $f, g \in R_\alpha[a,b]$. Then

(i) If $m \le f(x) \le M \quad \forall x \in [a,b]$, then $m(\alpha(b)-\alpha(a)) \le \int_a^b f d\alpha \le M(\alpha(b)-\alpha(a))$.

(ii) If $f \le g$ on $[a,b]$, then $\int_a^b f d\alpha \le \int_a^b g d\alpha$.

Proof. (i) Note that for any $P \in \Pi[a,b]$,

$$\int_a^b f d\alpha = L(f,\alpha) \ge L(f,\alpha,P)$$
$$\int_a^b f d\alpha = U(f,\alpha) \le U(f,\alpha,P).$$

In particular, for the partition $P = \{a,b\}$, we have

$$\int_a^b f d\alpha \ge L(f,\alpha,P) = \inf\{f(x) : x \in [a,b]\}(\alpha(b)-\alpha(a)) \ge m(\alpha(b)-\alpha(a))$$
$$\int_a^b f d\alpha \le U(f,\alpha,P) = \sup\{f(x) : x \in [a,b]\}(\alpha(b)-\alpha(a)) \le M(\alpha(b)-\alpha(a))$$

(ii) Let $h = g - f$. We have $h \ge 0$, so by part (i)

$$\begin{aligned} 0(\alpha(b)-\alpha(a)) &\le \int_a^b h d\alpha \\ &= \int_a^b g d\alpha - \int_a^b f d\alpha \\ \implies \int_a^b f d\alpha &\le \int_a^b g d\alpha. \end{aligned}$$

□

Theorem 2.3.4. ($|\int_a^b f d\alpha| \le \int_a^b |f| d\alpha$)
Suppose $f \in R_\alpha[a,b]$. Then

(i) $|f| \in R_\alpha[a,b]$.

(ii) $|\int_a^b f d\alpha| \le \int_a^b |f| d\alpha$.

Proof. (i) Since $\phi : \mathbb{R} \to \mathbb{R}$, $\phi(x) = |x|$ is continuous and f is Riemann-Stieltjes integrable, it follows that $|f| = \phi \circ f$ is Riemann-Stieltjes integrable. Hence $|f| \in R_\alpha[a,b]$.

(ii) Recall that $|t| \le s \iff -s \le t \le s$. So our goal is to show that

$$-\int_a^b |f| d\alpha \le \int_a^b f d\alpha \le \int_a^b |f| d\alpha$$

Clearly, (for any c, $-|c| \le c \le |c|$)

$$\begin{aligned} -|f(x)| \le f(x) \le |f(x)| \quad \forall x \in [a,b] \\ \implies -\int_a^b |f| d\alpha \le \int_a^b f d\alpha \le \int_a^b |f| d\alpha \quad \forall x \in [a,b] \end{aligned}$$

□

Theorem 2.3.5. (The Mean Value Theorem for Riemann-Stieltjes Integrals)
Suppose $f : [a,b] \to \mathbb{R}$ is continuous, $\alpha : [a,b] \to \mathbb{R}$ is increasing, and $\alpha(a) \ne \alpha(b)$. Then

$$\exists c \in [a,b] \text{ such that } f(x) = \frac{\int_a^b f d\alpha}{\alpha(b)-\alpha(a)}$$

Proof. Because f is continuous on $[a,b]$, and $[a,b]$ is compact, by the Extreme Value Theorem f attains its maximum and minimum on $[a,b]$. Let $m = \min\{f(x) : x \in [a,b]\}$ and $M = \max\{f(x) : x \in [a,b]\}$. We have

$$\begin{aligned} &\forall x \in [a,b] \quad m \le f(x) \le M \\ &\implies m(\alpha(b) - \alpha(a)) \le \int_a^b f d\alpha \le M(\alpha(b) - \alpha(a)) \\ &\implies m \le \frac{\int_a^b f d\alpha}{\alpha(b) - \alpha(a)} \le M. \end{aligned}$$

Since f is continuous, it follows from the Intermediate Value Theorem

$$\exists c \in [a,b] \text{ such that } f(c) = \frac{\int_a^b f d\alpha}{\alpha(b) - \alpha(a)}$$

□

Theorem 2.3.6. (Generalized Mean Value Theorem for Riemann-Stieltjes Integrals) Suppose $f : [a,b] \to \mathbb{R}$ is continuous, $\alpha : [a,b] \to \mathbb{R}$ is increasing, $g \in R_\alpha[a,b]$, and either $g \ge 0$ on $[a,b]$ or $g \le 0$ on $[a,b]$. Then

$$\exists c \in [a,b] \text{ such that } \int_a^b f g d\alpha = f(c) \int_a^b g d\alpha.$$

Proof. We will prove the claim for the case $g \ge 0$ on $[a,b]$. The proof for the case $g \le 0$ is analogous. Since f is continuous on $[a,b]$ and $[a,b]$ is compact, f contains its maximum and minimum on $[a,b]$. Let $M = \max\{f(x) : x \in [a,b]\}$ and $m = \min\{f(x) : x \in [a,b]\}$. Then, since $g \le 0$ on $[a,b]$,

$$\begin{aligned} &\forall x \in [a,b] \quad m \le f(x) \le M \\ \implies &\forall x \in [a,b] \quad mg(x) \le f(x)g(x) \le Mg(x) \\ \implies &\forall x \in [a,b] \quad m \int_a^b g d\alpha \le \int_a^b f g d\alpha \le M \int_a^b g d\alpha. \end{aligned}$$

Now, we may consider two cases:

Case 1: $\int_a^b g d\alpha = 0$ If $\int_a^b g d\alpha = 0$, it follows that $0 \le \int_a^b f g d\alpha \le 0$ and so $\int_a^b f g d\alpha = 0$. So in this case, the claim holds for every $c \in [a,b]$:

$$\int_a^b f g d\alpha = f(c) \int_a^b g d\alpha.$$

Case 2: $\int_a^b g d\alpha \ne 0$ In this case, since $g > 0$, it follows that $\int_a^b g d\alpha > 0$ and

$$m \le \frac{\int_a^b f g d\alpha}{\int_a^b g d\alpha} \le M.$$

Since f is continuous, it follows from the Intermediate Value Theorem that

$$\exists c \in [a,b] \text{ such that } f(x) = \frac{\int_a^b f g d\alpha}{\int_a^b g d\alpha}.$$

That is, $\int_a^b f g d\alpha = f(c) \int_a^b g d\alpha$.

□

Theorem 2.3.7. (Additivity for Riemann-Stieltjes Integrals) Suppose $f : [a,b] \to \mathbb{R}$ is continuous, $\alpha : [a,b] \to \mathbb{R}$ is increasing, and $c \in (a,b)$. Then

$$f \in R_\alpha[a,b] \iff f \in R_\alpha[a,c] \text{ and } f \in R_\alpha[c,b].$$

In this case, we have

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha.$$

□

2.4 The Fundamental Theorem of Calculus and Techniques of Integration

Lemma 2.4.1.
Suppose $f : [a,b] \to \mathbb{R}$ is bounded, $\alpha : [a,b] \to \mathbb{R}$ is increasing, $P = \{x_0, ..., x_n\} \in \Pi[a,b]$, and $R \in \mathbb{R}$.

(i) If for all tags $(s_k)_{a \le k \le n}$ of P we have $\sum_{k=1}^n f(s_k)\Delta\alpha_k \le R$, then $U(f,\alpha,P) \le R$.

(ii) If for all tags $(s_k)_{a \le k \le n}$ of P we have $R \le \sum_{k=1}^n f(s_k)\Delta\alpha_k$, then $L(f,\alpha,P) \ge R$.

Proof. (i) If α is constant, then $\sum_{k=1}^n f(s_k)\Delta\alpha_k = 0$ and $U(f,\alpha,P) = \sum_{k=1}^n M_k\Delta\alpha_k = 0$, so the claim obviously holds. Suppose $\alpha(b) \neq \alpha(a)$. It's enough to show

$$\forall \epsilon > 0 \;\; U(f,\alpha,P) \le R + \epsilon.$$

Let $\epsilon > 0$ be given. For each $k \in \{1, ..., n\}$ we have

$$M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\} \implies \exists s_k \in [x_{k-1}, x_k] \text{ such that } M_k - \frac{\epsilon}{\alpha(b)-\alpha(a)} < f(s_k).$$

We have

$$\begin{aligned} U(f,\alpha,P) &= \sum_{k=1}^n M_k\Delta\alpha_k < \sum_{k=1}^n \left[f(s_k) + \frac{\epsilon}{\alpha(b)-\alpha(a)}\right]\Delta\alpha_k \\ &= \sum_{k=1}^n f(s_k)\Delta\alpha_k + \frac{\epsilon}{\alpha(b)-\alpha(a)}\sum_{k=1}^n \Delta\alpha_k \\ &\le R + \epsilon \end{aligned}$$

as desired.

(ii) Completely analogous to (i). □

Theorem 2.4.1.
Suppose $f : [a,b] \to \mathbb{R}$ is bounded, $\alpha[a,b] \to \mathbb{R}$ is increasing, and $\alpha' \in R[a,b]$. Then

$$f \in R_\alpha[a,b] \iff f\alpha' \in R[a,b]$$

and in this case,

$$\int_a^b f d\alpha = \int_a^b f(x)\alpha'(x)dx.$$

Proof. It is enough to show that

$$\begin{aligned} U(f,\alpha) &= U(f\alpha') \\ L(f,\alpha) &= L(f\alpha') \end{aligned} \tag{$*$}$$

Indeed, if we prove $(*)$, then we can write

$$\begin{aligned} f \in R_\alpha[a,b] &\iff U(f,\alpha) = L(f,\alpha) \\ &\iff U(f\alpha') = L(f\alpha') \\ &\iff f\alpha' \in R[a,b]. \end{aligned}$$

Moreover,

$$\int_a^b f d\alpha = U(f,\alpha) \overset{(*)}{=} U(f\alpha') = \int_a^b f(x)\alpha'(x)dx.$$

In what follows, we will prove $U(f,\alpha) = U(f\alpha')$. The proof for $L(f,\alpha) = L(f\alpha')$ is completely analogous.

let $P = \{x_0, ..., x_n\}$ be any partition of $[a, b]$. Let (s_k) be any tag of P. We have

$$\begin{aligned}
\left|\sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k\right| &= \left|\sum_{k=1}^{n} f(s_k)\alpha'(t_k)\Delta x_k - \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k\right| \qquad \text{(MVT on } \Delta\alpha_k\text{)} \\
&= \left|\sum_{k=1}^{n} f(s_k)\left(\alpha'(t_k) - \alpha'(s_k)\right)\Delta x_k\right| \\
&\leq \sum_{k=1}^{n} |f(s_k)| \cdot |\alpha'(t_k) - \alpha'(s_k)|\,\Delta x_k \\
&\leq \hat{M}\sum_{k=1}^{n} |\alpha'(t_k) - \alpha'(s_k)|\,\Delta x_k \qquad (\hat{M} = \sup_{a\leq x\leq b}|f(x)|) \\
&\leq \hat{M}\sum_{k=1}^{n}\left[\sup_{I_k}\alpha' - \inf_{I_k}\alpha'\right]\Delta x_k \qquad \text{(Lemma 2.2.1)} \\
&= \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right]
\end{aligned}$$

So,

$$\left|\sum_{k=1}^{n} f(s_k)\Delta\alpha_k - \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k\right| \leq \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

Therefore,

$$\begin{aligned}
(1)\quad \sum_{k=1}^{n} f(s_k)\Delta\alpha_k &\leq \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right] \\
&\leq U(f\alpha', P) + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].
\end{aligned}$$

So, by Lemma 2.4.1,

$$U(f, \alpha, P) \leq U(f\alpha', P) + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

$$(2)\quad \sum_{k=1}^{n} f(s_k)\alpha'(s_k)\Delta x_k \leq \sum_{k=1}^{n} f(s_k)\Delta\alpha_k + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right]$$

So, by Lemma 2.4.1,

$$U(f\alpha', P) \leq U(f, \alpha, P) + \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

It follows from (1) and (2) that

$$(3)\quad |U(f, \alpha, P) - U(f\alpha', P)| \leq \hat{M}\left[U(\alpha', P) - L(\alpha', P)\right].$$

Note that

$$\begin{aligned}
U(f,\alpha) = \inf\{U(f,\alpha,P) : P \in \Pi\} &\implies \exists(P_n^{(1)}) \subseteq \Pi \text{ such that } \lim_{n\to\infty} U(f,\alpha,P_n^{(1)}) = U(f,\alpha) \\
U(f\alpha') = \inf\{U(f\alpha',P) : P \in \Pi\} &\implies \exists(P_n^{(2)}) \subseteq \Pi \text{ such that } \lim_{n\to\infty} U(f\alpha',P_n^{(2)}) = U(f\alpha') \\
\alpha' \in R[a,b] &\implies \exists(P_n^{(3)}) \subseteq \Pi \text{ such that } \lim_{n\to\infty}\left[U(\alpha',P_n^{(3)}) - L(\alpha',P_n^{(3)})\right] = 0.
\end{aligned}$$

For each $n \in \mathbb{N}$, let $P_n = P_n^{(1)} \cup P_n^{(2)} \cup P_n^{(3)}$. By the Squeeze Theorem,

$$\begin{aligned}
\forall n \geq 1 \;\; U(f,\alpha) \leq U(f,\alpha,P_n) \leq U(f,\alpha,P_n^{(1)}) &\implies \lim_{n\to\infty} U(f,\alpha,P_n) = U(f,\alpha) \\
\forall n \geq 1 \;\; U(f\alpha') \leq U(f\alpha',P_n) \leq U(f\alpha',P_n^{(2)}) &\implies \lim_{n\to\infty} U(f\alpha',P_n) = U(f\alpha').
\end{aligned}$$

Since P_n is a refinement of $P_n^{(3)}$,

$$0 \leq [U(\alpha', P_n) - L(\alpha', P_n)] \leq U(\alpha', P_n^{(3)}) - L(\alpha', P_n^{(3)}) \implies \lim_{n\to\infty}[U(\alpha', P_n) - L(\alpha', P_n)] = 0.$$

It follows from (3) that

$$\forall n \geq 1 \;\; 0 \leq |U(f,\alpha,P_n) - U(f\alpha',P_n)| \leq \hat{M}\left[U(\alpha',P_n) - L(\alpha',P_n)\right].$$

So, by the Squeeze Theorem,

$$\begin{aligned}
&\lim_{n\to\infty} |U(f,\alpha,P_n) - U(f\alpha',P_n)| = 0 \\
\implies &\left|\lim_{n\to\infty} (U(f,\alpha,P_n) - U(f\alpha',P_n))\right| = 0 \\
\implies &|U(f,\alpha) - U(f\alpha')| = 0 \\
\implies &U(f,\alpha) - U(f\alpha') = 0 \\
\implies &U(f,\alpha) = U(f\alpha')
\end{aligned}$$

□

Theorem 2.4.2. (Change of Variables for Integrals)
Let $f \in R_\alpha[a,b]$, and let $\phi : [A,B] \to [a,b]$ be onto and strictly increasing. If we let $g = f \circ \phi$ and $\beta = \alpha \circ \phi$, then

$$g \in R_\beta[A,B] \text{ and } \int_a^b f d\alpha = \int_A^B g d\beta.$$

Remark 2.4.1. Consider the special case where $\alpha(x) = x$ and $\phi' \in \mathbb{R}$. In this case, $\beta = \alpha \circ \phi = \phi$. So, the conclusion of the theorem can be written as

$$\int_a^b f(x)dx = \int_A^B (f \circ \phi)d\phi = \int_A^B f(\phi(y))\phi'(y)dy.$$

Proof. First, note that

$$\left.\begin{array}{l} \alpha : [a,b] \to \mathbb{R} \text{ is increasing} \\ \phi : [A,B] \to \mathbb{R} \text{ is increasing} \end{array}\right\} \implies \beta = \alpha \circ \phi \text{ is increasing on } [A,B]$$

and

$$\left.\begin{array}{l} f \in R_\alpha[a,b] \implies f \text{ is bounded on } [a,b] \\ \forall x \in [A,B]\ a \le \phi(x) \le b \implies \phi \text{ is bounded on } [A,B] \end{array}\right\} \implies g = f \circ \phi \text{ is bounded on } [A,B].$$

Also, note that there is a 1-1 correspondance between $\Pi[a,b]$ and $\Pi[A,B]$:

$$\begin{aligned} H : \Pi[a,b] &\to \Pi[A,B] \\ P = \{x_0,...,x_n\} &\mapsto Q = \{y_0,,...,y_n\} \quad (x_k = \phi(y_k)) \end{aligned}$$

Under this 1-1 correspondance, we have

$$\begin{aligned}
&\forall 1 \le k \le n \quad \phi([y_{k-1},y_k]) = [x_{k-1},x_k] \\
&\forall 1 \le k \le n \quad \Delta\beta_k = \beta(y_k) - \beta(y_{k-1}) = \alpha(\phi(y_k)) - \alpha(\phi(y_{k-1})) = \alpha(x_k) - \alpha(x_{k-1}) = \Delta\alpha_k \\
&\forall 1 \le k \le n \quad M_k^{(g)} = \sup\{g(y) : y \in [y_{k-1},y_k]\} = \sup\{f(x) : x \in [x_{k-1},x_k]\} = M_k^{(f)} \\
&\forall 1 \le k \le n \quad m_k^{(g)} = \inf\{g(y) : y \in [y_{k-1},y_k]\} = \inf\{f(x) : x \in [x_{k-1},x_k]\} = m_k^{(f)}.
\end{aligned}$$

Thus, under the above correspondace,

$$\begin{aligned} U(f,\alpha,P) &= U(g,\beta,Q) \\ L(f,\alpha,P) &= L(f,\alpha,Q) \end{aligned}$$

In order to show $g \in R_\beta[A,B]$, by the Cauchy Criterion, it is enough to prove that

$$\forall \epsilon > 0 \quad \exists Q \in \Pi[A,B] \text{ such that } U(g,\beta,Q) - L(g,\beta,Q) < \epsilon.$$

Let $\epsilon > 0$ be given. Since $f \in R_\alpha[a,b]$, $\exists P \in \Pi[a,b]$ such that

$$U(f,\alpha,P) - L(f,\alpha,P) < \epsilon.$$

We claim that $Q = H(P)$ is the partition that we're looking for. Indeed,

$$U(g,\beta,H(P)) - L(g,\beta,H(P)) = U(f,\alpha,P) - L(f,\alpha,P) < \epsilon.$$

Also,

$$\begin{aligned}\int_a^b f d\alpha = L(f,\alpha) &= \sup\{L(f,\alpha,P) : P \in \Pi[a,b]\} \\ &= \sup\{L(g,\beta,H(P)) : H(P) \in \Pi[A,B]\} \\ &= L(g,\beta) \\ &= \int_A^B g d\beta.\end{aligned}$$

□

Theorem 2.4.3. (The Fundamental Theorem of Calculus)

(i) Supose $f \in R[a,b]$ and $F : [a,b] \to \mathbb{R}$ satisfies $F'(x) = f(x) \ \ \forall x \in [a,b]$. Then

$$\int_a^b f = F(b) - F(a).$$

(ii) Suppose $g \in R[a,b]$ and $G : [a,b] \to \mathbb{R}$ is defined by $G(x) = \int_a^x g(t)dt$. Then

(1) G is uniformly continuous on $[a,b]$.

(2) If g is differentiable at $c \in [a,b]$, then G is differentiable at c and

$$G'(c) = g(c).$$

Proof. (i) In what follows, we'll show that

$$\forall P \in \Pi[a,b] \ \ L(f,P) \le F(b) - F(a) \le U(f,P). \qquad (*)$$

As a consequence of this:

(I) $F(b) - F(a)$ is an upper bound for $\{L(f,P) : P \in \Pi\}$, so

$$\sup\{L(f,P) : P \in \Pi\} \le F(b) - F(a) \implies L(f) \le F(b) - F(a)$$

(II) $F(b) - F(a)$ is a lower bound for $\{U(f,P) : P \in \Pi\}$, so

$$\inf\{U(f,P) : P \in \Pi\} \ge F(b) - F(a) \implies U(f) \ge F(b) - F(a).$$

Thus, $L(f) \le F(b) - F(a) \le U(f)$. Since $f \in R[a,b]$, $U(f) = L(f) = \int_a^b f d\alpha$, giving us

$$\int_a^b f d\alpha \le F(b) - F(a) \le \int_a^b f d\alpha.$$

Therefore, $\int_a^b f d\alpha$.

So, we just need to prove $(*)$. Let $P = \{x_0, ..., x_n\} \in \Pi[a,b]$. We have

$$\begin{aligned}F(b) - F(a) &= \sum_{k=1}^n F(x_k) - F(x_{k-1}) \\ &= \sum_{k=1}^n F'(t_k)(x_k - x_{k-1}) \text{for some } t_k \in [x_{k-1}, x_k] \qquad \text{(MVT for derivatives)} \\ &= \sum_{k=1}^n f(t_k)\Delta x_k.\end{aligned}$$

Therefore, it follows from

$$L(f,p) = \sum_{k=1}^n m_k \Delta x_k \le \sum_{k=1}^n f(t_k)\Delta x_k \le \sum_{k=1}^n M_k \Delta x_k = U(f,P)$$

that

$$L(f,P) \le F(b) - F(a) \le U(f,P).$$

(ii) (1) Our goal is to show G is uniformly continuous on $[a,b]$. That is, we need to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \forall x, y \in [a,b] \text{ if } |x-y| < \delta, \text{ then } |G(x) - G(y)| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ such that

$$\forall x, y \in [a,b] \text{ if } |x-y| < \delta, \text{ then } |G(x) - G(y)| < \epsilon. \qquad (*)$$

Let $x, y \in [a,b]$. If $x \geq y$,

$$\begin{aligned} |G(x) - G(y)| &= \left| \int_a^x g(t)dt - \int_a^y g(t)dt \right| \\ &= \left| \int_y^x g(t)dt \right| \\ &\leq \int_y^x |g(t)|dt \\ &\leq \int_y^x R dt \qquad \text{(Let } R = \sup_a \leq t \leq b |g|) \\ &= R(x-y) \\ &= R|x-y|. \end{aligned}$$

If $x < y$, then

$$\begin{aligned} |G(x) - G(y)| &= |G(y) - G(x)| \\ &= \left| \int_a^y g(t)dt - \int_a^x g(t)dt \right| \\ &= \left| \int_x^y g(t)dt \right| \\ &\leq \int_x^y |g(t)|dt \\ &\leq \int_x^y R dt \\ &= R(y-x) \\ &= R|x-y|. \end{aligned}$$

Thus, $\forall x, y \in [a,b]$

$$|G(x) - G(y)| \leq R|x-y|.$$

Hence, to ensure $|G(x) - G(y)| < \epsilon$ it is enough to make $R|x-y| < \epsilon$. That is, it is enough to make $|x-y| < \epsilon/R$. Choosing $\delta = \epsilon/R$ does the job, and so $(*)$ is satisfied.

(2) Suppose g is continuous at $c \in [a,b]$. Our goal is to show $G'(c) = g(c)$. That is, we want to show

$$\lim_{x \to c} \frac{G(x) - G(c)}{x - c} = g(c).$$

That is, our goal is to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \text{ if } 0 < |x-c| < \delta, \text{ then } \left| \frac{G(x) - G(c)}{x-c} - g(c) \right| < \epsilon.$$

Let $\epsilon > 0$ be given. Since g is continuous at c, for the given ϵ, there exists $\hat{\delta} > 0$ such that

$$\text{if } \underset{t \in [a,b]}{|t-c| < \hat{\delta}}, \text{ then } |g(t) - g(c)| < \epsilon.$$

We claim that this $\hat{\delta}$ can be used as the δ we're looking for. Indeed, if we let $\delta = \hat{\delta}$, then

Case 1: $0 < |x-c| < \hat{\delta}, x \in [a,b], x > c$

$$\begin{aligned}
\left|\frac{G(x)-G(c)}{x-c} - g(c)\right| &= \left|\frac{G(x)-G(c)-g(c)(x-c)}{x-c}\right| \\
&= \left|\frac{\int_a^x g(t)dt - \int_a^c g(t)dt - \int_a^x g(c)dt}{x-c}\right| \\
&= \left|\frac{\int_c^x g(t)dt - \int_c^x g(c)dt}{x-c}\right| \\
&= \left|\frac{\int_c^x g(t)-g(c)dt}{x-c}\right| \\
&= \frac{1}{|x-c|}\left|\int_c^x g(t)-g(c)dt\right| \\
&\leq \frac{1}{|x-c|}\int_c^x |g(t)-g(c)|dt \\
&\leq \frac{1}{x-c}\int_c^x \frac{\epsilon}{2}dt \\
&= \frac{\epsilon/2}{x-c}(x-c) < \epsilon.
\end{aligned}$$

Case 2: $0 < |x-c| < \hat{\delta}, x \in [a,b], x < c$

$$\begin{aligned}
\left|\frac{G(x)-G(y)}{x-c} - g(c)\right| &= \left|\frac{G(x)-G(y)-g(c)(x-c)}{x-c}\right| \\
&= \left|\frac{g(c)(c-x)-(G(c)-G(x))}{c-x}\right| \\
&= \left|\frac{\int_x^c g(c)dt - \int_a^c g(t)dt + \int_a^x g(t)dt}{c-x}\right| \\
&\vdots \\
&= \left|\frac{\int_x^c g(c)-g(t)dt}{c-x}\right| \\
&\leq \frac{1}{c-x}\int_x^c |g(c)-g(t)|dt \\
&\leq \frac{\int_x^c \frac{\epsilon}{2}dt}{c-x} \\
&< \epsilon.
\end{aligned}$$

□

Notation . Let I be a nonempty interval in $\mathbb{R}$.

$$C^0(I) = C(I) = \{f : I \to \mathbb{R} \mid f \text{ is continuous}\}$$

$$\forall n \geq 1 \quad C^n(I) = \left\{f : I \to \mathbb{R} \mid f, f', ..., f^{(n)} \text{ exist and are continuouts}\right\}$$

Theorem 2.4.4. (Integration by Parts)
Suppose $u : [a,b] \to \mathbb{R}$ and $v : [a,b] \to \mathbb{R}$ are differentiable and $u', v' \in R[a,b]$. Then

(i) $uv' \in R[a,b]$.

(ii) $u'v \in R[a,b]$.

(iii) $\int_a^b uv'dx = u(b)v(b) - u(a)v(a) - \int_a^b u'vdx$.

Proof. (i) $u : [a,b] \to \mathbb{R}$ is differentiable, so

$$\left.\begin{array}{r} u \in C[a,b] \implies u \in R[a,b] \\ u \in R[a,b] \\ v' \in R[a,b] \end{array}\right\} \implies uv' \in R[a,b]$$

(ii) Analogous to (i).

(iii) By the product rule,

$$(uv)' = u'v + uv'.$$

In particular, since $(uv)'$ is the sum of integrable functions it follows that $(uv)' \in R[a,b]$. Integrate both sides:

$$\int_a^b (uv)'dx = \int_a^b u'vdx = \int_a^b uv'dx \qquad (I)$$

By the Fundamental Theorem of Calculus part (i),

$$\int_a^b (uv)'dx = [uv]_{x=a}^{x=b} = u(b)v(b) - u(a)v(a) \qquad (II)$$

By (I) and (II), we have

$$\begin{aligned} u(b)v(b) - u(a)v(a) &= \int_a^b u'vdx + \int_a^b uv'dx \\ \implies \int_a^b u'vdx &= u(b)v(b) - u(a)v(a) - \int_a^b uv'dx \end{aligned}$$

□

Definition 2.4.1. (Unit Step Function)
The unit step function $I : \mathbb{R} \to \mathbb{R}$ is defined by

$$I(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}.$$

Note that for all $s \in \mathbb{R}$,

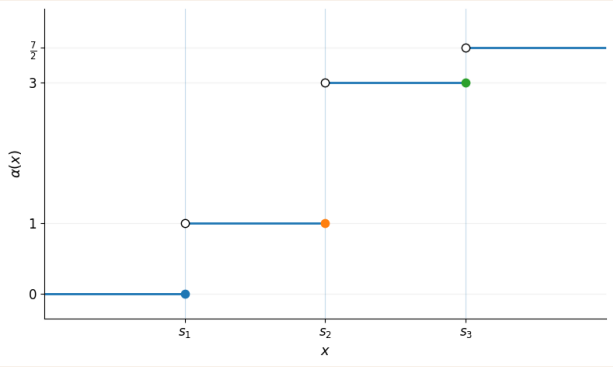
$$I(x-s) = \begin{cases} 0 & \text{if } x \leq s \\ 1 & \text{if } x > 2 \end{cases}.$$

Also, for all $c \neq 0$,

$$cI(x-s) = \begin{cases} 0 & \text{if } x \leq s \\ 1 & \text{if } x > s \end{cases}.$$

Example 2.4.1. Suppose $s_1 < s_2 < s_3$. Graph the function

$$\begin{aligned} \alpha(x) &= I(x-s_1) + 2I(x-s_2) + \frac{1}{2}I(x-s_3) \\ \implies \alpha(x) &= \begin{cases} 0+0+0 & \text{if } x \leq s_1 \\ 1+0+0 & \text{if } s_1 < x \leq s_2 \\ 1+2+0 & \text{if } s_2 < x \leq s_3 \\ 1+2+\frac{1}{2} & \text{if } s_3 < x \end{cases} \end{aligned}$$



Theorem 2.4.5.
Suppose $f : [a, b] \to \mathbb{R}$ is bounded, $s \in (a, b)$, f is right continuous at s, and $\alpha(x) = I(x - s)$. Then $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = f(s)$.

Proof. In what follows, we will introduce a sequence of partitions (P_n) in $\Pi[a, b]$ such that

$$\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = 0 \text{ and } \lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} L(f, \alpha, P_n) = f(s).$$

This, according to the sequential criterion of integrability, tells us that

$$f \in R_\alpha[a, b] \text{ and } \int_a^b f d\alpha = f(s).$$

Let N be large enough so that $s + \frac{1}{N} < b$. For each $n \geq N$, let

$$P_n = \left\{a, s, s + \frac{1}{n}, b\right\}.$$

We have

$$U(f, \alpha, P_n) = \sum_{k=1}^{3} M_k \Delta\alpha_k = M_2 \Delta\alpha_2 = \left[\sup_{s\leq x\leq s+\frac{1}{n}} f(x)\right](1) = \sup_{s\leq x\leq s+\frac{1}{n}} f(x)$$

$$L(f, \alpha, P_n) = \sum_{k=1}^{3} m_k \Delta\alpha_k = m_2 \Delta\alpha_2 = \left[\inf_{s\leq x\leq s+\frac{1}{n}} f(x)\right](1) = \inf_{s\leq x\leq s+\frac{1}{n}} f(x).$$

Therefore,

$$\lim_{n\to\infty} U(f, \alpha, P_n) = \lim_{n\to\infty} \sup_{s\leq x\leq s+\frac{1}{n}} f(x) = f(x)$$

$$\lim_{n\to\infty} L(f, \alpha, P_n) = \lim_{n\to\infty} \inf_{s\leq x\leq s+\frac{1}{n}} f(x) = f(x).$$

Here, we are using the fact that f is continuous at s, but right continuity would have been enough. We conclude that $\lim_{n\to\infty} [U(f, \alpha, P_n) - L(f, \alpha, P_n)] = f(s) - f(s) = 0$, so $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = \lim U(f, \alpha, P_n) = f(s)$. □

Theorem 2.4.6.
Suppose $\forall n \geq 1, c_n \geq 0$, $\sum_{n=1}^{\infty} c_n$ converges. Let $s_1 < s_2 < \ldots$ be points in $(a, b), \alpha : [a, b] \to \mathbb{R}, \alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$, and $f : [a, b] \to \mathbb{R}$ be continuous. Then

$$f \in R_\alpha[a, b] \text{ and } \int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$$

Chapter 3

Sequences of Functions

3.1 Sequences of Functions and Pointwise Convergence

Definition 3.1.1. (Sequences of Functions)
Let $A \neq$. If $\forall n \in \mathbb{N}$ we have $f_n : A \to \mathbb{R}$, then

$$f_1, f_2, f_3, \ldots$$

is a sequence of functions and is denoted

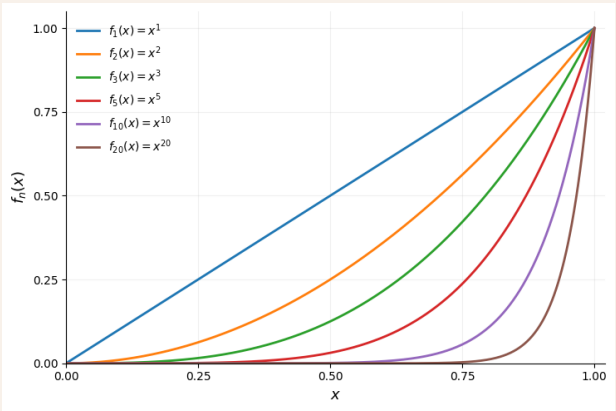
$$(f_n)_{n\geq 1}, (f_n), \text{ or } (f_n : A \to \mathbb{R})_{n\geq 1}.$$

Example 3.1.1. (i) $(f_n)_{n\geq 1} = \left(\frac{\sin(nx)}{n}\right)_{n\geq 1}$ on the set $\mathbb{R}$:

$$\begin{aligned} f_1 : \mathbb{R} \to \mathbb{R}, f_1(x) &= \sin x \\ f_2 : \mathbb{R} \to \mathbb{R}, f_2(x) &= \frac{1}{2}\sin 2x \\ f_3 : \mathbb{R} \to \mathbb{R}, f_3(x) &= \frac{1}{3}\sin 3x \\ &\vdots \end{aligned}$$

(ii) $(f_n)_{n\geq 1} = (x^n)$ on the set $[0, 1]$:

$$\begin{aligned} f_1 : [0,1] \to \mathbb{R}, f_1(x) &= x \\ f_2 : [0,1] \to \mathbb{R}, f_2(x) &= x^2 \\ f_3 : [0,1] \to \mathbb{R}, f_3(x) &= x^3 \\ &\vdots \end{aligned}$$



(iii) Let $f : [0, 1] \to \mathbb{R}$ be a given function. For all $n \in \mathbb{N}$, let $f_n(x) = f(x^n)$. Then (f_n) is a sequence of functions.

Remark 3.1.1. Suppose $(f_n : A \to \mathbb{R})_{n\geq 1}$ is a sequence of functions. Note that for each point $c \in A$, $(f_n(c))n \geq 1$ will be a sequence of numbers. For example, consider $(f_n(x) = x^n)_{n\geq 1}$ on the set $A = [0, 2]$:

$$\begin{aligned} x = 0: &\quad f_1(0) = 0, f_2(0) = 0, f_3(0) = 0, ... \text{ is a sequence of numbers} \\ x = 1: &\quad f_1\left(\frac{1}{2}\right) = \frac{1}{2}, f_2\left(\frac{1}{2}\right) = \frac{1}{2^2}, f_3\left(\frac{1}{2}\right) = \frac{1}{2^3}, ... \text{ is a sequence of numbers} \\ x = 2: &\quad f_1(1) = 1, f_2(1) = 1, f_3(1) = 1, ... \text{ is a sequence of numbers} \\ &\quad \vdots \end{aligned}$$

Definition 3.1.2. (Pointwise Convergent Sequence)
We say a sequence of functions $(f_n : A \to \mathbb{R})_{n\geq 1}$ is pointwise convergent if at any $c \in A$, the sequence $(f_n(c))$ converges.

Example 3.1.2. For all $n \geq 1$, $f_n : \{0, 1\} \to \mathbb{R}, f_n(x) = 2x + \frac{x}{n}$.

$c \in \{0, 1\}$:	$(f_n(c))$	Convergence
0	$(f_n(0)) = (0, 0, 0, ...)$	$f_n(0) \to 0$
1	$(f_n(1)) = (2 + 1, 2 + \frac{1}{2}, 2 + \frac{1}{3}, ...)$	$f_n(1) \to 2$

So, (f_n) is a pointwise convergent sequence.

Definition 3.1.3. (Pointwise Limit)
Let $(f_n : A \to \mathbb{R})_{n\geq 1}$ be a pointwise convergent sequence. Then we define a new function on A as follows:

$$f : A \to \mathbb{R}, \quad \forall c \in A \;\; f(c) = \lim_{n\to\infty} f_n(c).$$

We call f the pointwise limit of the sequence (f_n) on A. We say f_n converges pointwise to the function f on A.

Example 3.1.3. For all $n \geq 1$, $f_n : [-1, 1] \to \mathbb{R}$, $f_n(x) = x^2 + \frac{1}{n}$. Prove that $f_n \to f$ pointwise on $[-1, 1]$ where $f(x) = x^2$.
At each point $c \in [-1, 1]$, the corresponding sequence of numbers

$$(f_n(c))_n \geq 1 = (c^2 + 1, c^2 + \frac{1}{2}, c^2 + \frac{1}{3}, ...) = (c^2 + \frac{1}{n})_{n\geq 1}.$$

For each $c \in [-1, 1]$, we have

$$\lim_{n\to\infty} f_n(c) = \lim_{n\to\infty} \left(c^2 + \frac{1}{n}\right) = c^2 = f(c).$$

Thus, $f_n \to f$ pointwise where $f : [-1, 1] \to \mathbb{R}$ defined by $f(x) = x^2$.

Example 3.1.4. For all $n \geq 1$, $f_n : [0, 1] \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$. Provethat $f_n \to f$ where $f : [0, 1] \to \mathbb{R}$ is defined by $f(x) \equiv 0$.
At each point $c \in [0, 1]$, the corresponding sequence of numbers is

$$(f_n(c)) = \left(c, \frac{c}{2}, \frac{c}{3}, ...\right) = \left(\frac{c}{n}\right)_{n\geq 1}.$$

Thus for each $c \in [0, 1]$, we have

$$\lim_{n\to\infty} f_n(c) = \lim n \to \infty \frac{c}{n} = c \lim_{n\to\infty} \frac{1}{n} = c \cdot 0 = 0.$$

Example 3.1.5. $\forall n \geq 1$ $f_n : [0, 1] \to \mathbb{R}, f_n(x) = x^n$. Prove that $f_n \to f$ pointwise where $f : [0, 1] \to \mathbb{R}$ is defined by $f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}$.

At each point $c \in [0,1]$, the corresponding sequence of numbers is

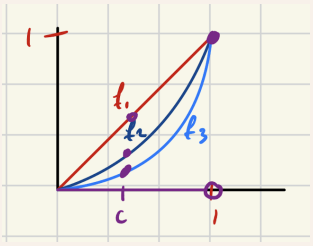
$$(f_c(c))_{n\geq 1} = (c^n)_{n\geq 1} = (c, c^2, c^3, ...).$$

Note that

$(*)$ if $0 \leq c < 1$, then $f_n(c) \to c^n = 0 = f(c)$.

$(*)$ if $c = 1$, then $f_n(c) \to 1 = f(c)$.

So, $f_n \to f$ pointwise.



Observation 3.1.1. In this example, each f_n is continuous but the pointwise limit f is not continuous.

Remark 3.1.2. The pointwise limit of a sequence of continuous functions is not necessarily continuous.

"Pointwise convergence does not preserve convergence."

Example 3.1.6. $\forall n \geq 1$ $f_n : [0,1] \to \mathbb{R}, f_n(x) = \begin{cases} x & \text{if } x \leq n \\ n & \text{if } x > n \end{cases}$. Prove $f_n \to f$ pointwise where $f : [0,\infty) \to \mathbb{R}$ is defined by $f(x) = x$.
At each point $c \in [0,\infty)$, the corresponding sequence of numbers is

$$(f_n(c))_{n\geq 1} = (f_1(c), f_2(c), f_3(c), ...).$$

Note that for fixed $c \in [0,\infty)$, $\exists N \in \mathbb{N}$ such that $c \leq N$, so clearly

$$c \leq N, \quad c \leq N+1, \quad c \leq N+2, ...$$

Thus

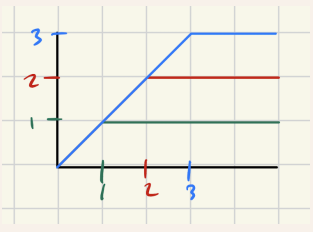
$$f_N(c) = c, f_{N+1}(c) = c, f_{N+2}(c) = c, ...$$

Therefore,

$$(f_n(c))_{n\geq 1} = (f_1(c), f_2(c), f_3(c), ..., f_{N-1}(c), c, c, c, ...).$$

We conclude that

$$\lim_{n\to\infty} f_n(c) = c.$$



Observation 3.1.2. Each f_n is bounded, but $f(x) = x$ is not bounded.

Remark 3.1.3. The pointwise limit of a sequence of bounded functions is not necessarily bounded.

"Pointwise convergence does not preserve boundedness."

Example 3.1.7. Let $f : [0,1] \to \mathbb{R}$ be a continuous, nowhere differentiable function. By the Weierstrass Approximation Theorem, there exists a sequence of polynomials $p_1(x), p_2(x), p_3(x), ...$ such that p_n converges to f pointwise.

Observation 3.1.3. Each p_n is differentiable, but the pointwise limit f is not differentiable.

Remark 3.1.4. The pointwise limit f of a sequence of differentiable functions is not necessarily differentiable. Also, if f' exists, it's not the case that $f_n' \to f'$.

"Pointwise convergence does not preserve differentiability."

Example 3.1.8. $\forall n \geq 4$, define $f_n : [0,1] \to \mathbb{R}$ as follows:

$$f_n(x) = \begin{cases} 0 & \text{if } 0 \leq x < \frac{1}{n} \\ n(nx-1) & \text{if } \frac{1}{n} \leq x < \frac{2}{n} \\ -n(nx-3) & \text{if } \frac{2}{n} \leq x < \frac{3}{n} \\ 0 & \text{if } \frac{3}{n} \leq x \leq 1 \end{cases}.$$

Prove that $f_n \to f$ pointwise where $f \equiv 0$.
At each fixed point $c \in [0,1]$, the corresponding sequence of numbers

$$(f_n(c))_{n\geq 4} = (f_4(c), f_5(c), f_6(c), ...).$$

Note that

$$\begin{aligned} c = 0: \quad (f_n(0))_{n\geq 4} = (0,0,0,0,...) &\implies \lim_{n\to\infty} f_n(0) = 0 \\ 0 < c \leq 1: \quad \exists N \in \mathbb{N} \text{ such that } \frac{3}{N} < c &\implies \frac{3}{N} < c, \frac{3}{N+1} < c, \frac{3}{N+2} < c, ... \\ &\implies f_N(c) = 0, f_{N+1}(c) = 0, f_{N+2}(c) = 0, ... \\ &\implies (f_n(c))_{n\geq 4} = (f_4(c), f_5(c), ..., f_{N-1}(c), 0, 0, ...) \\ &\implies \lim_{n\to\infty} f_n(c) = 0. \end{aligned}$$

$\forall c \in [0,1]$, $\lim_{n\to\infty} f_n(c) = 0 = f(c)$, so $f_n \to f = 0$ pointwise.

Observation 3.1.4. Each f_n, $\int_0^n f_n = 1$. However, $\int_0^1 f = 0$, so for every n $\int_0^1 f_n \neq \int_0^1 f$.

Remark 3.1.5. Pointwise convergence does not allow us to interchange limits and integrals.

"Pointwise convergence does not preserve integrals."

3.2 Uniform Convergence

Definition 3.2.1. (Uniform Convergence)
We say $(f_n : A \to \mathbb{R})$ converges uniformly to $f : A \to \mathbb{R}$ if

$$\forall \epsilon > 0 \;\; \exists N_\epsilon \text{ such that } \forall n > N_\epsilon \; \forall c \in A \;\; |f_n(c) - f(c)| < \epsilon.$$

Notation . $f_n \overset{u}{\to} f$.

Example 3.2.1. $\forall n \geq 1 \;\; f_n : \mathbb{R} \to \mathbb{R}, \; f_n(x) = \dfrac{1}{n(1+x^2)}$.

(i) Prove that $f_n \to f$ pointwise where $f : \mathbb{R} \to \mathbb{R}$ is defined by $f = 0$.
At each point $c \in \mathbb{R}$, the corresponding sequence of real numbers is

$$(f_n(c))_{n\geq 1} = \left(\frac{1}{n(1+c^2)}\right)_{n\geq 1} = \left(\frac{1}{1+c^2}, \frac{1}{2(1+c^2)}, \frac{1}{3(1+c^2)}, \ldots\right).$$

Thus for each $c \in \mathbb{R}$ we have

$$\lim_{n\to\infty} f_n(c) = \lim_{n\to\infty}\left[\frac{1}{n(1+c^2)}\right] = \frac{1}{1+c^2}\left(\lim_{n\to\infty}\frac{1}{n}\right) = 0 = f(c).$$

So, $f_n \to f$ pointwise where $f \equiv 0$.

(ii) Prove that $f_n \overset{u}{\to} f$.
Our goal is to show that

$$\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in \mathbb{R} \; |f_n(c) - 0| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find N such that

$$\forall n > N \;\; \forall c \in \mathbb{R} \; \frac{1}{n(1+c^2)} < \epsilon.$$

By Archimedean property, $\exists \hat{N}$ such that $\frac{1}{\hat{N}} < \epsilon$. Then $\forall n > \hat{N}$ and $\forall c \in \mathbb{R}$, we have

$$\frac{1}{n(1+c^2)} \leq \frac{1}{n} < \frac{1}{\hat{N}} < \epsilon$$

as desired.

Example 3.2.2. $\forall n \geq 1$, $f_n : [0,1] \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$. Prove that $f_n \to 0$ uniformly on $[0,1]$.
Our goal is to the show that

$$\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \;\; \forall c \in [0,1] \;\; |f_n(0) - 0| < \epsilon.$$

Let $\epsilon > 0$ be given. By the Archimedean Property, $\exists \hat{N}$ such that $\frac{1}{\hat{N}} < \epsilon$. We claim that this $\hat{N}$ can be used as the N that we are looking for. Indeed, $\forall n > \hat{N}$ and $\forall c \in [0,1]$ we have

$$\frac{c}{n} \leq \frac{1}{n} < \frac{1}{N} < \epsilon$$

as desired.

Example 3.2.3. $\forall n \geq 1$, $f_n : [0,\infty) \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$.

(i) Prove that $f_n \to 0$ pointwise on $[0,\infty)$.
At each point $c \in [0,\infty)$, the corresponding sequence of real numbers is

$$(f_n(c))_{n\geq 1} = \left(\frac{c}{n}\right)_{n\geq 1} = \left(c, \frac{c}{2}, \frac{c}{3}, \frac{c}{4}, \ldots\right).$$

Thus, for each $c \in [0, \infty)$,

$$\lim_{n\to\infty} f_n(c) = \lim_{n\to\infty} \frac{c}{n} = c\left(\lim_{n\to\infty} \frac{1}{n}\right) = 0.$$

So, $f_n \to f$ pointwise where $f \equiv 0$.

(ii) Prove that the convergence is not uniform on $[0, \infty)$.

Assume for contradiction that $f_n \overset{u}{\to} 0$ on $[0, \infty)$. Then

$$\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in [0, \infty) \;\; |f_n(c) - 0| < \epsilon.$$

In particular, for $\epsilon = 1$, $\exists N \in \mathbb{N}$ such that

$$\forall n > N \; \forall c \in [0, \infty) \;\; \frac{c}{n} < 1.$$

Letting $n = N + 1$, we get

$$\forall c \in [0, \infty) \;\; \frac{c}{N+1} < 1.$$

Thus,

$$\forall c \in [0, \infty) \;\; c < N + 1.$$

This is clearly false.

Theorem 3.2.1. (Uniform Converges Preserves Boundedness)
Suppose A is not empty, and for each $n \in \mathbb{N}$ let $f_n : A \to \mathbb{R}$ be bounded, and $f_n \overset{u}{\to} f$ on A. Then $f : A \to \mathbb{R}$ is bounded.

Proof. Our goal is to show that

$$\exists M \in \mathbb{R} \quad \text{such that } \forall c \in A \;\; |f(c)| \leq M.$$

Since $f_n \overset{u}{\to} f$ on A, we have

$$\begin{aligned} &\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in A \;\; |f_n(c) - f(c)| < \epsilon \\ \implies &\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in A \;\; |f_n(c)| - |f(c)| < \epsilon \\ \implies &\forall \epsilon > 0 \;\; \exists N \text{ such that } \forall n > N \; \forall c \in A \;\; |f(c)| < \epsilon + |f_n(c)|. \end{aligned}$$

In particular, for $\epsilon = 1$, $\exists N \in \mathbb{N}$ such that

$$\forall n > N \; \forall c \in A \;\; |f(c)| < 1 + |f_n(c)|.$$

Letting $n = N + 1$, we get

$$\forall c \in A \;\; |f(c)| < 1 + |f_{N+1}(c)|. \tag{1}$$

Since, by assumption, f_{N+1} is bounded, $\exists \hat{M}_{N+1}$ such that

$$\forall c \in A \;\; |f_{N+1}(c)| \leq \hat{M}_{N+1}. \tag{2}$$

It follows from (1) and (2) that

$$\forall c \in A \;\; |f(c)| < 1 + \hat{M}_{N+1}.$$

Clearly, we can use $1 + \hat{M}_{N+1}$ as the M we were looking for. □

Theorem 3.2.2. (Uniform Convergence Preserves Continuity)
Let (X, d) be a metric space. Suppose $A \subseteq (X, d)$, $c \in A$, for each $n \in \mathbb{N}$ $f_n : A \to R$ is continuous at c, and $f_n \overset{u}{\to} f$ on A. Then $f : A \to \mathbb{R}$ is continuous at c.

Proof. Our goal is to show that

$$\forall \epsilon > 0 \;\; \exists \delta > 0 \text{ such that } \text{ if } d(x, c) < \delta \text{ for } x \in A, \text{ then } |f(x) - f(c)| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find $\delta > 0$ satisfying

$$\text{if } d(x,c) < \delta, \text{ then } |f(x) - f(c)| < \epsilon.$$

Informal Discussion

Recall that our general strategy to ensure that a quantity of interest is less than ϵ is as follows:

Step 1: Bound quantity 1 with quantity 2, which is a combination of terms that are easier to deal with or are terms over which we can control.

Step 2: Make sure quantity 2 is less than ϵ.

According to the hypothesis of this theorem, here is a list of quantities over which we have control:

(i) $f_n \overset{u}{\to} f$ on $A \implies$ we can make $|f_n(x) - f(x)|$ small $\forall x$.

(ii) f_n is continuous at $c \implies$ we can make $|f_n(x) - f(c)|$ small for each n.

We have

$$\begin{aligned} |f(x) - f(c)| &= |f(x) - f_n(x) + f_n(x) - f_n(c) + f_n(c) - f(c)| \\ &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(c)| + |f_n(c) - f(c)|. \end{aligned}$$

So, in order to ensure $|f(x) - f(c)| < \epsilon$, it is enough to make each piece on the RHS less than $\frac{\epsilon}{3}$. We will first find f_n that forces the first term and the third term to be less than $\frac{\epsilon}{3}$. Once we establish which f_n we want, we can use the continuity of f_n at c to produce a δ that makes the middle term be less than $\frac{\epsilon}{3}$ for $x \in N_\delta(c) \cap A$.

Since $f_n \overset{u}{\to} f$ on A,

$$\exists N \in \mathbb{N} \text{ such that } \forall n > N \ \forall z \in A \ \ |f_n(z) - f(z)| < \frac{\epsilon}{3}. \tag{1}$$

Also, since f_{N+1} is continuous at c,

$$\exists \hat{\delta} > 0 \text{ such that } \forall x \in N_{\hat{\delta}}(c) \cap A \ \ |f_{N+1}(x) - f_{N+1}(c)| < \frac{\epsilon}{3}. \tag{2}$$

We claim that this $\hat{\delta}$ can be used as the δ we are looking for. For all $x \in N_\delta(c) \cap A$, we have

$$\begin{aligned} |f(x) - f(c)| &= |f(x) - f_{N+1}(x) + f_{N+1}(x) - f_{N+1}(c) + f_{N+1}(c) - f(c)| \\ &\leq |f(x) - f_{N+1}(x)| + |f_{N+1}(x) - f_{N+1}(c)| + |f_{N+1}(c) - f(c)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \qquad ((1) \text{ and } (2)) \\ &= \epsilon \end{aligned}$$

as desired. □

Observation 3.2.1. Let (a_n) be a sequence of real numbers. Suppose $a_n \to a \in \mathbb{R}$. Suppose there exists N such that

$$\forall m, n > N \ \ |a_n - a_m| \leq \frac{1}{3}.$$

Note that, for each fixed $n > N$, we have

$$\forall m > N \ \ |a_n - a_m| \leq \frac{1}{3}.$$

So, by taking the limit as $m \to \infty$, it follows from the Order Limit Theorem that for each $n > N$

$$\lim_{m\to\infty} |a_n - a_m| \leq \lim_{m\to infty} \frac{1}{3}.$$

Therefore,

$$\forall n > N \ \ |a_n - a| \leq \frac{1}{3}.$$

More generally, given $\epsilon > 0$, if there exists N such that

$$\forall n, m > N \ \ |a_n - a_m| \leq \epsilon$$

then by taking the limit as $m \to \infty$, we obtain

$$\forall n > N \quad |a_n - a| \leq \epsilon.$$

Theorem 3.2.3. (Cauchy Criterion for Uniform Convergence)
Let $A \neq \emptyset$. Suppose for each $n \in \mathbb{N}$, $f_n : A \to \mathbb{R}$ is a function. Then

$$(f_n)_{n\geq 1} \text{ converges uniformly } \iff \forall \epsilon > 0 \ \exists N \text{ such that } \forall n, m > N \ \forall c \in A \quad |f_n(c) - f_m(c)| < \epsilon.$$

Proof. ($\implies$) Suppose $\exists f : A \to \mathbb{R}$ such that $f_n \overset{u}{\to} f$ on A. Let $\epsilon > 0$ be given. Our goal is to find N such that

$$\forall n, m > N \ \forall c \in A \quad |f_n(c) - f_m(c)| < \epsilon.$$

Since $f_n \overset{u}{\to} f$ on A, for the given $\epsilon > 0$ $\exists \hat{N}$ such that

$$\forall k > \hat{N} \ \forall c \in A \quad |f_k(c) - f(c)| < \frac{\epsilon}{2}.$$

We claim this $\hat{N}$ can be used as the N we're looking for. Indeed, if $m, n > \hat{N}$ and $c \in A$ then

$$|f_n(c) - f_m(c)| \leq |f_n(c) - f(c)| + |f_m(c) - f(c)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

($\impliedby$) Suppose $\forall \epsilon > 0$ $\exists N$ such that $\forall n, m > N$ $\forall c \in A$ $|f_n(c) - f_m(c)| < \epsilon$. It follows from the assumption that at each point $c \in A$, the sequence of real numbers $(f_n(c))_{n\geq 1}$ is a Cauchy sequence in $\mathbb{R}$. Since $\mathbb{R}$ is complete, we can conclude that for each point $c \in A$, the sequnce of real numbers $(f_n(c))_{n\geq 1}$ converges. This tells us that the sequence of functions $(f_n)_{n\geq 1}$ is pointwise convergent on A. Let's denote the pointwise limit of $(f_n)_{n\geq 1}$ by $f : A \to \mathbb{R}$. In what follows, we will prove $f_n \overset{u}{\to} f$ on A. To this end, we need to show

$$\forall n > N \ \forall c \in A \quad |f_n(c) - f(c)| < \epsilon.$$

It follows from the assumption that for the given $\epsilon > 0$, $\exists \hat{N}$ such that

$$\forall n, m > \hat{N} \ \forall c \in A \quad |f_n(c) - f_m(c)| < \frac{\epsilon}{2}.$$

Then, if $n > \hat{N}$ and $c \in A$ we have

$$\begin{aligned} &\forall m > \hat{N} \quad |f_n(c) - f_m(c)| < \frac{\epsilon}{2} \\ \implies &\lim_{n\to\infty} |f_n(c) - f_m(c)| \leq \lim_{n\to\infty} \frac{\epsilon}{2} \\ \implies &|f_n(c) - f(c)| \leq \frac{\epsilon}{2} < \epsilon. \end{aligned}$$

□

Theorem 3.2.4. (Uniform Continuity and Differentiability)
Suppose for each $n \in \mathbb{N}$, $f_n : [a, b] \to \mathbb{R}$ is differentiable, $f_n \to f$ pointwise on $[a, b]$, and $f_n' \overset{u}{\to} g$ on $[a, b]$. Then f is differentiable on $[a, b]$ and $f' = g$.

Proof. Our goal is to show that $\forall c \in [a, b]$, $f'(c) = g(c)$. Let $c \in [a, b]$. We want to show

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} = g(c).$$

That is, we want to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that if } 0 < \underset{x\in[a,b]}{|x - c|} < \delta, \text{ then } \left| \frac{f(x) - f(c)}{x - c} - g(c) \right| < \epsilon. \qquad (*)$$

Let $\epsilon > 0$ be given.

Informal Discussion

Strategy: Bound $\left|\frac{f(x)-f(c)}{x-c} - g(c)\right|$ with a combintation of terms over which we control.

$*)$ f_n is differentiable $\implies$ we control $\left|\frac{f_n(x)-f_n(c)}{x-c} - f_n'(c)\right|$.

$*)$ $f_n' \overset{u}{\to} g \implies$ we control $|f_n'(z) - g(z)|$ for all $z \in [a,b]$.

$*)$ $f_n \to f$ pointwise $\implies$ we control $|f_n(z) - f(z)|$ at each $z \in [a,b]$.

Note that:

$$\begin{aligned}\left|\frac{f(x)-f(c)}{x-c} - g(c)\right| &= \left|\frac{f(x)-f(c)}{x-c} - \frac{f_n(x)-f_n(c)}{x-c} + \frac{f_n(x)-f_n(c)}{x-c} - f_n'(c) + f_n'(c) - g(c)\right| \\ &\le \left|\frac{f(x)-f(c)}{x-c} - \frac{f_n(x)-f_n(c)}{x-c}\right| + \left|\frac{f_n(x)-f_n(c)}{x-c} - f_n'(c)\right| + |f_n'(c) - g(c)|.\end{aligned}$$

In order to ensure $\left|\frac{f(x)-f(c)}{x-c} - g(c)\right| < \epsilon$, it's enough to make each term less than $\epsilon/3$. We will find an f_n that forces the first and third term to be less than $\epsilon/3$. We use the differentiability of f_n at c to find a δ that makes the middle term less than $\epsilon/3$ for $x \in [a,b]$ with $0 < |x-c| < \delta$.

(1) Since $f_n' \overset{u}{\to} g$, $\exists N_1$ such that $\forall n > N_1 \quad \forall z \in [a,b] \quad |f_n'(z) - g(z)| < \epsilon/3$.
(2) Since $f_n' \overset{u}{\to} g$, by the Cauchy Criterion $\exists N_2$ such that $\forall n, m > N_2 \quad \forall z \in [a,b] \quad |f_n'(z) - f_m'(z)| < \epsilon/3$.
Let $N = \max\{N_1, N_2\} + 1$.
(3) f_n is differentiable at c, so for the given ϵ

$$\exists \hat{\delta} > 0 \text{ such that if } x \in [a,b] \text{ and } 0 < |x-c| < \hat{\delta}, \text{ then } \left|\frac{f_N(x)-f_N(c)}{x-c} - f_N'(c)\right| < \epsilon/3.$$

We claim this $\hat{\delta}$ can be used as to satisfy $(*)$. Indeed, if $x \in [a,b]$ and $0 < |x-c| < \hat{\delta}$, then

$$\begin{aligned}\left|\frac{f(x)-f(c)}{x-c} - g(c)\right| &\le \left|\frac{f(x)-f(c)}{x-c} - \frac{f_N(x)-f_N(c)}{x-c}\right| + \left|\frac{f_N(x)-f_N(c)}{x-c} - f_N'(c)\right| + |f_N'(c) - g(c)| \\ &< \left|\frac{(f-f_N)(x)-(f-f_N)(c)}{x-c}\right| + \epsilon/3 + \epsilon/3. \qquad ((1) \text{ and } (3))\end{aligned}$$

We consider two cases: $x < c$ and $x > c$.

Case 1: $x \in [a,b]$, $x < c$.
For every $m > N$, apply the Mean Value Theorem to $f_m - f_N$ on $[x,c]$:

$$\begin{aligned}\forall m > N \ \exists \alpha_m \in [x,c] \text{ such that } (f_m - f_N)'(\alpha_m) &= \frac{(f_m-f_N)(c)-(f_m-f_N)(x)}{c-x} \\ \implies \left|\frac{(f_m-f_N)(c)-(f_m-f_N)(x)}{c-x}\right| &= |f_m'(\alpha_m) - f_N'(\alpha_m)| \\ &< \epsilon/3. \qquad ((2))\end{aligned}$$

By the order limit theorem with $m \to \infty$, we have

$$\begin{aligned}&\left|\frac{(fm-f_N)(c)-(fm-f_N)(x)}{c-x}\right| \le \epsilon/3 \\ \implies &\left|\frac{(f-f_N)(c)-(f-f_N)(x)}{c-x}\right| \le \epsilon/3,\end{aligned}$$

as desired.

Case 2: $x \in [a,b]$, $x > c$. Analogous to case 1.

□

Lemma 3.2.1. Let $A \neq \emptyset$ and. $f : A \to \mathbb{R}$. Suppose $(f_n : A \to \mathbb{R})$ is a sequence of functions. The following are equivalent:

(i) $f_n \overset{u}{\to} f$ on A.

(ii) $\lim_{n\to\infty} \sup_{x\in A} |f_n(x) - f(x)| = 0$.

Lemma 3.2.2. Suppose $\alpha : [a, b] \to \mathbb{R}$ is increasing, $f : [a, b] \to \mathbb{R}$ and $g : [a, b] \to \mathbb{R}$ are bounded, and $f \leq g$. Then

$$L(f, \alpha) \leq L(g, \alpha)$$
$$U(f, \alpha) \leq U(g, \alpha)$$

Theorem 3.2.5. (Uniform Continuity and Integrability)
Suppose $\alpha : [a, b] \to \mathbb{R}$ is increasing. For each $n \in \mathbb{N}$, let $f_n \in R_\alpha[a, b]$ with $f_n \overset{u}{\to} f$ on $[a, b]$. Then $f \in R_\alpha[a, b]$ and $\int_a^b f d\alpha = \lim_{n\to\infty} \int_a^b f_n d\alpha$.

Proof. Since uniform convergence preserves boundedness, we can conclude that $f : [a, b] \to \mathbb{R}$ is bounded. Now, in order to show that $f \in R_\alpha[a, b]$, it is enough to prove that $L(f, \alpha) = U(f, \alpha)$. For each $n \in \mathbb{N}$, let

$$r_n = x \in \sup_{[a,b]} |f_n(x) - f(x)|.$$

Since $f_n \to f$ uniformly, we know that $\lim_{n\to\infty} r_n = 0$. For each $n \in \mathbb{N}$, we have

$$r_n = \sup_{x\in[a,b]} |f_n(x) - f(x)| \implies \forall x \in [a, b] \;\; |f(x) - f_n(x)| \leq r_n$$
$$\implies \forall x \in [a, b] \;\; f_n(x) - r_n \leq f(x) \leq f_n(x) + r_n. \qquad (*)$$

It follows from lemma 3.2.2 that

$$L(f_n - r_n, \alpha) \leq L(f, \alpha) \leq U(f, \alpha) \leq U(f_n + r_n, \alpha).$$

Thus

$$0 \leq U(f, \alpha) - L(f, \alpha) \leq U(f_n + r_n, \alpha) - L(f_n - r_n, \alpha).$$

Note that

$$\begin{aligned} U(f_n + r_n, \alpha) - L(f_n - r_n, \alpha) &= \int_a^b f_n + r_n d\alpha - \int_a^b f_n - r_n d\alpha \qquad (f_n\text{'s are integrable}) \\ &= \int_a^b (f_n + r_n) - (f_n - r_n) d\alpha \\ &= 2r_n \int_a^b d\alpha \\ &= 2r_n[\alpha(b) - \alpha(b)]. \end{aligned}$$

So,

$$0 \leq U(f, \alpha) - L(f, \alpha) \leq 2r_n[\alpha(b) - \alpha(a)] \to 0[\alpha(b) - \alpha(a)] = 0 \text{ as } n \to \infty.$$

By the squeeze theorem, $U(f, \alpha) = L(f, \alpha)$. From $(*)$,

$$\int_a^b f_n - r_n d\alpha \leq \int_a^b f d\alpha \leq \int_a^b f_n + r_n d\alpha,$$

so

$$\int_a^b -r_n d\alpha \leq \int_a^b f d\alpha - \int_a^b f_n d\alpha \leq \int_a^b r_n d\alpha.$$

Thus,

$$-r_n[\alpha(b) - \alpha(a)] \leq \int_a^b f d\alpha - \int_a^b f_n d\alpha \leq r_n[\alpha(b) - \alpha(a)].$$

By the squeeze theorem, $\lim_{n\to\infty} \left[\int_a^b f d\alpha - \int_a^b f_n d\alpha\right] = 0$. That is,

$$\lim_{n\to\infty} \int_a^b f_n d\alpha = \int_a^b f d\alpha$$

□

3.3 Series of Functions

Definition 3.3.1. (Series of Functions)
Let $(f_n : A \to \mathbb{R})_{n\geq 1}$ be a sequence of functions defined on $A \neq \emptyset$.

(i) An expression of the form

$$\sum_{n=1}^{\infty} f_n = f_1 + f_2 + ...$$

is called an infinite series of functions.

(ii) The functions $f_1, f_2, ...$ are the terms of the series.

(iii) The corresponding sequences of partial sums is defined by

$$s_m = \sum_{n=1}^{m} f_n.$$

That is,

$$\begin{aligned} s_1 &= f_1 \\ s_2 &= f_1 + f_2 \\ s_3 &= f_1 + f_2 + f_3 \\ &\vdots \\ s_m &= f_1 + ...f_m. \end{aligned}$$

(iv) We say the series converges pointwise on A to $f : A \to \mathbb{R}$ $\left(\sum_{n=1}^{\infty} f_n = f \text{ pointwise}\right)$ if

$$\lim_{n\to\infty} s_m = f \text{ pointwise on } A.$$

(v) We say that the series converfes uniformly on A to $f : A \to \mathbb{R}$ $\left(\sum_{n=1}^{\infty} f_n = f \text{ uniformly}\right)$ if

$$s_m \to f \text{ uniformly on } A.$$

Observation 3.3.1. Note that the convergence of $\sum_{n=1}^{\infty} f_n$ is defined in terms of the convergence of the corresponding convergence of partial sums. So, all questions about the convergence of series of functions can be viewed as the convergence of sequences of functions.
As we shall now see, every question about convergence of a sequence can also be translated into a question about convergence of a series.

Consider the sequence $(g_n : A \to \mathbb{R})_{n\geq 1}$. For each n, define

$$\begin{aligned} f_1 &= g_1 \\ f_2 &= g_2 - g_1 \\ f_3 &= g_3 - g_2 \\ &\vdots \\ f_n &= g_n - g_{n-1} \\ &\vdots \end{aligned}$$

Now, consider the series $\sum_{n=1}^{\infty} f_n$. The sequence of partial sums for $\sum_{n=1}^{\infty} f_n$ is exactly the same as (g_n):

$$\begin{aligned} s_1 &= f_1 = g_1 \\ s_2 &= f_1 + f_2 = g_2 \\ s_3 &= f_1 + f_2 + f_3 = g_3 \\ &\vdots \\ s_m &= f_1 + \ldots + f_m = g_m \\ &\vdots \end{aligned}$$

So, we may conclude that

$$\sum_{n=1}^{\infty} f_n \text{ converges} \iff (s_n) \text{ converges} \iff (g_n) \text{ converges}$$

Example 3.3.1. Let $g_n(x) = \dfrac{\sin(nx)}{n}$.

$$\begin{aligned} f_1 &= g_1 = \sin x \\ f_2 &= g_2 - g_1 = \frac{\sin 2x}{2} - \sin x \\ f_3 &= g_3 - g_2 = \frac{\sin 3x}{3} - \frac{\sin 2x}{2} \\ &\vdots \\ f_n &= g_n - g_{n-1} = \frac{\sin(nx)}{n} - \frac{\sin[(n-1)x]}{n-1} \end{aligned}$$

Then

$$(g_n) \text{ converges} \iff \sum_{n=1}^{\infty} f_n \text{ converges}$$

This observation allows us to translate all the interesting examples we studied for sequences of functions into examples of series of functions. For example, we discussed a sequence of continuous functions (g_n) such that $g_N \to g$ pointwise but g is not continuous:

$$g_n : [0,1] \to \mathbb{R},\ g_n(x) = x^n,\ g(x) \begin{cases} 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1 \end{cases}$$

Now, defining

$$\begin{aligned} f_1 &= g_1 = x \\ f_2 &= g_2 - g_1 = x^2 - x \\ f_3 &= g_3 - g_2 = x^3 - x^2 \\ &\vdots \end{aligned}$$

then $\sum_{n=1}^{\infty} f_n$ is a series of continuous functions that converges to $f(x) = \begin{cases} 0 & \text{if } 0 \le x < 1 \\ 1 & \text{if } x = 1 \end{cases}$, which is NOT continuous.

Observation 3.3.2. Consider the infinite series $\sum_{n=1}^{\infty} f_n$. Let (s_m) be the corresponding sequence of partial sums. Recall that

$$\sum_{n=1}^{\infty} f_n \text{ converges} \iff (s_m) \text{ converges.}$$

Note that

$*$) It follows from the algebraic continuity theorem that if each f_n is continuous, then the partial sums $s_1, s_2, s_3, \ldots$ will be continuous.

$*$) It follows from the algebraic differentiability theorem that if each f_n is differentiable, then the partial sums $s_1, s_2, s_3, ...$ will be differentiable.

$*$) It follows from the algebraic properties of integrals that if each $f_n \in R_\alpha[a,b]$, then the partial sums $s_1, s_2, s_3, ...$ will be $R_\alpha[a,b]$.

As a result, we can immediately translate the claims about series of functions into claims of sequences of functions and then prove the claims using the results we already proved for sequences.

Theorem 3.3.1. (Term-by-term Continuity Theorem)
Suppose $A \subseteq (X,d)$ is nonempty. For each $n \in \mathbb{N}$, let $f_n : A \to \mathbb{R}$ is continuous and suppose $\sum_{n=1}^{\infty} f_n$ converges uniformly to $f : A \to \mathbb{R}$. Then $f : A \to \mathbb{R}$ is continuous.

Proof. Apply the corresponding theorem for sequences of functions to the sequence of partial sums $s_m = f_1 + ... + f_m$. □

Theorem 3.3.2. (Term-by-term Differentiability Theorem)
Suppose for each $n \in \mathbb{N}$, $f_n : [a,b] \to \mathbb{R}$ is differentiable. Suppose $\sum_{n=1}^{\infty} f_n = f$ pointwise on $[a,b]$, and that $\sum_{n=1} \infty f_n$ converges uniformly on $[a,b]$. Then f is differenitable on $[a,b]$ and $(\sum_{n=1}^{\infty} f_n)' = \sum_{n=1}^{\infty} f_n'$.

Proof. Apply the corresponding theorem for sequences of functions to the sequence of partial sums. □

Theorem 3.3.3. (Term-by-term Integrability Theorem)
Suppose $\alpha : [a,b] \to \mathbb{R}$ is increasing. For each $n \geq 1$, suppose $f_n \in R_\alpha[a,b]$. Suppose $\sum_{n=1}^{\infty} f_n$ converges uniformly to $f : [a,b] \to \mathbb{R}$. Then $f \in R_\alpha[a,b]$ and $\int_a^b \sum_{n=1}^{\infty} f_n d\alpha = \sum_{n=1}^{\infty} \int_a^b f_n d\alpha$.

Proof. Apply the corresponding theorem for sequences of functions to the sequence of partial sums. □

Theorem 3.3.4. (Cauchy Criterion for Uniform Convergence of Series of Functions)
Let $A \neq \emptyset$ and suppose for each $k \in \mathbb{N}$, $f_k : A \to \mathbb{R}$. Then

$$\sum_{k=1}^{\infty} f_k \text{ converges uniformly} \iff \forall \epsilon > 0 \ \exists N \text{ such that } \forall n > m > N \ \forall c \in A \ \left| \sum_{k=m+1}^{n} f_k(c) \right| < \epsilon.$$

Proof. Let $s_m = f_1 + ... + f_m$. We have

$$\begin{aligned}
\sum_{k=1}^{\infty} f_k \text{ converges uniformly} &\iff (s_m) \text{ converges uniformly} \\
&\iff (s_m) \text{ satisfies Cauchy Criterion for sequences of functions} \\
&\iff \forall \epsilon > 0 \ \exists N \text{ such that } \forall n, m > N \ \forall c \in A \ \ |s_n(c) - s_m(c)| < \epsilon \\
&\iff \forall \epsilon > 0 \ \exists N \text{ such that } \forall n > m > N \ \forall c \in A \ \ |s_n(c) - s_m(c)| < \epsilon \\
&\iff \forall \epsilon > 0 \ \exists N \text{ such that } \forall n > m > N \ \forall c \in A \ \left| \sum_{k=m+1}^{n} f_k(c) \right| < \epsilon.
\end{aligned}$$

□

Theorem 3.3.5. (Weierstrass M-Test)

Version 1: Let $A \neq \emptyset$ and $\forall n \in \mathbb{N}$ let $f_n : A \to \mathbb{R}$. Suppose $\forall n \in \mathbb{N} \ \exists M_n$ such that $\forall c \in A \ |f_n(c)| \leq M_n$, and $\sum_{n=1}^{\infty} M_n$ converges. Then $\sum_{n=1}^{\infty} f_n$ converges uniformly on A.

Version 2: Let $A \neq \emptyset$ and suppose $\forall n \in \mathbb{N} \ \ f_n : A \to \mathbb{R}$ is bounded. $\forall n \in \mathbb{N}$, let $M_n = \sup_{x \in A} |f_n(x)|$ and suppose $\sum_{n=1}^{\infty} M_n$ converges. Then $\sum_{n=1}^{\infty} f_n$ converges uniformly on A.

Proof. We prove version 1. According to the Cauchy Criterion of uniform convergence of series of functions, it is enough to show that

$$\forall \epsilon > 0 \ \exists N \text{ such that } \forall n > m > N \ \forall c \in A \quad \left| \sum_{k=m+1}^{n} f_k(c) \right| < \epsilon \tag{$*$}$$

Since, by assumption, $\sum_{n=1}^{\infty} M_n$ converges, for this ϵ there exists $\hat{N}$ such that

$$\forall n > m > \hat{N} \quad \left| \sum_{k=m+1}^{\infty} M_k \right| < \epsilon.$$

We claim we can use this $\hat{N}$ as the N we're looking for. Indeed, if $N = \hat{N}$, then $(*)$ holds:

$$\forall n > m > \hat{N} \ \forall c \in A \quad \left| \sum_{k=m+1}^{\infty} f_k(c) \right| \leq \sum_{k=m+1}^{\infty} |f_k(c)| \leq \sum_{k=m+1}^{n} M_k = \left| \sum_{k=m+1}^{n} M_k \right| < \epsilon.$$

□

Example 3.3.2. (i) Prove $\sum_{n=1}^{\infty} \frac{\cos(2^n x)}{2^n}$ converges uniformly on $\mathbb{R}$.

$$\left.\begin{array}{l} \forall n \quad \left|\frac{\cos(2^n x)}{2^n}\right| \leq \frac{1}{2^n} \\ \sum_{n=1}^{\infty} \frac{1}{2^n} \text{ converges} \end{array}\right\} \implies \sum_{n=1}^{\infty} \frac{\cos(2^n x)}{2^n} \text{ converges uniformly by the Weierstrass M-test.}$$

(ii) Let $f_n : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = \sum_{n=1} \infty \frac{\cos(2^n x)}{2^n}$. Explain why f is continuous.

$$\left.\begin{array}{l} \forall n \quad \frac{\cos(2^n x)}{2^n} \text{ is continuous on } \mathbb{R} \\ \sum_{n=1}^{\infty} \frac{\cos(2^n x)}{2^n} \text{ converges uniformly} \end{array}\right\} \implies f(x) = \sum_{n=1}^{\infty} \frac{\cos(2^n x)}{2^n} \text{ is continuous.}$$

3.4 Arzela-Ascoli Theorem

Does continuity preserve uniform convergence? No. Consider

$$\forall n \in \mathbb{N} \quad f_n : (0,1) \to \mathbb{R}, \ f_n(x) = \frac{1}{nx+1}.$$

Then

$*$) $f_n \to 0$ pointwise.

$*$) Each f_n is continuous.

$*$) 0 is continuous.

But, $f_n \to f$ is not uniform. Note

$$f_n \overset{u}{\to} 0 \iff \lim_{n\to\infty} \sup_{x\in(0,1)} |f_n(x) - 0| = 0.$$

However,

$$\begin{aligned} &\forall n \quad f_n\left(\frac{1}{n}\right) = \frac{1}{2} \\ \implies &\forall n \geq 2 \quad \sup_{x\in(0,1)} |f_n(x)| \geq f_n\left(\frac{1}{n}\right) = \frac{1}{2} \\ \implies &\lim_{n\to\infty} \sup_{x\in(0,1)} |f_n(x) - 0| \neq 0. \end{aligned}$$

Theorem 3.4.1. (Dini's Theorem)
Suppose (X,d) is a metric space, $K \subseteq X$ is compact, for each $n \in \mathbb{N}$ $f_n : K \to \mathbb{R}$ is continuous, $f_n \to f$ pointwise on K where $f : K \to \mathbb{R}$ is a function, and $\forall n \in \mathbb{N} \quad f_{n+1} \leq f_n$. Then $f_n \overset{u}{\to} f$ on K.

Proof. Let $\epsilon > 0$ be given. Our goal is to show there exists N such that

$$\forall n > N \ \forall x \in K \quad |f_n(x) - f(x)| < \epsilon.$$

For each $n \in \mathbb{N}$, let $g_n = f_n - f$. So, our goal is to show there exists N such that

$$\forall n > N \ \forall x \in K \quad |g_n| < \epsilon.$$

Observation 1: $\forall n \ g_n \geq 0$.
For each $x \in K$, $(f_n(x))_{n\geq 1}$ is a decreasing sequence of real numbers that converges to $f(x)$. It follows from the monotone convergence theorem that

$$f(x) = \inf\{f_n(x) : n \in \mathbb{N}\}.$$

Thus

$$\forall n \quad f(x) \leq f_n(x).$$

Therefore, for all n, $g_n \geq 0$. So, our goal is to show $\exists N$ such that

$$\forall n > N \ \forall x \in K \quad g_n(x) < \epsilon.$$

That is, our goal is to show $\exists N$ such that

$$\forall n > N \quad \{x \in K : g_n(x) \geq \epsilon\} = \emptyset.$$

Our goal is to show $\exists N$ such that

$$\forall n > N \quad g_n^{-1}([\epsilon, \infty)) = \emptyset.$$

For each $n \in \mathbb{N}$, let $K_n = g_n^{-1}([\epsilon, \infty))$. Our goal is to show $\exists N$ such that

$$\forall n > N \quad K_n = \emptyset.$$

Observation 2: For each n, K_n is compact.
For each n,

$$\left.\begin{array}{l} g_n : K \to \mathbb{R} \text{ is continuous} \\ [\epsilon, \infty) \text{ is closed} \end{array}\right\} \implies K_n = g_n^{-1}([\epsilon, \infty)) \text{ is closed in } K.$$

Thus,

$$\left.\begin{array}{l} K \text{ is compact} \\ K_n \subseteq K \\ K_n \text{ is closed} \end{array}\right\} \implies K_n \text{ is compact.}$$

Observation 3: $K_1 \supseteq K_2 \supseteq K_3 \supseteq \ldots$ $(\forall n \;\; k_{n+1} \subseteq K_n)$.

$$x \in K_{n+1} \implies g_{n+1}(x) \geq \epsilon \implies g_n(x) \geq \epsilon \implies x \in K_n.$$

So, it is enough to show $\exists N \in \mathbb{N}$ such that $K_N = \emptyset$. $(\forall n > N, K_n \subseteq K_N = \emptyset)$. Assume for contradiction that $\forall n$, $K_n \neq \emptyset$. Then

$$\left.\begin{array}{l} K_1 \supseteq K_2 \supseteq K_3 \supseteq \ldots \\ \forall n \;\; K_n \text{ is compact} \\ \forall n \;\; K_n \neq \emptyset \end{array}\right\} \implies \bigcap_{n=1}^{\infty} K_n \neq \emptyset \qquad \text{(Nested compact set property)}$$

Therefore, $\exists x \in \bigcap_{n=1}^{\infty} K_n$, that is, $\exists x \in K$ such that

$$\forall n \in \mathbb{N} \;\; g_n(x) \geq \epsilon.$$

This contradicts the fact that $g_n(x) \to 0$ pointwise. □

Example 3.4.1. For all $n \in \mathbb{N}$ let $f_n : [0, 2\pi] \to \mathbb{R}$, $f_n(x) = \sin(nx)$. Clearly,

$$\forall n \in \mathbb{N} \; \forall x \in [0, 2\pi] \;\; |f_n(x)| = |\sin(nx)| \leq 1.$$

So, $(f_n)_{n\geq 1}$ is a uniformly bounded sequence of functions. In homework, we proved that $(f_n)_{n\geq 1}$ does not have any subsequence $(f_{n_k})_{k\geq 1}$ that converges.

Definition 3.4.1. (Equicontinuous Sequence of Functions)
Let $A \subseteq (X, d)$. A sequence of functions $(f_n : A \to \mathbb{R})_{n\geq 1}$ is said to be equicontinuous if

$$\forall \epsilon > 0 \; \exists \delta_\epsilon > 0 \text{ such that } \forall n \in \mathbb{N} \; \forall x, c \in A \text{ if } d(x, c) < \delta_\epsilon \text{ then } |f_n(x) - f_n(c)| < \epsilon.$$

Fact 3.4.1. If $(f_n : K \to \mathbb{R})$ converges uniformly (K is compact) and each f_n is continuous, then (f_n) is equicontinuous.

Fact 3.4.2. If $(f_n : K \to \mathbb{R})$ is equicontinuous and $f_n \to f$ pointwise, then $f_n \xrightarrow{u} f$.

Example 3.4.2. For each $n \in \mathbb{N}$, let $f_n : [0, 1] \to \mathbb{R}$, $f_n(x) = \frac{x}{n}$. Previously, we proved $f_n \xrightarrow{u} 0$. We claim that f_n is equicontinuous on $[0, 1]$. We need to show that

$$\forall \epsilon > 0 \; \exists \delta > 0 \text{ such that } \forall n \in \mathbb{N} \; \forall x, c \in [0, 1] \text{ if } |x - c| < \delta \text{ then } |f_n(x) - f_n(c)| < \epsilon.$$

Let $\epsilon > 0$ be given. Our goal is to find δ such that

$$\forall n \in \mathbb{N} \; \forall x, c \in [0, 1] \text{ if } |x - c| < \delta \text{ then } \frac{|x - c|}{n} < \epsilon. \qquad (*)$$

Clearly, $(*)$ holds for $\delta = \epsilon$.

Example 3.4.3. For all $n \geq 1$, let $f_n : [0, 1] \to \mathbb{R}$, $f_n(x) = x^n$. We claim that $(f_n)_{n\geq 1}$ is not equicontinuous. For the sake of contradiction, suppose $(f_n)_{n\geq 1}$ is equicontinuous. Then

$$\forall \epsilon > 0 \; \exists \delta > 0 \text{ such that } \forall n \in \mathbb{N} \; \forall x, c \in [0, 1] \text{ if } |x - c| < \delta \text{ then } |f_n(x) - f_n(c)| < \epsilon.$$

In particular, for $\epsilon = \frac{1}{2}$, ther exists $\delta > 0$ such that

$$\forall n \in \mathbb{N} \; \forall x, c \in [0, 1] \text{ if } |x - c| < \delta \text{ then } |x^n - c^n| < \frac{1}{2}.$$

In particular, if we let $c = 1$, we get

$$\forall n \in \mathbb{N}\ \forall x \in [0,1] \text{ if } |x-1| < \delta \text{ then } |x^n - 1| < \frac{1}{2}.$$

Let $x_0 \in (0,1)$ such that $|x_0 - 1| < \delta$. Then

$$\begin{aligned} &\forall n \in \mathbb{N} \quad |x_0^n - 1| < \frac{1}{2} \\ \implies &\lim_{n\to\infty} |x_0^n - 1| \leq \lim_{n\to\infty} \frac{1}{2} \\ \implies &|0-1| \leq \frac{1}{2} \\ \implies &1 \leq \frac{1}{2}, \end{aligned}$$

a contradiction.

Theorem 3.4.2. (The Arzela-Ascoli Theorem)
Suppose (X,d) is a metric space, $K \subseteq X$ is infinite and compact, $(f_n : K \to \mathbb{R})_{n\geq 1}$ is uniformly bounded, $(f_n : K \to \mathbb{R})_{n\geq 1}$ is equicontinuous. Then (f_n) has a uniformly convergent subsequence.

Proof. Our proof consists of two main steps.
Step 1: Let E be a countable dense subset of K. We will use the assumption that (f_n) is uniformly bounded to show that there exists a subsequence (f_{n_k}) of (f_n) that converges to each point of E. To simplify the notation, we let $g_k = f_{n_k}$.
Basic Idea of Proof of Step 1: Cantor's diagonal process.
Step 2: We will use the assumption that (f_n) is equicontinuous to prove that (g_k) is uniformly convergent on K.
Basic Idea of Proof of Step 2: (g_k) fulfills the Cauchy Criterion for uniform convergence:

$$\forall \epsilon > 0\ \exists N \text{ such that } \forall m,n > N\ \forall x \in K \quad |g_m(x) - g_n(x)| < \epsilon.$$

Note that for each $x \in K$ and $r \in E$ we have

$$|g_m(x) - g_n(x)| \leq |g_m(x) - g_m(r)| + |g_m(r) - g_n(r)| + |g_n(r) - g_n(x)|.$$

Let's begin with the claim of **step 2.** Suppose $(g_k)_{k\geq 1}$ is equicontinuous, E is a countable dense subset of K, and for each $r \in E$ the sequence of numbers $(g_k(r))_{k\geq 1}$ converges. Let $\epsilon > 0$ be given. Our goal is to find N such that

$$\forall m,n > N,\ \forall x \in K \quad |g_m(x) - g_n(x)| < \epsilon. \tag{$*$}$$

(i) (g_k) is equicontinuous, so for the given $\epsilon\ \exists \delta > 0$ such that

$$\forall k \in \mathbb{N},\ \forall x,c \in K,\ \text{if } d(x,c) < \delta, \text{ then } |g_k(x) - g_k(c)| < \epsilon/3.$$

(ii) For each $r \in E$, the sequence of numbers $(g_k(r))$ is convergent, and so it's Cauchy. Hence

$$\forall r \in E\ \exists N_r \text{ such that } \forall m,n > N_r \quad |g_m(r) - g_n(r)| < \epsilon/3.$$

Notice that, since E is dense in K,

$$K \subseteq \bigcup_{r\in E} B_\delta(r).$$

So, $\{B_\delta(r)\}_r \in E$ is an open cover of K. Since K is compact, this open cover has a finite subcover: $\exists r_1, ..., r_l \in E$ such that $K \subseteq [B_\delta(r_1) \cup ... \cup B_\delta(r_l)]$ (so, every point of K is within δ of at least one $r_1, ..., r_l$). We claim that $\max\{N_{r_1}, ..., N_{r_l}\}$ can be used as the N we are looking for. Indeed, if we let $N = \max\{N_{r_1}, ..., N_{r_l}\}$, then $(*)$ holds. Suppose $m,n > N$ and $x \in K$. Since $x \in K$, $\exists i \in \{1, ..., l\}$ such that $x \in B_\delta(r_i)$. We have

$$\begin{aligned} |g_m(x) - g_n(x)| &= |g_m(x) - g_m(r_i) + g_m(r_i) - g_n(r_i) + g_n(r_i) - g_n(x)| \\ &\leq |g_m(x) - g_m(r_i)| + |g_m(r_i) - g_n(r_i)| + |g_n(r_i) - g_n(x)| \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 \\ &= \epsilon. \end{aligned}$$

as desired.
Now, we prove **step 1**.
Suppose E is a countable dense subset of K and (f_n) is uniformly bounded. Our goal is to show there is a subsequence of (f_n) that converges at each point of $r \in E$. In what follows, we will prove a more general statement: suppose $E = \{x_1, x_2, x_3, ...\}$ is some countable subset of K and $(f_n : K \to \mathbb{R})_{n\geq 1}$ is pointwise bounded. Then (f_n) has a subsequence that converges at each point $r \in E$.

$*$) The sequence of real numbers $(f_n(x_1))_{n\geq 1}$ is bounded. By Bolzano-Weierstrass, it has a subsequence $(f_{n_k}(x_1))_{n\geq 1}$ that converges. To emphasize that this sequence is generated by considering the values at x_1, we use the notation

$$f_{1,k} = f_{n_k}.$$

$*$) Since $(f_{1,k}(x_2))_{k\geq 1}$ is bounded, by Bolzano-Weierstrass it has a convergent subsequence $(f_{2,k}(x_2))_{k\geq 1}$.

$*$) Since $(f_{2,k}(x_3))_{k\geq 1}$ is bounded, by Bolzano-Weierstrass it has a convergent subsequence $(f_{3,k}(x_2))_{k\geq 1}$.

In general, $\forall m \geq 2$ the sequence $(f_{m-1,k})_{k\geq 1}$ has a convergent subsequence $(f_{m,k}(x_m))_{k\geq 1}$. In this way, we obtain the following table of functions:

$$\begin{array}{cccc} f_{1,1} & f_{1,2} & f_{1,3} & \dots \quad \text{(Converges at } x_1\text{)} \\ f_{2,1} & f_{2,2} & f_{2,3} & \dots \quad \text{(Converges at } x_2\text{)} \\ f_{3,1} & f_{3,2} & f_{3,3} & \dots \quad \text{(Converges at } x_3\text{)} \\ \vdots & \vdots & \vdots & \\ f_{m,1} & f_{m,2} & f_{m,3} & \dots \quad \text{(Converges at } x_m\text{)} \\ \vdots & \vdots & \vdots & \end{array}$$

(Note that if $\hat{m} > m$, then $(f_{\hat{m},k})_{k\geq 1}$ is a subsequence of $(f_{m,k})_{k\geq 1}$). Now, consider the diagonal sequence $f_{1,1}, f_{2,2}, f_{3,3}, ...$ Let $g_k = f_{k,k}$ for all $k \in \mathbb{N}$. We claim (g_k) converges at x_i forall $i \in N$. Indeed, let $i \in \mathbb{N}$. Note that

$$g_{i+1}, g_{i+2}, g_{i+3}, \cdots$$

is a subsequence of $f_{i,1}, f_{i,2}, f_{i,3}, ...$ By contsruction, $f_{i,1}(x_i), f_{i,2}(x_i), f_{i,3}(x_i), ...$ converges. Since $g_{i+1}(x_i), g_{i,2}(x_i), g_{i,3}(x_i), ...$ is a subsequence of the above sequence, it also converges. Thus

$$g_1(x_i), g_2(x_i), g_3(x_i), ..., g_i(x_i), g_{i+1}(x_i), g_{i+2}(x_i), ...$$

also converges. □

Remark 3.4.1. Let $A \subseteq (X, d)$ be nonempty. A family $\mathcal{F}$ of real-valued functions defined on A is said to be equicontinuous on A if

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \forall f \in \mathcal{F} \ \forall x, c \in A \text{ if } d(x, c) < \delta_\epsilon \text{ then } |f(x) - f(c)| < \epsilon.$$

Note that each member of an equicontinuous family is uniformly continuous.

Remark 3.4.2. Let X be any infinite set of functions and equip X with a metric d. Each such metric "d" introduces a mode of convergence for sequences of functions in X. The sequence $(f_n)_{n\geq 1}$ in X converges to $f \in X$ if and only if $\forall \epsilon > 0 \ \exists N$ such that $\forall n > N \ \ d(f_n, f) < \epsilon$.

Example 3.4.4. Let $X = C([a, b]; \mathbb{R}) = \{f : E \to \mathbb{R} : f \text{ is continuous an bounded}\}$. The following are examples of metrics on X.

(i) $d_1(f, g) = \int_a^b |f - g| dx$. ($L^1$ convergence)

(ii) $d_2(f, g) = \left(\int_a^b |f - g|^2 dx\right)^{\frac{1}{2}}$ .. (L^2 convergence)

More generally, if $p \geq 1$ then

$$d_p(f,g) = \left(\int_a^b |f-g|^p dx \right)^{\frac{1}{p}} \qquad (L^p \text{ convergence})$$

Example 3.4.5. Let (E,d) be a metric space. Let $X = BC(E;\mathbb{R}) = \{f : E \to \mathbb{R} : f \text{ is continuous and bounded}\}$ and $d_\infty(f,g) = \sup_{x\in E}|f(x)-g(x)|$. Note that if $(f_n)_{n\geq 1}$ is a sequence in $BC(E;\mathbb{R})$ and $f \in BC(E;\mathbb{R})$, then

$$\begin{aligned} f_n \to f \text{ in } BC(E;\mathbb{R}) &\iff \forall \epsilon > 0\ \exists N \text{ such that } \forall n > N \quad d_\infty(f_n,f) < \epsilon \\ &\iff \forall \epsilon > 0\ \exists N \text{ such that } \forall n > N \quad \sup_{x\in E}|f_n(x)-f(x)| < \epsilon \\ &\iff f_n \to f \text{ uniformly on the set } E \end{aligned}$$

Note . If E is compact, then by the extreme value theorem

$$BC(E;\mathbb{R}) = C(E;\mathbb{R})$$

Theorem 3.4.3. ($BC(E;\mathbb{R})$ is Complete)
Let (E,d) be a metric space. Let $X = BC(E;\mathbb{R})$ be equipped with the d_∞ metric. Then (X,d_∞) is a complete metric space.

Proof. Let $(f_n)_{n\geq 1}$ be a cauchy sequence in (X,d_∞). Our goal is to show that $(f_n)_{n\geq 1}$ is convergent in (X,d_∞). We have

$$\begin{aligned} (f_n)_{n\geq 1} \text{ is cauchy} &\iff \forall \epsilon > 0\ \exists N \text{ such that } \forall n,m > N \quad d_\infty(f_n,f_m) < \epsilon \\ &\iff \epsilon > 0\ \exists N \text{ such that } \forall n,m > N \quad \sup_{x\in E}|f_n(x)-f_m(x)| < \epsilon \\ &\iff \forall \epsilon > 0\ \exists N \text{ such that } \forall n,m > N\ \forall x \in E \quad |f_n(x)-f_m(x)| < \epsilon \\ &\iff (f_n)_{n\geq 1} \text{ is uniformly convergent on } E \\ &\qquad\qquad \text{(Cauchy criterion for uniform convergence)} \end{aligned}$$

Let $f = \lim_{n\to\infty} f_n$. Since for each n, f_n is bounded and continuous, we conclude that f is bounded and continous. Thus $f \in X$. We have

$$\left.\begin{array}{l} (f_n) \subseteq X \\ f \in X \\ f_n \xrightarrow{u} f \end{array}\right\} \implies f_n \to f \text{ in } X.$$

□

3.5 Weierstrass Approximation Theorem

Definition 3.5.1. (Modulus of Continuity)
Let $g : [a, b] \to \mathbb{R}$ be a bounded function. The modulusof continuity of g is defined as follows

$$\forall 0 < r \leq b - a \quad w_g(r) = \sup_{\substack{|x-z| \leq r \\ x,z \in [a,b]}} |g(x) - g(z)|$$

Fact 3.5.1. If $g : [a, b] \to \mathbb{R}$ is continuous, then $w(r) \to 0$ as $r \to 0^+$.

Proof. Our goal is to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \text{ if } 0 < r < \delta \text{ then } |w(r) - 0| < \epsilon.$$

That is, we want to show

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \text{ if } 0 < r < \delta, \text{ then for } x, z \in [a, b] \text{ such that } |x-z| \leq r, \ |\sup |g(x) - g(z) - 0|| < \epsilon.$$

To this end, it is enough to show that

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \text{ if } 0 < r < \delta \text{ then for all } x, z \in [a, b] \text{ with } |x - z| \leq r, \quad |g(x) - g(z)| < \epsilon.$$

But this is a direct consequence of uniform continuity of g on $[a, b]$:

$$\forall \epsilon > 0 \ \exists \delta > 0 \text{ such that } \forall x, z \in [a, b] \text{ if } |x - z| < \delta, \text{ then } |g(x) - g(z)| < \epsilon.$$

□

Fact 3.5.2. Let $g : [a, b] \to \mathbb{R}$ be a bounded function. Then

$$\forall x, z \in [a, b], \forall 0 < r \leq b - a \quad |g(x) - g(z)| \leq \left[\frac{|x - z|}{r} + 1\right] w(r)$$

Proof. **Case 1:** $|x - z| \leq r$

$$|g(x) - g(z)| \leq w(r) \leq \left[\frac{|x - z|}{r} + 1\right] w(r)$$

Case 2: $|x - z| > r$ (without loss of generality, $x < z$)
Let $m \in \mathbb{N}$ be such that $x + mr \leq z < x + (m + 1)r$. We have

$$\begin{aligned}
|g(x) - g(z)| &= |g(x) - g(x + r) + g(x + r) - g(x + 2r) + ... + g(x + mr) - g(z)| \\
&\leq |g(x) - g(x + r)| + |g(x + r) - g(x + 2r)| + ... + |g(x + mr) - g(z)| \\
&\leq w(r) + w(r) + ... + w(r) \\
&= (m + 1)w(r) \\
&\leq \left[\frac{|z - x|}{r} + 1\right] w(r). \qquad (x + mr \leq z \implies m \leq \tfrac{z-x}{r})
\end{aligned}$$

□

Theorem 3.5.1. (Weierstrass Approximation Theorem)
If $f : [a, b] \to \mathbb{R}$ is continuous, then there exists a sequence of $p_1(x), p_2(x), p_3(x), ...$ of polynomials such that $d_\infty(p_n, f) \to 0$. (That is, $p_n \xrightarrow{u} f$ on $[a, b]$)

Proof. We will consider two cases: Case 1: $[a, b] = [0, 1]$.
Case 2: $[a, b] \neq [0, 1]$.

Case 1: Our proof is constructive. For each $n \in \mathbb{N}$, we will construct a polynomial $p_n(x)$ in $\mathbb{P}_n$ (the vectors space of all polynomials of degree at most n, together with the 0 polynomial) as follows:

(i) Consider $n+1$ equally spaced meshpoints in the interval $[0,1]$ as follows:

$$0 = x_{0,n} < x_{1,n} < x_{2,n} < ... < x_{n,n} = 1, \quad x_{j,n} = \frac{j}{n}$$

(ii) Associate the jth meshpoint with a polynomial $\phi_{j,n}(x) \in \mathbb{P}_n$ such that

(1) $\phi_{j,n}(x) \geq 0 \quad \forall x \in [0,1]$

(2) $\displaystyle\sum_{j=0}^{n} \phi_{j,n}(x) = 1 \quad \forall x \in [0,1]$

(3) $\displaystyle\sum_{j=0}^{n} [x - x_{j,n}]^2 \, \phi_{j,n}(x) \leq \frac{M}{n}$ for some M independent of x and n.

(iii) Define $p_n(x) = \sum_{j=0}^{n} f(x_{j,n})\phi_{j,n}(x)$.

Claim 1: There exists polynomials $\phi_{0,n}(x), ..., \phi_{n,n}(x)$ in $\mathbb{P}_n$ that satisfy properties (1) and (3). Indeed, in HW 7, you will show that the functions:

$$\phi_{j,n}(x) = \binom{n}{j} x^j (1-x)^{n-j}$$

satisfies properties (1) to (3).

Claim 2: $d_\infty(f, p_n) \to 0$ as $n \to \infty$.
We have for all $x \in [0,1]$ and $n \in \mathbb{N}$,

$$\begin{aligned}
|f(x) - p_n(x)| &= |f(x) - \sum_{j=0}^{n} f(x_{j,n})\phi_{j,n}(x)| \\
&= \left| f(x) \sum_{j=0}^{n} \phi_{j,n}(x) - \sum_{j=0}^{n} f(x_{j,n})\phi_{j,n}(x) \right| && \text{(By (2) } \textstyle\sum_{j=0}^{n} \phi_{j,n}(x)) \\
&= \left| \sum_{j=0}^{n} f(x)\phi_{j,n}(x) - \sum_{j=0}^{n} f(x_{j,n})\phi_{j,n}(x) \right| \\
&= \left| \sum_{j=0}^{n} (f(x) - f(x_{j,n}))\phi_{j,n}(x) \right| \\
&\leq \sum_{j=0}^{n} |f(x) - f(x_{j,n})| \cdot \phi_{j,n}(x) && \text{(By (1), } \phi_{j,n} \geq 0)
\end{aligned}$$

Therefore, for all $x \in [0,1]$, $n \in \mathbb{N}$ and $0 < r \leq 1$ we have

$$\begin{aligned}
|f(x) - p_n(x)| &\leq \sum_{j=0}^{n} \left[\frac{|x - x_{j,n}|}{r} + 1 \right] w(r)\phi_{j,n}(x) \\
&= w(r) \left(\sum_{j=0}^{n} \frac{|x - x_{j,n}|}{r} \phi_{j,n}(x) + \sum_{j=0}^{n} \phi_{j,n}(x) \right) \\
&= \frac{w(r)}{r} \sum_{j=0}^{n} |x - x_{j,n}| \phi_{j,n}(x) + w(r)
\end{aligned}$$

Since each $\phi_{j,n}$ is greater than or equal to zero,we write

$$\begin{aligned}
\sum_{j=0}^{n} |x - x_{j,n}|\phi_{j,n}(x) &= \sum_{j=0}^{n} |x - x_{j,n}|\phi_{j,n}^{\frac{1}{2}}(x)\phi_{j,n}^{\frac{1}{2}}(x) \\
&\leq \left[\sum_{j=0}^{n} |x - x_{j,n}|^2\phi_{j,n}(x)\right]^{\frac{1}{2}} \left[\sum_{j=0}^{n} \phi_{j,n}(x)\right]^{\frac{1}{2}} && \text{(Cauchy-Schwarz)} \\
&= \sqrt{\sum_{j=0}^{n} |x - x_{j,n}|^2\phi_{j,n}(x) \cdot 1} \\
&\leq \sqrt{\frac{M}{N}} && \text{(By (3))}
\end{aligned}$$

Thus for all $x \in [0,1]$, $n \in \mathbb{N}$, $0 < r \leq 1$, we have

$$\begin{aligned}
|f(x) - p_n(x)| &\leq \frac{w(r)}{r}\sqrt{\frac{M}{N}} + w(r) \\
&= \left[\frac{\sqrt{M}}{r\sqrt{n}} + 1\right] w(r)
\end{aligned}$$

In particular, for $r = \frac{1}{\sqrt{n}}$, we will have

$$\begin{aligned}
&\forall x \in [0,1]\ \forall n \in \mathbb{N}\ \ |f(x) - p_n(x)| \leq \left[\sqrt{M} + 1\right] w\left(\frac{1}{\sqrt{n}}\right) \\
\implies &\forall n \in \mathbb{N}\ \ \max_{0\leq x\leq 1} |f(x) - p_n(x)| \leq \left[\sqrt{m} + 1\right] w\left(\frac{1}{\sqrt{n}}\right) \\
\implies &\ 0 \leq d_\infty(f, p_n) \leq \left[\sqrt{m} + 1\right] w\left(\frac{1}{\sqrt{n}}\right)
\end{aligned}$$

It follows from the squeeze theorem that $\lim_{n\to\infty} d_\infty(f, p_n) = 0$, as desired.

Case 2: $[a,b] \neq [0,1]$

Define $T : [0,1] \to [a,b]$ by $T(t) = a + t(b-a)$. Clearly, T and T^1 are both polynomials. Suppose $f : [a,b] \to \mathbb{R}$ is a continuous function. Then $f \circ T$ will be a continuous function on $[0,1]$. By case 1, there exists a sequence $q_1(x), q_2(x), q_3(x), \ldots$ of polynomials on $[0,1]$ such that $d_\infty(q_n, f \circ T) \to 0$. For each $n \in \mathbb{N}$, let $p_n = q_n \circ T^{-1}$. Since both q_n and T^{-1} are polynomials, we can conclude that p_n is also a polynomial on $[a,b]$. We have

$$\begin{aligned}
d_\infty(p_n, f) &= \max_{a\leq x\leq b} |p_n(x) - f(x)| \\
&= \max_{a\leq x\leq b} |(q_n \circ T^{-1}) - f(x)| \\
&= \max_{a\leq x\leq b} |q_n(T^{-1}) - f(x)| \\
&= \max_{a\leq x\leq b} |q_n(T^{-1}(x)) - (f \circ T)(T^{-1}(x))| \\
&= \max_{0\leq z\leq 1} |q_n(z) - (f \circ T)(z)| && \text{(Let } z = T^{-1}(z)\text{)} \\
&= d_\infty(q_n, f \circ T) \to 0 \text{ as } n \to \infty.
\end{aligned}$$

□